

# Chapter 1 : Basics for Electronics

## 1.1 Introduction

Electronics has been the key to the development of modern technology based civilization. The electronics experts have great contribution to build up this technology based civilization. A lot of researches are going on to improve the performance of electronics based systems and to invent more and more new electronics systems. Different institutions in Bangladesh and abroad, teaches electronics courses both theoretical and practical. The aim of this book is to discuss the theoretical and practical topics of electronics for undergraduate levels. This chapter does not include the main theories of electronics, rather discusses some topics that are directly or indirectly related to electronics and helps understanding the main theories of electronics.

## 1.2 Different Kinds of Formulas

In science, a formula is a concise way of expressing information symbolically, as in a *mathematical formula* or a *chemical formula*. The informal use of the term *formula* refers to the general relationship between some quantities. The formula may be equality, and inequality, or other mathematical description. In this book, we will find many formulas used for different purposes. Mainly formulas are used in three ways as described below.

### 1.2.1 Definition

When we study electricity and electronics, we come across a lot of new words like *current*, *voltage*, and *resistance*, and so on. Only a verbal explanation of these words is not enough to understand them clearly and apply them for problems solving. So, we use verbal description as well as mathematical formula to define them, for better understanding. These mathematical formulas help use to use the new terms to solve related problems.

Here is an example of a definition. In our earlier courses, we already learnt that the capacitance is defined as the ability to store charge for a particular voltage. The formula for this definition will be:

$$C = \frac{Q}{V} \quad (1-1)$$

Where,  $C$  is the capacitance,  $Q$  is the charge stored in the capacitor, and  $V$  is the potential difference across the capacitor. This formula is a definition. It tells us what capacitance  $C$  is and how to calculate it. It also helps us to calculate the other parameters as:

$$V = \frac{Q}{C} \quad (1-2)$$

or, 
$$Q = CV \quad (1-3)$$

In summary, definitions are formulas that a researcher creates. They are based on scientific observation and form the basis for the study of electronics, and, of course, in other fields.

### 1.2.2 Law

Scientific laws are natural phenomena that the scientific community has found to be true (up to the present time). Generally, laws describe what will happen in a given situation as demonstrable by a mathematical equation or description. Generally, scientific laws are described by theories. A scientific theory is a description of the natural world that scientists have proven through rigorous testing, studying and observing. As understood within the scientific community, a theory explains how nature behaves under specific conditions, that is, theories describe natural laws.

Here is an example of a law:

$$F = k \frac{Q_1 Q_2}{d^2} \quad (1-4)$$

where,

$F$  = electrostatic force between two point charges

$k$  = a constant of proportionality

$Q_1$  = first charge

$Q_2$  = second charge

$d$  = distance between the point charges

This equation is called **Coulomb's law**. It explains the nature of force between two point charges. That is, the force of attraction or repulsion between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

This is an important equation (law), because, it is the foundation of electricity and electronics. All the variables and the interaction of them, explained in this law, were present in the nature, before its discovery. Coulomb was able to prove that behavior experimentally and express it by an equation. Although earlier researchers could measure  $F$ ,  $Q_1$ ,  $Q_2$ , and  $d$ . Coulomb discovered the law relating the quantities and wrote the formula for it.

Before discovering a law, someone may have a hunch that such a relationship exists. After a number of experiments, the researcher writes a formula that summarizes the discovery. When enough people confirm the discovery through experiments, the formula becomes a law. A law is true because we can verify it with experiments, even for different values of the parameters.

### 1.2.3 Derivation

Given an equation like this:

$$y = 9x$$

We can add 5 to both sides and get:

$$y + 5 = 9x + 5$$

The new equation is true because both sides are still equal. There are many other operations like subtraction, multiplication, division, factoring, and substitution that preserve the equality of both sides of the equation. Thus, we may derive many new formulas using algebraic rules.

**A derivation** is a formula that we can get from other formulas. This means that we start with one or more formulas and, using mathematics, arrive at a new formula not in our original set of formulas. A derivation is also true, because mathematics preserves the equality of both sides of every equation between the starting formula and the derived formula.

For instance, George Ohm was conducting experiments with conductors. He discovered that the ratio of voltage to current was a constant. He named this constant **resistance** and wrote the following formula for it:

$$R = \frac{V}{I} \quad (1-5)$$

This is the original form of **Ohm's law**. By rearranging it, we can get:

$$I = \frac{V}{R} \quad (1-6)$$

This is a derivation. It is not the original form of Ohm's law, but it is converted to another equation, which is also true.

Here is another example of derivation. The definition for capacitance is:

$$C = \frac{Q}{V} \quad (1-7)$$

This equation can be rearranged as,

$$Q = CV \quad (1-8)$$

This new equation is a derivation. It says that the charge on a capacitor will be equal to the product of its capacitance and the voltage across it.

Summary:

**Definition:** A formula invented for a new concept or a new term.

**Law:** A formula to represent the behavior of a natural phenomenon.

**Derivation:** A formula found mathematically altering the definitions or laws.

## 1.3 Approximations

We use approximations all the time in our everyday life. In mathematics, a result that is not exact but close to the exact value, is called an **approximate value** and the process is called **approximation**. As shown in Fig.1-1, the exact length of the line is  $L = 2.95$  cm, but for convenience, we may say it  $L = 3$  cm. Here, the exact value is 2.95 cm and the approximate value is 3 cm. If the difference does not affect the other results significantly, we can use this approximate value.

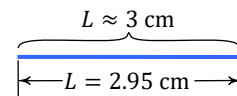


Fig.1-1: Length of a line

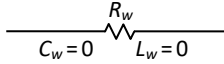
### 1.3.1 The Ideal Approximation

As will be described in **Section 1.25**, the size of copper wire is represented using American Standard Wire Gauge (AWG) or British Standard Wire Gauge (SWG). 1 foot of AWG-22 wire has a resistance ( $R_w$ ) of 16.14 mΩ, and inductance ( $L_w$ ) of 0.24 μH. If the wire is placed 1 inch above the chassis

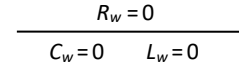
(the circuit board), it will produce a capacitance ( $C_w$ ) of 1.2 pF. If we had to include the effects of resistance, inductance, and capacitance in every calculation of current, the calculation will be complicated and hence, we have to spend too much time. For this reason, everybody ignores the resistance, inductance, and capacitance of connecting wires in most situations. If the wire is considered just as a connecting path, neglecting the resistance, inductance, and capacitance, it is called **ideal approximation**. The ideal approximation, sometimes, is called the **first approximation**. It gives the simplest (ideal) equivalent circuit for a device. As already discussed, the ideal approximation of a piece of wire is just a conductor of zero resistance, zero inductance and zero capacitance.



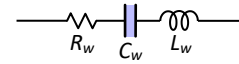
(a): Practical Wire



(c): Equivalent circuit (second approximation)



(b): Equivalent circuit (first approximation)



(d): Equivalent circuit (third approximation)

Fig.1-2: Equivalent circuits of a connecting wire

The ideal approximation is adequate for everyday electronics work. The exception occurs at higher frequencies, where we have to consider the inductance and capacitance of the wire. Suppose a 1 meter wire has an inductance of  $0.15 \mu\text{H}$  and a capacitance of  $0.2 \text{ pF}$ . At  $10 \text{ MHz}$ , the inductive reactance of the wire is  $15.71 \Omega$ , and the capacitive reactance is  $16 \Omega$ . For this frequency, the reactance can no longer be neglected. So, the ideal approximation, will not work at the limit of  $10 \text{ MHz}$ . As a thumb rule, we can say, if the frequency of the circuit is below  $1 \text{ MHz}$ , we can use the ideal approximation. Otherwise, the calculated result will be much deviated from the actual one.

### 1.3.2 Second Approximation

As we have seen above that, beyond a limit, the ideal approximation does not work. To get a more accurate theoretical result, in addition to the ideal approximation, some other mostly effective parameters have to be considered. The second approximation adds one or more components to the ideal approximation. For example, if the frequency of the current passing through the wire is less than  $10 \text{ MHz}$ , but the value of current is very high, we can neglect the capacitive and inductive reactances but we have to consider the resistance of the wire. Thus the equivalent circuit of the wire will be as shown in Fig.1-2(c). This is the **second approximation** of the wire.

### 1.3.3 Third Approximation and Beyond

The third approximation includes some other components in the equivalent circuit of the device, to get more accurate results. For example, if the frequency of the current passing through the wire is quite greater than  $10 \text{ MHz}$ , and the value of current is very high, we have to consider both the inductive reactances and the resistance of the wire. Thus, the equivalent circuit of the wire will be as shown in Fig.1-2(d). This is the third approximation of the wire.

## 1.4 Concept of Voltage and Current

All materials are made of atoms and atoms are made of negatively charged electrons ( $Q_e = -1.60217663 \times 10^{-19} \text{ Coulombs}$ ), positively charged protons ( $Q_p = +1.60217663 \times 10^{-19} \text{ Coulombs}$ ) and neutrons (chargeless). Normally, atoms possess equal number of electrons and

protons. For this reason, any materials, like- gold (Au), silver (Ag) copper (Cu), aluminum (Al) etc., have equal amount of positive charge and negative charge. As a result, normally each material is electrically neutral and has no potential difference. The situation is depicted in Fig.1-3 using water tank analogy. In Fig.1-3(a), there are two water tanks connected by a pipe. As the water levels are same in both the tanks, there is no potential (mechanical) difference, and hence, no flow of water. Similarly, in Fig.1-3(b), there are two metals, connected by a conducting wire. As the metals have equal number of positive and negative charges, so, there is no potential difference between them. Hence, there is no flow of charges and no current.

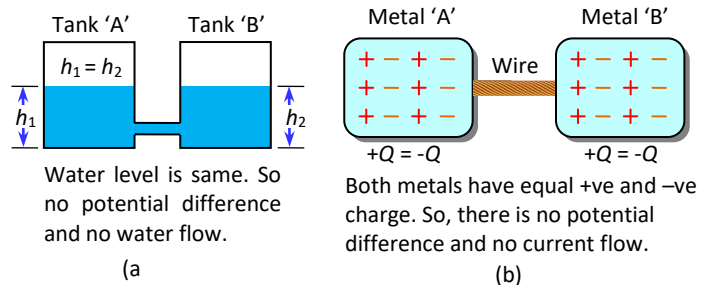


Fig.1-3: Concept of potential (a) Mechanical potential, and (b) Electrical potential(a)(a)

But as shown in Fig.1-4(a), the water level in tank 'A' is higher than that in tank 'B'. So, there will have a mechanical potential difference, and hence, water will flow from tank 'A' to tank 'B' (higher potential to lower potential). Similarly, by any mechanism, if the positive charges and negative charges of metal 'A' and 'B' [Fig.1-4(b)] are separated, there exists a potential difference. As shown in Fig.1-4(b), metal 'A' is at higher potential (higher positive charges) compared to metal 'B'. Therefore, positive charges will flow from metal 'A' to metal 'B' (actually electrons flow in opposite direction).

Potential difference tells us how much electrical energy is available to push electric charges through a circuit. The unit to measure electric potential difference is volt (V). Electric potential difference is also referred to as simply **voltage**.

Potential difference (or voltage) is measured by an instrument called **Voltmeter**. It is always connected in parallel to the points where potential difference is to be measured as given in Fig.1-7(a).

As shown in Fig.1-4(a), due to the potential difference between tank 'A' and 'B', water will flow through the pipe connecting them. Similarly, due to the

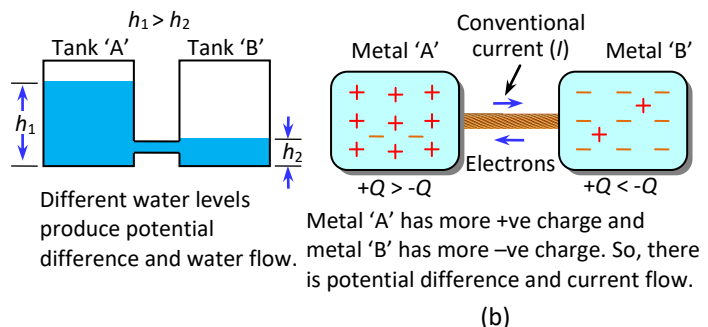


Fig.1-4: Concept of potential (a) Mechanical potential, and (b) Electrical potential

potential difference between metals 'A' and 'B', electric charge will flow from 'A' to 'B'. As positive charge cannot move, electrons will flow from metal 'B' to metal 'A' [Fig.1-4(b)]. The flow of charge will stop when the potential difference is zero. This flow of charge is called the **current**. Actually the rate at which the electric charges flow through a device/wire is called current. The unit used to measure the current is called **Ampere** (A). 1 ampere is equivalent to a flow of 1 Coulomb per second through the cross-section of a conductor or device [further discussed in Section]. Although current was originally thought to be a flow of positive charge, we now know that in most cases it is

the flow of electrons. The direction of conventional current is just opposite to the direction of electrons flow. The instrument used to measure the current is **ammeter**. Unlike the voltmeter, ammeter is connected in series, so that, the full current can flow through the meter as shown in Fig.1-13(b).

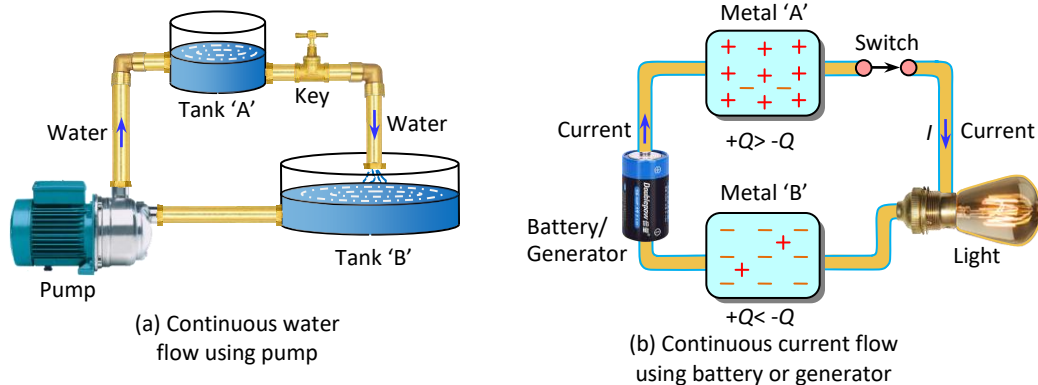


Fig.1-5: Continuous flow of water and current with the help of external energy source

As shown in Fig.1-5(a), if we want continuous flow of water, we have to use a pump that will maintain the potential difference. Similarly, to get continuous current, we must use some source of electrical energy (battery or generator). The **friction** of the pipe that opposes the flow of water is analogous to the **resistance** in electrical circuit that opposes the flow of current.

## 1.5 AC and DC Currents

There are two types of voltage and current. These are: 1) **DC voltage** and **current**, 2) **AC voltage** and **current**. The examples of DC voltage sources are dry cell batteries, lead-acid batteries, DC power supply etc. The main property of these sources is that the voltage and current do not change with time (do not alternate). Fig.1-6(a) shows the graphs of DC voltage and DC current. The voltage produced by a generator is AC. The power coming from grid-line in our house is the example of AC voltage and current. The main property of these sources is that the voltage and current change with time (alternate). Fig.1-6(b) shows the graphs of AC voltage and AC current. The DC voltage and current is represented by their values only, whereas, the AC voltage and current are represented by their RMS (root-mean-square) values and frequency. In Bangladesh, the grid power used in households has 220 V (RMS) and 50 Hz.

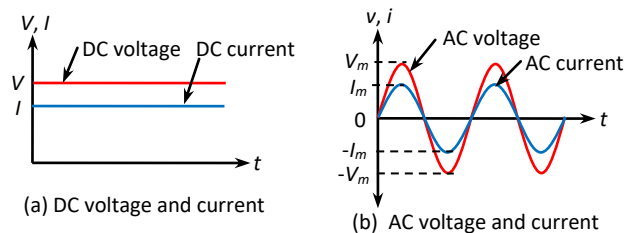


Fig.1-6: AC and DC voltage and current

## 1.6 Voltage Sources

An ideal DC voltage source produces a constant voltage at the output. The simplest example of an ideal DC voltage source is a remote battery (dry cell) whose internal resistance is zero. Fig.1-7(a) shows an ideal voltage source connected to a variable load resistance of  $1\ \Omega$  to  $1\ \text{M}\Omega$ . The voltmeter reads  $10\ \text{V}$ , exactly same as the source voltage, though the load resistance is varied. This type of output voltage is possible if the internal resistance of the battery is zero.

Fig.1-7(b) shows the graph of output voltage (load voltage) versus load resistance. As shown in this graph, the load voltage remains fixed at  $12\ \text{V}$  though the load (output) resistance changes from  $1\ \Omega$  to  $1\ \text{M}\Omega$ . In other words, an ideal DC voltage source

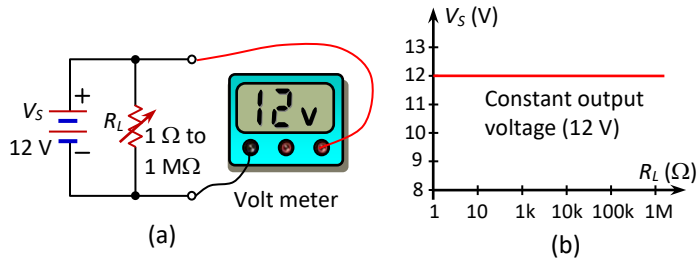


Fig.1-7: Ideal voltage source: (a) Circuit and (b) Variation of voltage with  $R_L$

produces a constant load voltage, regardless the value of load resistance. For an ideal voltage source, only the load current changes with the changes in load resistance as shown in Fig.1-7(b).

However, it is difficult to make an ideal voltage source. The practically available voltage sources [Fig.1-8] has a finite internal resistance though its value is very small. For this internal resistance the output voltage will decrease with the decrease of the load resistance. If the load resistance decreases, the output current increases and the voltage drops across the internal resistance ( $r_s$ ) will increase and hence the output voltage decreases. The output voltage can be calculated if the current is known as,

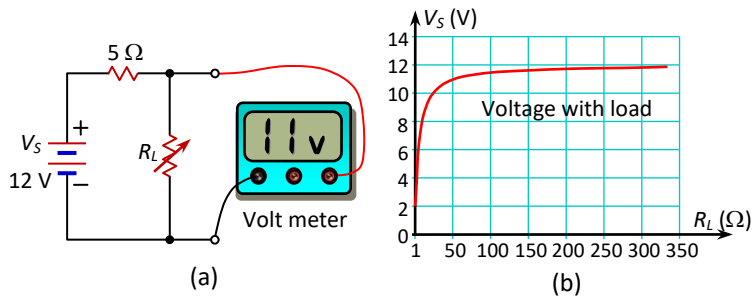


Fig.1-8: Practical voltage source: (a) Circuit and (b) Variation of voltage with  $R_L$

the voltage drops across the internal resistance ( $r_s$ ) will increase and hence the output voltage decreases. The output voltage can be calculated if the current is known as,

$$V_o = V_s - I_L r_s \quad (1-9)$$

### Example 1-1

The open circuit (output) voltage of a voltage source is  $12.2\ \text{V}$ . If a  $220\ \Omega$  load is connected at the output the source voltage falls to  $12\ \text{V}$ . Determine the internal resistance of the voltage source.

### Solution:

If a  $220\ \Omega$  load is connected at the output the source voltage

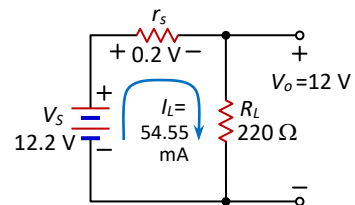


Fig.1-9: Circuit for Example 1-1

falls to 12 V. So the current through the load resistance will be

$$I_L = \frac{V_L}{R_L} = \frac{V_o}{R_L} = \frac{12 \text{ V}}{220 \Omega} \approx 54.55 \text{ mA}$$

Therefore, when 54.55 mA current is drawn from the voltage source, the output voltage becomes 12 V. That is,  $12.2 \text{ V} - 12 \text{ V} = 0.2 \text{ V}$  drops across the internal resistance, when 54.55 mA current flows through it. So, the internal resistance of the voltage source will be,

$$r_s = \frac{0.2 \text{ V}}{54.55 \text{ mA}} \approx 3.67 \Omega \text{ [Ans.]}$$

**Comments:** The internal resistance of a voltage source is, always, very small.

### 1.6.1 Stiff Voltage Source

We can ignore the internal resistance of the voltage source when it is very smaller than the load resistance, say at least 100 times smaller than the load resistance. Any source that satisfies this condition is called a **stiff voltage source**.

As shown in Fig.1-10(b), when the load resistance becomes 100 times the source resistance ( $R_L = 500 \Omega$ ) the output voltage becomes almost equal to the source voltage ( $V_L = 11.88 \text{ V}$ ). After this point, the output voltage is almost constant, that is independent of the load

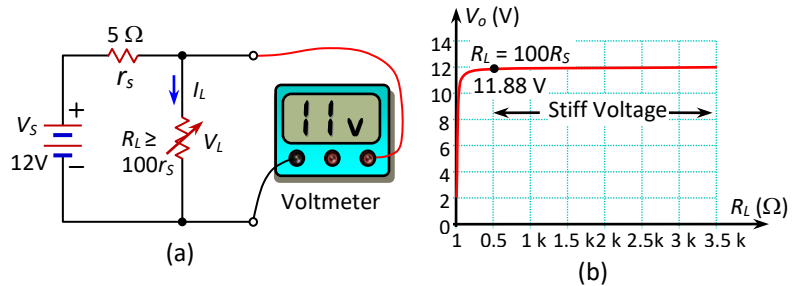


Fig.1-10: Practical voltage source: (a) Circuit and (b) Variation of voltage with  $R_L$

resistance (and load current). This portion in the graph is called **stiff region**. Stiff voltage is an important term for AC voltage of grid supply.

The internal resistance of the voltage source of Fig.1-10(a) is  $5 \Omega$ . So, if the value of the load resistance is greater than  $100 \times 5 \Omega = 500 \Omega$ , the source will work as a stiff voltage source. Thus, as long as the load resistance is greater than  $100r_s$ , the source is a stiff voltage source. When the load resistance equals this limiting value (worst-case), the calculated error from ignoring the source resistance is 1%, small enough to ignore practically.

### 1.7 Current Sources

A DC voltage source (ideal) produces a constant output (load) voltage regardless the load resistance, and hence the load current. Similarly, a DC current source gives a constant current, regardless the output load resistor and hence the output voltage. This type of current source is called **ideal current source**. On the other hand, a practical current source has a very large internal resistance that comes in parallel with the source. The schematic symbols of current sources are shown in Fig.1-11. Fig.1-11(a) shows the symbol of an ideal current source whose internal resistance is infinite (open circuit) and Fig.1-11(b) shows a practical current source. The finite source resistance is connected in parallel with the ideal source. The symbol shown in Fig.1-11(a) is

the first approximation and that in Fig.1-11(b) is the second approximation. The variation of the load (output) current with the variation of output load resistance is shown in Fig.1-11(d).

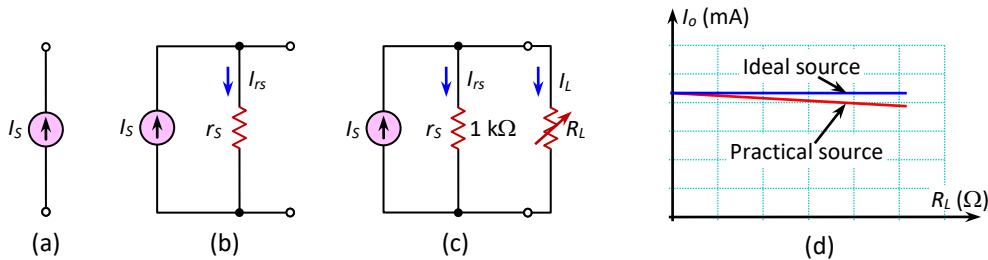


Fig.1-11: Practical current source: (a) Symbol (b) Equivalent circuit, (c) Equivalent circuit with variable load  $R_L$  and (d) Variation of output voltage with  $R_L$

### Example 1-2

The rated current capacity of a practical current source is 500 mA, but when a 1 k $\Omega$  load resistance is connected at the output, 490 mA current flows through it. Determine the internal resistance of the current source.

#### Solution:

When a 1 k $\Omega$  load resistance is connected at the output, 450 mA current flows through it. Thus, 500 mA – 490 mA = 10 mA current flows through the internal resistance of the source. As  $r_s$  and  $R_L$  are in parallel, the voltage drop across these resistors will be,

$$V_L (= V_o) = I_L R_L = 450 \text{ mA} \times 1 \text{ k}\Omega = 450 \text{ V}$$

Therefore, when 50 mA flows through the internal resistance 450 V drops across it. So the value of  $r_s$  will be,

$$r_s = \frac{450 \text{ V}}{10 \text{ mA}} \approx 45 \text{ k}\Omega \text{ [Ans.]}$$

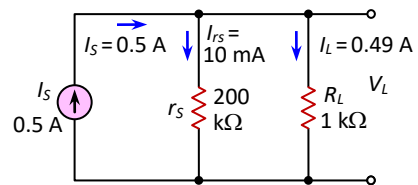


Fig.1-12: Circuit for Example 1-2

**Comments:** The internal resistance of a current source is always very large.

Current sources are constructed using transistors or some other electronic devices. We can convert a DC voltage source into a current source by adding a large resistor in series. But the problem is, it needs very large source voltage. Fig.1-13(a) shows such a current source constructed using a voltage source. The source voltage is assumed to be 100 V DC and the series resistance is 200 k $\Omega$ . Using Norton's theorem, we can convert this voltage source into a current source which is shown in the shaded area of Fig.1-13(b). The variation of output current with load resistance  $R_L$  is shown in Fig.1-13(c).

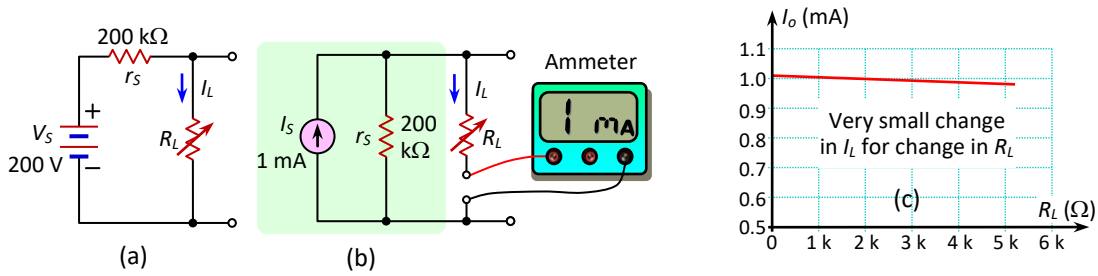


Fig.1-13 (a): Voltage source converted into current source, (b) Norton's equivalent circuit, and (c) Variation of voltage with load resistance  $R_L$

### 1.7.1 Stiff Current Source

If the source resistance is at least 100 times greater than the load resistance, then we can ignore the effect of the source resistance. The current source that satisfies this condition is called **stiff current source**. Thus, for a stiff current source,  $r_s \geq 100 \times R_L$  [Fig.1-14(a)]. The worst case limit is  $r_s = 100 \times R_L$ . For this limiting case, if the source resistance is neglected, the calculated result will be 1% smaller than the actual one.

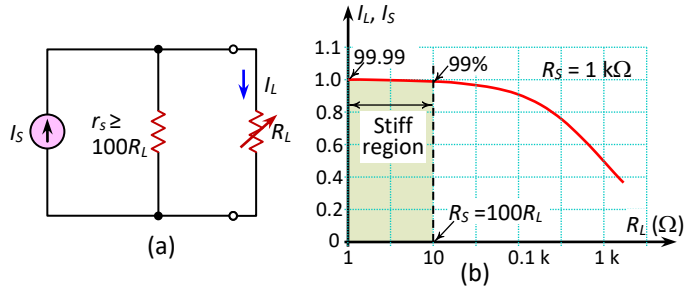


Fig.1-14: Stiff current source: (a) Symbol (b) Variation of output voltage with  $R_L$

## 1.8 Motions of Charge Carriers

Though in the early age, it was assumed that current was produced due to the movement of positive charges, later it was proved that the current, in a conductor, was produced due to the movement of negatively charged electrons, not the positive charges. Thus, in conductors, electrons are the only charge carriers. In semiconductors, both electrons and holes are the charge carriers. In solutions/liquids (like electrolytes of batteries), negative and positive ions can move and produce current. Here, we will discuss the motion of electrons and the motion of holes has been discussed in Chapter 2, Section 2.14.

### 1.8.1 Motion of Electrons without any Potential Difference

Let us consider a section of a conducting material (like Cu) as shown in Fig.1-15. All conductors have a number of free electrons, which are essentially free from their original atoms while the positively charged atomic ions are held in place by the bonding forces which create the solid (conductor). When a conductor is kept in room temperature, thermal energy will be added to the conductor. So electrons will gain thermal energy and will produce **random motions** or **Brownian motions** [Fig.1-15(a)]. The speed of the electrons is called **thermal speed** and is denoted by  $V_T$ . The value of the thermal speed is given by

$$V_T = \sqrt{\frac{3kT}{m_e}} \quad (1-10)$$

where,  $k = \text{Boltzmann constant} (= 1.38 \times 10^{-23} \text{ J/K})$ ,  $T = \text{absolute temperature}$ , and  $m_e = \text{mass of electrons} (= 9.1 \times 10^{-31} \text{ kg})$ .

At room temperature ( $T = 300 \text{ K}$ ) the value of  $V_T$  will be  $\approx 10^5 \text{ m/s}$ . All electrons will have the same thermal speed, but the thermal velocity will be different for different electrons, due to different motions in different directions. The average value of thermal velocity ( $\bar{A}_T$ ) will be zero, and hence, the net current flow will also be zero [Fig1-15(b,c)]. In a time interval ( $\Delta t$ ) the same number of electrons will move from left to right and vice versa [Fig1-15(b)]. Therefore, the net current flow will be certainly zero.

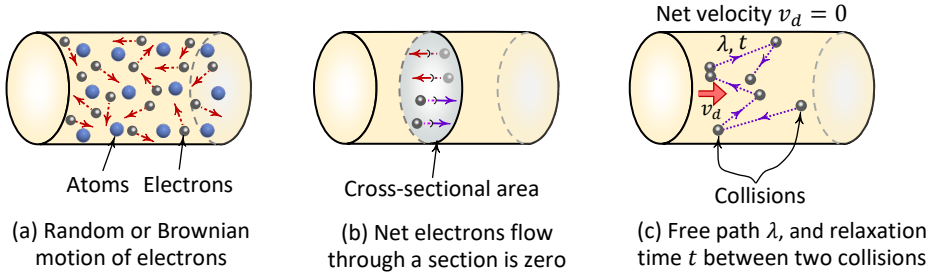


Fig.1-15: Motion of electrons without potential difference

### 1.8.2 Motion of Electrons with Potential Difference

When a potential difference is applied across a conductor, a drift velocity is developed in addition to the thermal speed. Due to this drift velocity, electrical current flows through the conductor. It is instructive to calculate the effective velocity of the electrons when a current flows, as the result is rather unexpected.

Let us first calculate the density of free (electrons in conduction band) electrons in copper. One mole of copper has a mass of  $63.5 \times 10^{-3} \text{ kg}$  and, by definition, this contains  $N_A = 6.02 \times 10^{23}$  atoms (Avogadro's number), and the density of copper is  $\rho = 8920 \text{ kg m}^{-3}$ . So, the volume,  $V$ , of one mole of copper is:

$$V = \frac{\text{mass } (m)}{\text{density } (\rho)} = \frac{63.5 \times 10^{-3} \text{ kg}}{8920 \text{ kg m}^{-3}} = 7.12 \times 10^{-6} \text{ m}^3$$

For copper, each atom contributes one free electron so the density of free electrons in copper will be:

$$n = \frac{N_A}{V} = \frac{6.02 \times 10^{23}}{7.12 \times 10^{-6} \text{ m}^3} = 8.46 \times 10^{28} \text{ m}^{-3}$$

The current density  $j_c$  is related to the drift velocity  $v_d$  by the following equation:

$$j_c = nQ_e v_d \quad (1-11)$$

$$\text{or,} \quad v_d = \frac{j_c}{nQ_e} \quad (1-12)$$

Now let us consider a current of, say,  $1 \text{ A}$  through a wire of cross-sectional area  $1 \text{ mm}^2 (= 10^{-6} \text{ m}^2)$ . Therefore, the drift velocity of the charge carriers (electrons), for this current, will be:

$$v_d = \frac{j_c}{nQ_e} = \frac{1/10^{-6}\text{m}^2}{8.46 \times 10^{28}\text{m}^{-3} \times 1.6 \times 10^{-19}\text{C}} = 74 \times 10^{-6}\text{ms}^{-1} \quad (1-13)$$

This is a very small velocity indeed. A free electron takes about 3.75 h to travel a distance of 1 m. But we find that when a switch is turned ON, a light connected at the end of a wire, glows immediately. This is due to the effect of electric field.

When the switch is ON, due to the pressure of the voltage source, electrons start to move. If an electron moves a very small distance, it will create a force on the next electron, the next electron will produce a pressure on the adjacent electron and so on. Thus, the electrons, near the light, will immediately start to move and the light will lit immediately. The situation is illustrated in Fig.1-16. To understand the process let us consider a tube full of marbles. If we apply pressure at one end, it will move a little distance, but the marble at the opposite end will fall off immediately.

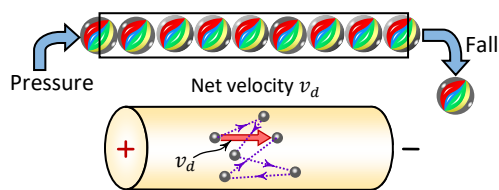


Fig.1-16: Motion of electrons with potential difference

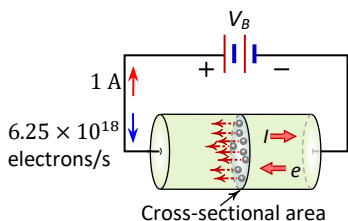


Fig.2-1: Current flows through a wire

### Example 1-2

As shown in Fig.2-, a wire is carrying 1 A of current. Determine the average number of electrons passing through a cross-sectional area in 1 s.

**Solution:**

We know,  $I = \frac{Q}{t}$

or,  $Q = It$

or,  $Q = (1 \text{ A}) \times (1 \text{ s}) = 1 \text{ Coulomb}$

Again we know, 1 electron has a charge of  $1.6 \times 10^{-19}\text{Coulomb}$ . Therefore, the number of electrons passing through the cross-section should be

$$n_e = \frac{1 \text{ C}}{1.6 \times 10^{-19}\text{C}} = 6.25 \times 10^{18} \text{ [Ans.]}$$

**Comments:** The number of electrons passing through a conductor for 1 A of current is huge! This is due to the very small amount of charge associated with an electron.

## 1.9 Current in Capacitors

From the construction of capacitor described in Section 1.16, we know that the space between two conducting plates is either empty, or filled up with insulating dielectric material. When a capacitor is connected to a voltage source, +ve charge accumulates in one plate, and -ve charge accumulates in the other plate, until the potential difference between the plates becomes equal to the source voltage. Conduction current (due to the movement of electrons) flows through connecting wires, and capacitor leads but not through the gap (or dielectric) between the capacitor plates. We know that in a single loop circuit, the value of current should be equal at every point of the loop. To describe the continuity of this current, Maxwell introduced a current called **displacement current** through vacuum or any dielectric material (insulator). This current is developed due to the variation of electric field in the space between the capacitor plates. As shown in Fig.1-17, when +ve and -ve charges accumulate in the plates, the electric field strength between the plates, increases gradually. The displacement current is directly proportional to the rate of change of electric field. Mathematically, we can define the displacement current as:

$$I_D = \epsilon S \frac{dE}{dt} = \epsilon \frac{d\phi_E}{dt} \quad [\because \phi_E = SE] \quad (1-14)$$

$$\text{or,} \quad j_D = \frac{I_D}{S} = \epsilon \frac{dE}{dt} \quad (1-15)$$

where,

$I_D$  = Displacement current

$j_D$  = Displacement current density

$\epsilon$  = Permittivity of the material used in between the capacitor plates

$E$  = Electric field-strength

$\phi_E$  = Electric flux, and

$S$  = area of the capacitor plates.

Displacement currents play an important role in the propagation of electromagnetic radiation, such as light and radio waves, through empty space. A time varying magnetic field is always associated with a time varying electric field that may be conceived in terms of a displacement current.

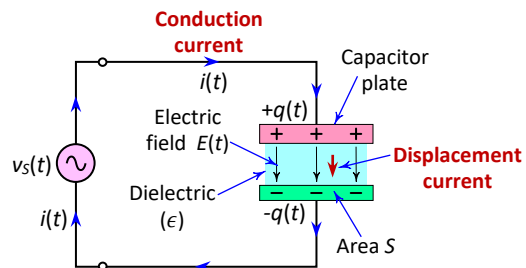


Fig.1-17: Displacement current in capacitor

## 1.10 Resistors

The word Resistor is born out of the word “resist”, which means to withstand the effect. A **resistor** resists, or regulates the flow of electrons that passes through it. The device resistor was invented by Boykin, an American engineer, in 1959. These little devices regulate, attenuate, or reduce

voltage and current, but do not have a source of power in their own. The current flows through them in a regulated manner leading to a loss of energy in the form of heat.

Although any conducting wire of circuits produces unwanted resistance to the flow of current, we always add some resistance in circuits to control the flow of current. A resistor is a passive two-terminal electrical component that provides resistance (opposition) to the flow of charges. The ability of a resistor to resist the current is called **resistance** and is denoted by  $R$ .

In electronic circuits, resistors are used to reduce current flow, adjust signal levels, to divide voltage, and bias active elements (like diodes and transistors). Resistance of a conducting wire is given by,

$$R = \rho \frac{l}{A} \quad (1-16)$$

where,  $\rho$  is the resistivity of the material of the wire,  $l$  is the length of the wire, and  $A$  is its cross-sectional area. Different materials have different values of  $\rho$ . Silver has the lowest value of  $\rho$ . The values of resistivity of some common materials are given in Table 1-1.

Table 1-1: Resistivity and conductivity of some common materials

| Sl. No. | Material        | Resistivity ( $\rho$ ) $\Omega \cdot \text{m}$ | Conductivity ( $\sigma$ ) S/m | Sl. No. | Material        | Resistivity ( $\rho$ ) $\Omega \cdot \text{m}$ | Conductivity ( $\sigma$ ) S/m |
|---------|-----------------|------------------------------------------------|-------------------------------|---------|-----------------|------------------------------------------------|-------------------------------|
| 1       | Silver          | $1.59 \times 10^{-8}$                          | $6.30 \times 10^7$            | 9       | Nickel          | $6.99 \times 10^{-8}$                          | $1.43 \times 10^7$            |
| 2       | Copper          | $1.68 \times 10^{-8}$                          | $5.96 \times 10^7$            | 10      | Lithium         | $9.28 \times 10^{-8}$                          | $1.08 \times 10^7$            |
| 3       | Annealed copper | $1.72 \times 10^{-8}$                          | $5.80 \times 10^7$            | 11      | Iron            | $1.0 \times 10^{-7}$                           | $1.0 \times 10^7$             |
| 4       | Gold            | $2.44 \times 10^{-8}$                          | $4.10 \times 10^7$            | 12      | Platinum        | $1.06 \times 10^{-7}$                          | $9.43 \times 10^6$            |
| 5       | Aluminum        | $2.82 \times 10^{-8}$                          | $3.50 \times 10^7$            | 13      | Tin             | $1.09 \times 10^{-7}$                          | $9.17 \times 10^6$            |
| 6       | Calcium         | $3.36 \times 10^{-8}$                          | $2.98 \times 10^7$            | 14      | Lead            | $2.2 \times 10^{-7}$                           | $4.55 \times 10^6$            |
| 7       | Tungsten        | $5.60 \times 10^{-8}$                          | $1.79 \times 10^7$            | 15      | Titanium        | $4.22 \times 10^{-7}$                          | $2.38 \times 10^6$            |
| 8       | Zinc            | $5.90 \times 10^{-8}$                          | $1.69 \times 10^7$            | 16      | Stainless steel | $6.9 \times 10^{-7}$                           | $1.45 \times 10^6$            |

To construct circuits, resistors of different values and shapes are used. Different resistors have different physical shapes, colors, resistance and power rating. Some of the resistors commercially available in the market are shown in Fig.1-18.

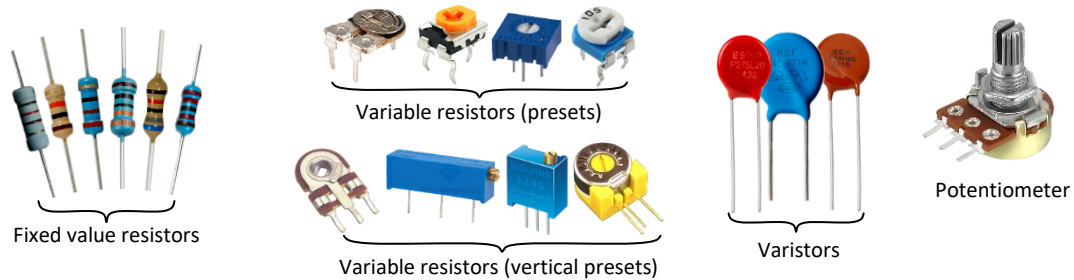


Fig.1-18: Photographs of different types of resistors

## 1.11 Different Types of Resistors

There is an endless list of different types of resistors in the context of composition type, film type, wire-wound type, resistors characteristics like - physical size, reliability, temperature rating, noise, temperature coefficient, and voltage coefficient and so on. The chart in Fig.1-19, shows some common types of resistors.

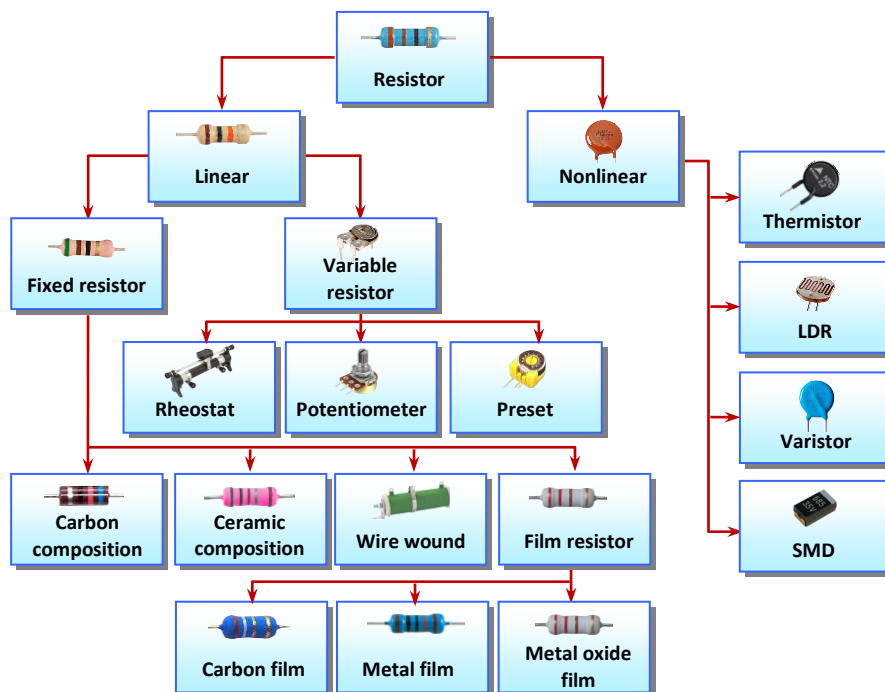


Fig.1-19: Classifications of resistors

### 1.11.1 Fixed Value Resistors

Fixed value resistors are constructed to provide a fixed resistance in the circuit. Their resistance cannot be changed by the users. Due to the change in operating temperature their resistance may vary by a small amount. These are two-terminal components (leaded or SMD- Surface Mounted Device). There are many types of fixed value resistors:

#### Composition Resistors

These resistors are made of a mixture of conducting and non-conducting materials. The most commonly available types of composition resistors are:

- Carbon Composition Resistors
- Ceramic Composition Resistors

#### Carbon Composition Resistors

**Carbon composition resistors** are the oldest type resistors. These resistors are most commonly used as they are easy to manufacture, inexpensive and are suitable for most of the common applications. These resistors are made up of a mixture of carbon powder and ceramic filler material. The mixture is molded into axial resistor package enclosed by an insulating coating with two leads attached at the two ends of the package [Fig.1-20]. The ratio of carbon to the ceramic material and distance between the end terminals determine its resistance. **The greater is the proportion of carbon in the mixture, the lesser is its resistivity, and vice versa.** The ceramic filler used in the mixture is mostly phenolic, which is similar to the plastic. These resistors are available in an axial package with color-coding [discussed later] for an indication of resistor value and tolerance.

Carbon composition resistors can have very low resistance ( $1\ \Omega$ ) to extremely high resistance ( $22\ \text{M}\Omega$ ). The power rating of these resistors depends on the physical size of the resistor. **Greater is the physical size, greater is the power rating.** These resistors are commonly available in 1/4 and 1/2 Watt series. These resistors have wide tolerance levels (deviation from the nominal resistance value) from 5% to 20%. Due to wide tolerance, these resistors are not suitable for precision applications.

The operating temperature range of carbon composition resistors typically ranges from  $-55^\circ\text{C}$  to  $125^\circ\text{C}$ .

The carbon composition resistors offer quite pure resistance and almost negligible inductance or capacitance. So, they typically have good frequency response, low noise, and acceptable stability. This makes these resistors suitable for high-frequency applications as well as low power circuits.

#### Ceramic Composition Resistors

The ceramic composition resistors are similar to carbon composition resistors except the materials [Fig.1-21]. They are composed of a mixture of a finely ground ceramic and conductor which is

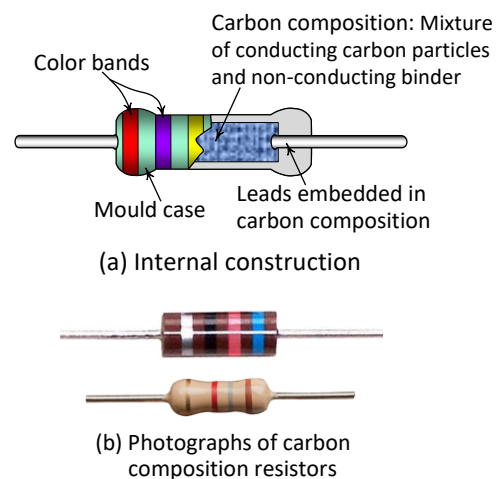


Fig.1-20: Carbon composition resistors

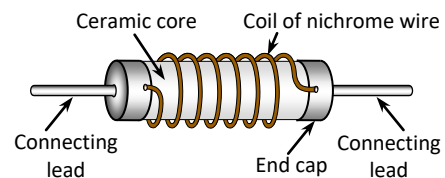
compressed into a cylindrical shape. This improves their pulse stability and operating temperature range. The ceramic resistors can operate at ambient temperatures up to 250° C. These are typically available in 1/2 Watt, 1 Watt and 2 Watt series with resistance values ranging from 3.3  $\Omega$  to 1 M $\Omega$ . There are ceramic composition resistors available with a power rating up to 1000 Watt. The ceramic resistors offer the same advantages as the carbon composition resistors (pure resistance, stability, reliability, and pulse stability). These resistors are available in many types of packages like axial, tubular, slab, disk and washer, encapsulated, load bank, water-cooled and various custom packages.



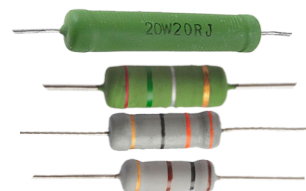
Fig.1-21: Ceramic composition resistors

### Wire Wound Resistors

Wire wound resistors are made by winding metal wire on a former (core). A Nichrome (Nickel/Chromium) or Manganin (Copper/Nickel/Manganese) are commonly used as the metal wires, because they provide high resistance to the electric current and operates at high temperature. Most commonly used core materials include plastic, fiberglass, or ceramic. The wire has a specific fixed-value resistance. The entire component is insulated by a coat of vitreous (opaque) enamel. These resistors are shown in Fig.1-22. The resistance value is dependent on the resistivity of the wire, the cross section, and the length. Since these parameters can be accurately controlled, a high precision resistance can be achieved. Wire wound resistors are mainly produced with alloys, since pure metals have a high **temperature coefficient of resistance (TCR)**. However, for high temperatures, pure metals such as tungsten are often used. The temperature coefficient is a measure of how much the resistance will change as the temperature changes. TCR is measured in units of ppm/°C. If a manufacturer rates a resistor at 50 ppm/°C, the resistor will not change more than 50  $\Omega$  in resistance for each 1 M $\Omega$  of the resistors given value, for a temperature change of 1° C.



(a) Internal construction



(b) Photographs of wire wound resistors

Fig.1-22: Wire wound resistor

Wire wound resistors are low resistance devices, commonly manufactured in sizes from 5 to 200 watts. The wattage chosen depends on the physical size of the resistor.

### Power wire wound resistors

Power wire wound resistors exist for very high power applications. The range varies from 0.5 W to more than 1000 W. Power wire wound resistors can be divided into types according to their coating types.

Silicone resins are used for lowest power dissipation levels. These are compact resistors that can withstand temperature up to 300° C above the ambient temperature.

Another type of coating is vitreous enamel. This traditional coating has good insulating properties at low temperatures, but the insulating capability is reduced when operated at the full rated temperature.

The majority of the power wire wound resistors have a ceramic core and a ceramic coating to protect the winding. The ceramic coating combines a high insulation and physical protection with good heat dissipation. Typical power ratings are between 4 and 15 W.

For the highest power dissipation values, wire wound resistors have an aluminum case with fins. These fins give a larger surface area to dissipate heat, letting the resistor handle more power without being damaged. These resistors have a ceramic core and a silicone resin coating, encased in an aluminum extrusion. The surface is anodized to maintain a good insulation resistance. These power wire wound resistors have a typical power rating of 25 to 50W.

### Film resistor

Along with the composition and wire-wound resistors, film resistors are widely used. These resistors are made by depositing a resistive material (called film) on a substrate in a zigzag pattern. The film may be a **Carbon Film**, **Metal Film**, or **Metal Oxide Film**. The substrate is made from glass, ceramic, or silicon, and it is used as an insulating base for the resistor. Also a fine spiral groove cut along the film-rod using a laser or diamond cutter results in a long spiral strip, which forms the resistor (Fig.1-24 & 25). By precisely varying the thickness, zigzag pattern, and the composition of the material the desired resistance value is achieved. The resistor materials are made from metals (nickel alloy) or carbon or metal oxide (tin oxide) deposited with precise control. The deposited layer may be a thin film (few micrometers)- called **thin film resistor** or thick film (10 to 50 micrometers)- called **thick film resistor**.

A protective coating is used to cover the resistor material deposited on the substrate. The end caps and the connecting terminals are made of metal such as nickel or silver, or they can be made from an alloy, and attached to the resistors.

Film technology allows producing resistor networks on a single chip [Fig.1-23]. A resistor network consists of two or more independent resistors contained in one single surface- mounted package.

### Carbon Film Resistors

**Carbon film** resistors have the carbon film on the ceramic former and they are covered with insulation coating material like epoxy (for protection). After that leads connection are added and a helix cut is given to adjust the value of resistance. Carbon film resistors are a significant improvement of carbon composition resistors.

Due to the helical resistive path, these resistors develop some capacitance and inductance. These small resistors have a capacitance of approximately 0.5 pF and the self-induction is around 0.01  $\mu$ H for uncut resistors, and up to several  $\mu$ H for spiral cut resistors. These resistors are available in values between 1  $\Omega$  – 10,000 M $\Omega$  and have power ratings of 1/16 to 3 W.

### Manufacturing

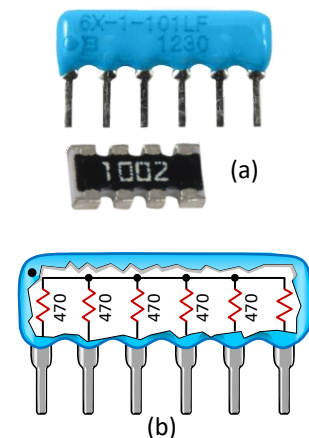
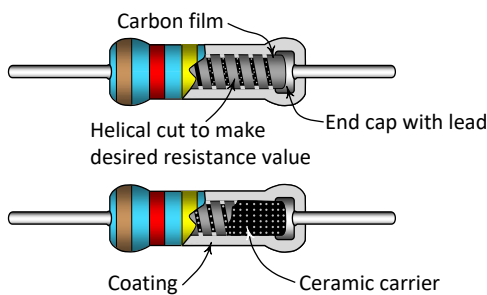


Fig.1-23: Resistor package: (a) Photographs (b) Cutaway diagram

Carbon film resistors are made with a deposition process. At high temperature and under a high pressure, a ceramic carrier is held in hydrocarbon gas. The gas (methane or benzene) is cracked at a temperature of  $1000^{\circ}\text{C}$ . The crystalline carbon is pyrolytically deposited on the ceramic substrate. The desired resistance value can be obtained by choosing the right layer thickness, and by cutting a spiral shape in the carbon layer. The helical cut in the film increases the length of the current path. By decreasing the pitch of the helix, the length of the resistive path increases, and therewith the resistance value increases. Carbon film resistors provide better tolerance value in the range of 1 % to 5 %. These resistors can have resistance in range from a few milliohms to  $100\text{ M}\Omega$ .



(a) Internal construction



(b) Photographs of carbon film resistors

Fig.1-24: Carbon film resistor

Because of the use of pure carbon, the carbon film resistor has a higher negative temperature coefficient than carbon composition. The operating temperature range of carbon film resistors is similar to carbon composition resistors. It typically varies from  $-55^{\circ}\text{C}$  to  $155^{\circ}\text{C}$ .

### Metal Film Resistors

**Metal film resistors** are constructed by depositing a film of metal (Nichrome or Tantalum Nitride) on a cylindrical ceramic substrate or rod. Like carbon film resistors, the coating is cut to a long helical resistive path by laser.

These resistors are available for a wide range of resistances- from few milliohms to  $60\text{ M}\Omega$ . The value of the resistance depends on the thickness and length of the resistive path. Metal film resistors are available for a wide range of power ratings-  $1/20\text{ Watt}$  to  $20\text{ Watt}$ . Unlike carbon film resistors, metal film resistors have better temperature stability, with 1 % to 2 % tolerance. There are also precise metal film resistors available that have tolerance as low as 0.01%. However, metal film resistors having precise tolerance often have a lower power rating.

The operating temperature range of metal film resistors range from  $-65^{\circ}\text{C}$  to  $250^{\circ}\text{C}$ . They have a lower temperature coefficient compared to carbon composition and carbon film resistors which gives better temperature stability. Important thing is that these resistors have a positive temperature coefficient.

### Metal Oxide Film Resistors

**Metal-oxide film resistors** are fixed value, axial resistors and are similar to metal film resistors. They are made of ceramic rod that is coated with a thin film of metal oxides, such as tin oxide. Also, antimony oxide is added to yield better resistance. **The more the antimony, the more is the resistance.** The width of the helical metal oxide film cut and its thickness are inversely proportional to the resistance. The metal oxide film is most often produced using chemical deposition methods.

Metal oxide film resistors exceed the performance of metal film and carbon film resistors for the following properties: power rating, voltage rating, overload capabilities, surge capacity, and high temperature operation. These resistors have a higher operating temperature range (up to 450°C) and the temperature coefficient is around 300 ppm/°C, which is higher than for metal film types. These resistors are typically available in power ratings from 1/4 W to 140 W with tolerance typically 0.5 % to 20 %.

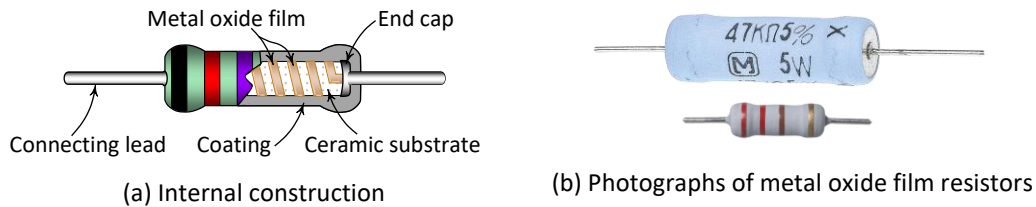


Fig.1-25: Metal oxide film resistors

Another type of film resistor commonly known as a **Thick Film Resistor** is manufactured by depositing a much thicker conductive paste of **ceramic** and **metal**, called **cermet**, onto an alumina ceramic substrate. The resistive path is drawn like zigzag lines between the terminals of the resistor. Cermet resistors have similar properties to metal film resistors and are generally used for making small surface mount chip type resistors, multi-resistor networks in one package for PCB's and high frequency resistors.

### SMD Resistors

So far we have discussed leaded, through-hole mounting resistors. Through-hole mounting refers to installing electronic components on a PCB (printed circuit board) by inserting their leads through pre-drilled holes on the board and soldering them in place. To save PCB space and to avoid the

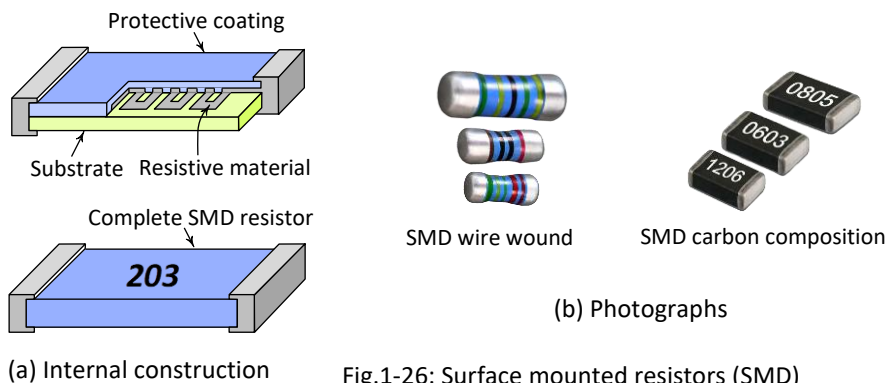


Fig.1-26: Surface mounted resistors (SMD)

nuisance of drilling, surface-mount devices (SMD) are introduced. A surface mount resistor is a tiny rectangular ceramic body with silver conductive edges on either end. An SMD resistor offers advantages in saving space on printed circuit boards (PCBs). Solder paste is printed onto the mounting pads of a PCB, and the resistors are placed automatically onto these pads by pick and place equipment. The whole board is then passed through a reflow oven which melts the solder into place, making the contacts.

A surface mount resistor has a lower power dissipation capacity. Surface mount resistors can be thick film, which is most commonly used, thin film, which offers accuracy and stability, metal strip, for current sensing, or wire-wound, which is molded construction as opposed to a flat chip [Fig.1-26].

Surface mount resistors are used in compact PCBs like- telecommunication, automotive, medical equipment, personal devices, displays, and advanced technology research instruments.

### Variable Resistors

**Variable resistors** are made with such an arrangement that the user can change its resistance if necessary. The resistance value depends on the position of a wiper (a moveable contact) or other natural parameters (temperature, light etc.). Variable resistors are used in systems, where the user has frequent requirement for adjusting the resistance, as in tone, volume, focus, and brightness control circuits. There are three basic types of variable resistors: **Potentiometer** (abbreviated as Pot), **preset** (also called trimmer), and **rheostat**. All of them have three terminals, two fixed and one moveable. Light dependent resistor, thermistors are the examples of variable resistors that depends on natural parameters.

### Potentiometer

Potentiometers (or Pots) are most commonly used variable resistors. They are used in fan-regulators, variable power supplies, different audio controlling systems, etc. Potentiometers are

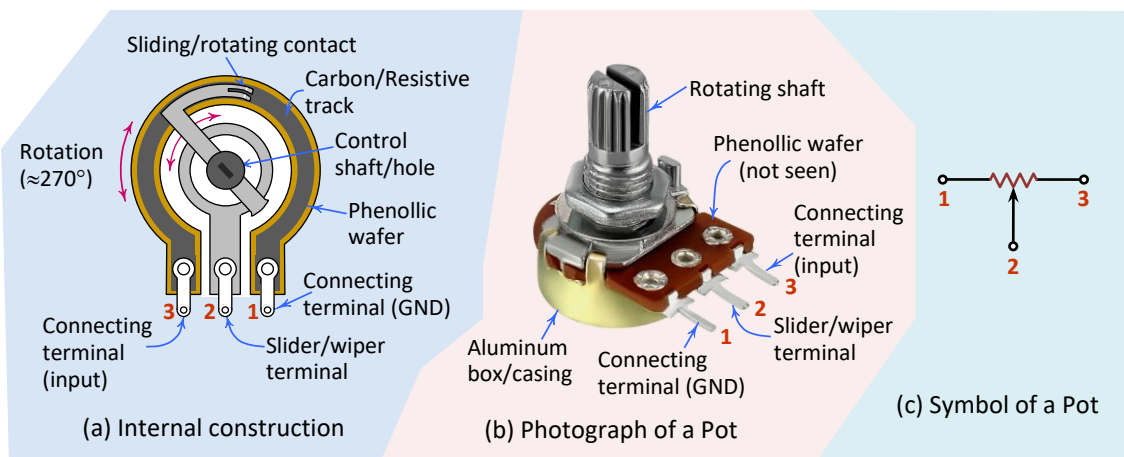


Fig.1-27: Potentiometer (Pot)

made with either composition or wire-wound resistance elements. The basic components of the most widely used types of potentiometers are shown in the Fig.1-27. Although the shapes and sizes

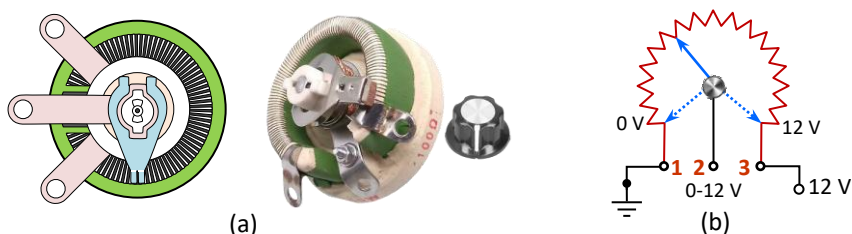


Fig.1-28: Wire wound potentiometer: (a) Construction (b) Application of Pot

of the parts may vary with the wattage rating and with the particular production techniques of various manufacturers, the general arrangement is nearly the same. The only fundamental difference is the types of resistance elements used. In potentiometer, there are three terminals and a rotating shaft which, when rotated, changes the value of resistance. The resistive element may be a strip of composition materials or wire-wound.

The shaft can be turned so that the contact arm (wiper) can be set to any desired position on the resistance track. As is the case with fixed resistors, there are a limited number of different wattage ratings available for potentiometer.

Potentiometers are mainly used as a voltage divider network as shown in Fig.1-28(b). The terminal-1 is connected to the ground (Zero voltage), the input voltage (or signal) is applied to terminal-3, and the output is taken from terminal-2. The output can have any value between zero to full input voltage. Clockwise rotation of the knob will increase the output voltage, and vice-versa.

However, all of these three-terminal Pots can be used as variable resistors. We know that a variable resistor has two terminals. So we have to use any one of the fixed terminals (1 or 3) and the variable terminal (2). We can short the variable terminal with the remaining third terminal.

### Example 1-3

As shown in Fig.1-29, the wiper of a 10 kΩ potentiometer is set a point A. The resistance between terminal 1 and point A is 4 kΩ. If a 12 V battery is connected to terminal 3, and the terminal 1 is grounded, what will be the output voltage at terminal 2?

#### Solution:

The total resistance of the potentiometer is 10 kΩ, and the resistance from terminal 1 to point A is 4 kΩ. Hence, the resistance between point A and terminal 3 will be  $10\text{ k}\Omega - 4\text{ k}\Omega = 6\text{ k}\Omega$ . The voltage at terminal 2 (wiper) will be same as the voltage at terminal A. Here, the potentiometer will work as a voltage divider circuit as shown in Fig.1-29(b). Therefore, using voltage divider rule,

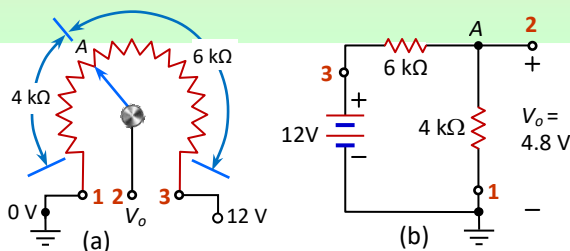


Fig.1-29: Circuit for Example 1-3

$$V_o = 12\text{ V} \times \frac{4\text{ k}\Omega}{4\text{ k}\Omega + 6\text{ k}\Omega} = 3.6\text{ V}$$

**Comments:** The output voltage at terminal 2 can be changed by changing the position of the wiper (rotating the knob clockwise or anticlockwise).

### Slider Potentiometer

A potentiometer which is adjusted by sliding the wiper along a straight resistance element instead of rotating shaft is called a *slider potentiometer*. These types of pots are used in large systems like sound mixer control panels, faders, graphic equalizers, etc.

The main advantage of the slider Pot is a large number of them can be installed side-by-side in a small area. The main disadvantages of a slider potentiometer is that they have a long open slot to allow the wiper lug to move freely up and down along the full length of the resistive tracks (Fig.1-30). This open slot makes the resistive tracks susceptible to contamination from dust and dirt, or by sweat and grease from the users hands.

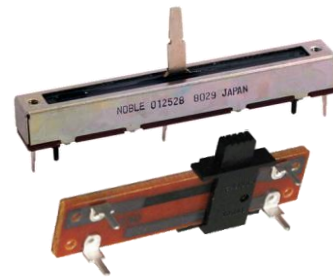


Fig.1-30: Sliding potentiometer

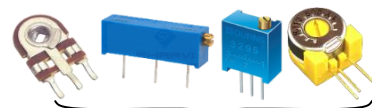
### Presets

There are applications, where the required exact resistance in the circuit cannot be predicted during design. Moreover, some circuits need a precise value of resistance that cannot be achieved with the fixed value resistors. In such applications, it is helpful to use a variable resistor, called *presets*, which can be adjusted to the exact required value when the equipment is aligned (tuned) after manufacturing, but the users need not to adjust it. Actually a preset resistor is a smaller PCB mounted version of a potentiometer.

Different types of preset potentiometers are shown in Fig.1-31. A screw driver is used to rotate the wiper and to change the resistance of these potentiometers.



Variable resistors (presets)



Variable resistors (vertical presets)

Fig.1-31: Different types of presets

### Rheostat

A rheostat is nothing but a variable resistor. The main function of a rheostat is to control the flow of current by changing the resistance value. By increasing or decreasing the resistance value we can increase or decrease the flow of current. Generally, rheostats are used for high voltage and high current applications. Rheostat can be used either as a variable resistance (two terminals are used), or a potentiometer (three terminals are used).

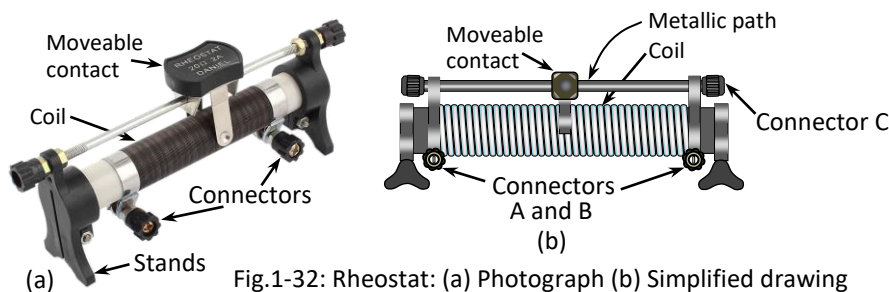


Fig.1-32: Rheostat: (a) Photograph (b) Simplified drawing

As shown in Fig.1-32, a coil is wound on an insulating tube. The two terminals of the coil are attached to two fixed connectors, A and B. Terminal C (variable connector) is attached to the metallic path on which the wiper moves. So when we move the wiper, we will get different resistance values between terminal A and terminal C (also B and C). Remember that this coil is not insulated. By moving the wiper we mainly increase or decrease the turns. So, increasing the turns resistance will increase, and decreasing the turns resistance will decrease. Like the any other potentiometers, if we use this rheostat as variable resistor, we have to use connectors A and C or B and C. But if we want to use as potentiometer, we have to use all the three terminals.

### Non-Linear Resistor

Non-linear resistors are those types of resistors where the electric current flowing through it changes with the change in applied voltage or temperature and does not change according to Ohm's law. There are several types of non-linear resistors, but the most commonly used types are described below.

### Thermistors

Thermistor is a type of variable resistor that is very sensitive to temperature. The sensitivity of thermistors to temperature is called **temperature coefficient**. There are two types of temperature coefficients- PTC (**Positive Temperature Coefficient**), and NTC (**Negative Temperature Coefficient**). In PTC thermistors, resistance increases with the increase in temperature. As shown in Fig.1-33 symbol of PTC thermistors has  $+t^{\circ}$ . On the other hand, in NTC thermistors, resistance decreases with the increase in temperature. Symbol of NTC thermistors has  $-t^{\circ}$ .

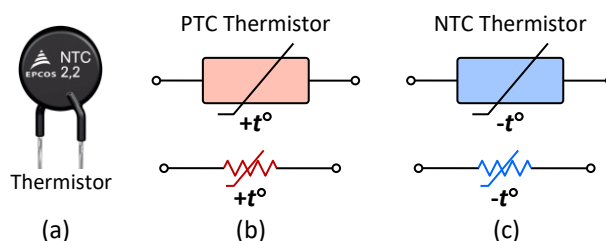


Fig.1-33: Thermistors: (a) Photograph, (b) Symbol of PTC thermistor, and (c) Symbol of NTC thermistor

The temperature of thermistors can be changed in two ways. First one is by the change in atmospheric temperature and the second one is by changing current in the thermistors. By increasing the current, there will be a heating effect in the resistor and thus it will change its resistance by its own heat. This is called the self-heating effect.

Thermistors are used as temperature sensors in fire alarms, ovens and refrigerators. They are also used in digital thermometers and in many automotive applications to measure temperature. Thermistors are often used in temperature-dependent time delay circuits.

### Varistor Resistors

A varistor is a non-linear resistor made of semiconductor. The word **varistor** comes from two words: **variable resistor**. The resistance of a varistor is variable and depends on the voltage applied. MOV or metal oxide varistor is the most commonly used form of varistor resistors. Their resistance decreases when the voltage increases. In case of excessive voltage rise, their resistance drops dramatically. This behavior makes them suitable to protect circuits during voltage surges. Causes of a surge can include lightning strikes and electrostatic discharges.

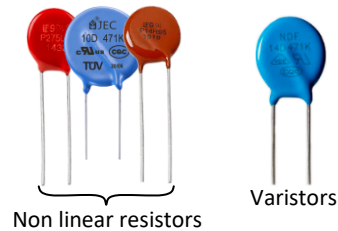


Fig.1-34: Photographs of different varistor

### Photoresistors or Light Dependent Resistors (LDR)

**Photo resistor** or **LDR** (Light Dependent Resistors) or **Photo Conductive Cell** is a variable resistor whose resistance is changed depending of the light falling on its surface. Light Dependent Resistors, LDRs are also called photo resistors. These resistors are made up of high resistance semiconductors which are capable of absorbing photons. Depending on the intensity and frequency of incident light, bound electrons in the semiconductor jump into the conduction band, therefore decreasing the resistance. The semiconductors commonly used to make photo resistors include cadmium sulfide, cadmium selenide, lead sulfide, lead selenide, and indium antimonide. Several countries have now banned photo resistors made from cadmium or lead as these photo resistors may be harmful to the environment. The IEC standard symbol, and its construction are shown in Fig.1-35.

LDRs are used in light-sensitive applications like in smart street lights, camera, etc.

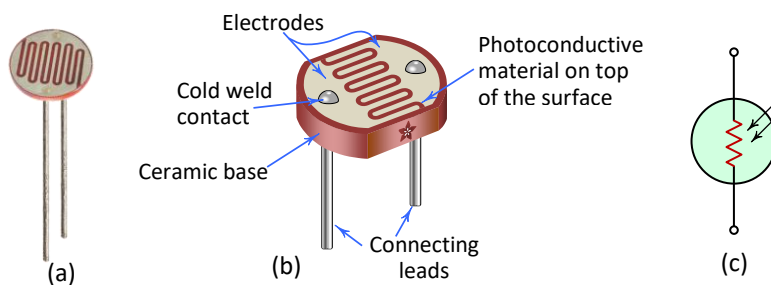


Fig.1-35: LDR (a) Photograph, (b) Different components, and (c) Symbol

### 1.11.2 Color Code Marking of Leaded Resistors

Larger resistors are usually marked with their numerical resistance value printed directly on the side of the resistor. However, this type of labeling is not always practical, especially on small resistors. The resistor color code system was developed for this purpose. The value of resistance of a resistor is written on it using different colors called **color code**. The resistance color code consists of three or four color bands and is followed by a band representing the tolerance (variation of resistance). Another color is sometimes used which is the temperature coefficient band. If provided, it is to the right of the tolerance band. Thus, the total color bands may vary from 4 to 6 (Fig.1-36).

In the resistance color codes, the first two or three colors give the significant figures of the resistance value (in ohms), followed by a multiplier color. This is a factor of 10 by which the significant figure must be multiplied (i.e. the number of zeros to be added after the significant figures) to find the actual resistance value. The next color is the tolerance of the resistance. The tolerance of a resistor is the maximum difference between its actual value and the written value. For example, If a 1 k $\Omega$  resistor has a 10 % tolerance, the resistor's value can vary by  $\pm 100 \Omega$ . In this case, the resistor can have a true resistance value of 900  $\Omega$  to 1100  $\Omega$ . Another color may be added to the right of tolerance band is the temperature coefficient of resistance (TCR). TCR tells how much resistance changes as its temperature changes. It is usually expressed in ppm/ $^{\circ}\text{C}$  (parts per million per degree Celsius) units.

As shown in Fig.1-36, the colors on a resistor represent a number (decimal digit). We can remember these numbers using a popular sentence: "B B ROY Good Boy Very Good Worker" [Fig.1-37]. First "B" is black and its value is "0" and the second "B" is brown = "1".

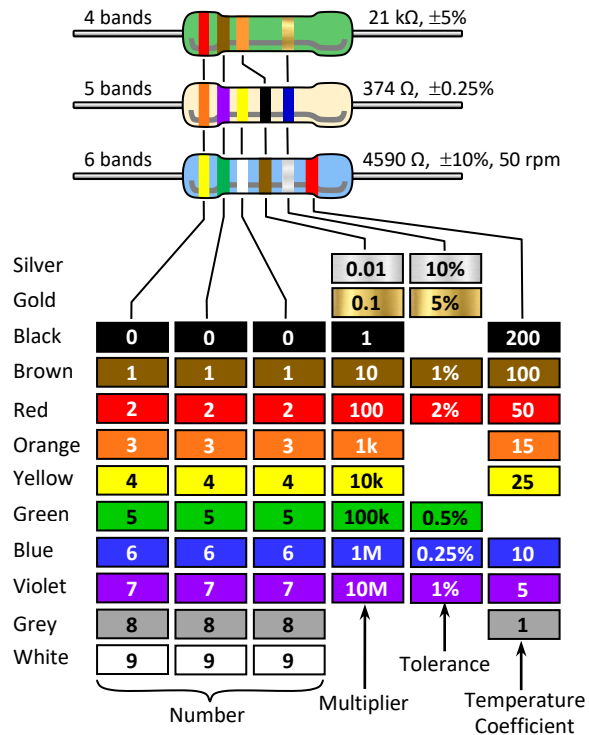


Fig.1-36: Process to read color codes of resistors

**B B ROY Good Boy Very Good Worker**  
 0 1 2 3 4 5 6 7 8 9

Fig.1-37: Process to memorize values of colors

#### Example 1-4

Determine the resistance value of the resistor shown in Fig.1-38.

**Solution:**

Here, the first three colors are Red (2) – Violet (7) – Yellow (4). So the significant figure will be 274. The multiplier color is Gold = 0.1, and the tolerance is Brown = 1 %. Therefore, the value of this resistor will be  $274 \times 0.1 \pm 1\% = 27.4 \Omega \pm 1\%$ . Moreover, the Red color on the rightmost position represents 50 ppm/°C TCR.

**Comments:** 50 ppm/°C TCR means, its resistance will not change more than 0.00005 ohms (50/1,000,000) per ohm per degree Celsius temperature change (within the rated temperature range of  $-55$  to  $+145^\circ\text{C}$ , measured from  $25^\circ\text{C}$  room temperature.)



Fig.1-38: Resistor for Example 1.4

## 1.12 Resistors in Series and Parallel

In building circuits, sometimes we have to connect resistors in series or parallel or both. We connect resistors to increase or decrease the resistance or to increase the power capacity of the combination. Here the series and parallel combinations are discussed.

### 1.12.1 Resistor in Series

When resistors are connected in series, the same current flows through them. But the voltage dropped across individual resistors will be different. In fact, the supplied voltage will be divided among the resistors. The total resistance of a number of resistors in series is equal to the sum of all individual resistances. Mathematically,

$$R_{eq} = R_1 + R_2 + \cdots R_N \quad (1-17)$$

#### Resistors in Parallel

Resistors are said to be in parallel when one end of some resistors (more than one) are connected to one point (or node) and the other end of the resistors are also connected to another point as shown in Fig.1-40. The potential drop across each resistor is the same, but the current through each resistor can be found using Ohm's law  $I = V/R$ , where the voltage is constant across each resistor. The current flowing from the voltage source in Fig.1-40 depends on the voltage supplied by the voltage source and the equivalent resistance of the circuit, and can be calculated using current divider rule,

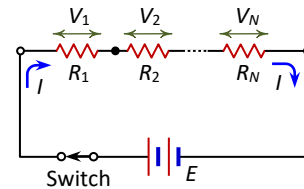


Fig.1-39: Series connected resistors:  
Same current different voltage

$$I_1 = I_s \times \frac{R_2}{R_1 + R_2} \quad (1-18)$$

$$I_2 = I_s \times \frac{R_1}{R_1 + R_2}$$

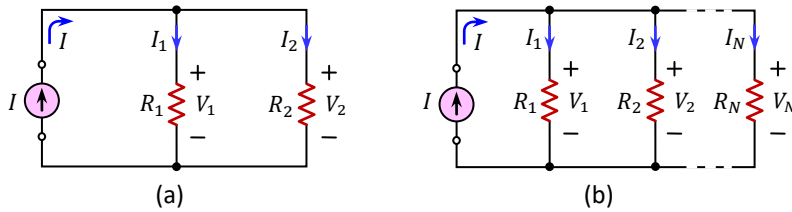


Fig.1-40: Parallel connected resistors: Same voltage different current (a): Two resistors, and (b) N number of resistors

If only two resistors are connected in parallel, the equivalent resistance of the combination will be,

$$R_{eq} = \left( \frac{1}{R_1} + \frac{1}{R_2} \right)^{-1} = \frac{R_1 R_2}{R_1 + R_2} \quad (1-19)$$

But if there are more than two resistors (say  $N$  resistors) the equivalent resistance will be,

$$R_{eq} = \left( \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_2} \dots \dots + \frac{1}{R_N} \right)^{-1} = \left( \sum_{i=1}^N \frac{1}{R_i} \right)^{-1} \quad (1-20)$$

We use resistors in parallel to build a lower value resistance from higher value resistances. In some cases, it helps us to make a precision resistor. For example, suppose we need a 3.1 K resistor but the standard available values are 3.0 k and 3.3 k. 6.2 k is also a standard value. So by connecting two 6.2 k resistors in parallel, we can get a 3.1 k resistor easily.

### Equal Resistors in Parallel

Adding  $N$  number of same-valued resistors ( $R$  is the resistance of each resistor) in parallel gives us  $R/N \Omega$ . Let's say we need a 2.5 k $\Omega$  resistor, but all we've got is a drawer full of 10 k $\Omega$ . Combining four of them in parallel gives us  $(10 \text{ k}\Omega)/4 = 2.5 \text{ k}\Omega$ .

### Tolerance

Resistor tolerance is the deviation from the nominal resistance value. It is expressed as a  $\pm \%$ , measured at 25° C with no load applied. Though most common resistors have  $\pm 10 \%$  tolerance, precision resistors are made with tolerance as low as  $\pm 0.005 \%$ . When resistors are parallelly connected, the actual value of the resistance will also deviate from the exact calculated value. The tolerance level will be same as the tolerance of individual resistors. However, if the component resistors have different tolerance level, the tolerance of the equivalent resistor will have the maximum tolerance of the component resistors. For example, if three resistors of values  $15 \text{ k}\Omega \pm 10\%$ ,  $15 \text{ k}\Omega \pm 5\%$ , and  $15 \text{ k}\Omega \pm 1\%$  are connected in parallel, the equivalent resistance will be  $5 \text{ k}\Omega \pm 10\%$ . This rule is independent of the number of resistors connected in parallel.

### Example 1-5

Suppose three resistors  $R_1$ ,  $R_2$ , and  $R_3$  are connected in parallel. The values of the resistors are  $15 \text{ k}\Omega \pm 10\%$ ,  $15 \text{ k}\Omega \pm 5\%$ , and  $15 \text{ k}\Omega \pm 1\%$ , respectively. Calculate the value of equivalent resistor with its tolerance.

**Solution:**

To calculate the tolerance of the equivalent resistor, we have to consider the two extreme values of the resistors. That is,

$$R_1 = 15 \text{ k}\Omega \pm 10\% = 15 \text{ k}\Omega \pm 1.5 \text{ k}\Omega = 16.5 \text{ k}\Omega \text{ and } 13.5 \text{ k}\Omega$$

$$R_2 = 15 \text{ k}\Omega \pm 5\% = 15 \text{ k}\Omega \pm 0.75 \text{ k}\Omega = 15.75 \text{ k}\Omega \text{ and } 14.25 \text{ k}\Omega$$

$$R_3 = 15 \text{ k}\Omega \pm 1\% = 15 \text{ k}\Omega \pm 0.15 \text{ k}\Omega = 15.15 \text{ k}\Omega \text{ and } 14.85 \text{ k}\Omega$$

So, the maximum value of the equivalent resistance will be,

$$R_{eq(max)} = \left( \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)^{-1} = \left( \frac{1}{16.5 \text{ k}\Omega} + \frac{1}{15.75 \text{ k}\Omega} + \frac{1}{15.15 \text{ k}\Omega} \right)^{-1} = 5.26 \text{ k}\Omega$$

and the minimum value of the equivalent resistance will be

$$R_{eq(min)} = \left( \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right)^{-1} = \left( \frac{1}{13.5 \text{ k}\Omega} + \frac{1}{14.25 \text{ k}\Omega} + \frac{1}{14.85 \text{ k}\Omega} \right)^{-1} = 4.726 \text{ k}\Omega$$

The theoretical equivalent resistance should be  $15 \text{ k}\Omega / 3 = 5 \text{ k}\Omega$

Thus, the tolerance level of the equivalent resistor will be,

$$\text{Tolerance}_{eq} = \frac{(5.26 \text{ k}\Omega - 4.726 \text{ k}\Omega)}{5 \text{ k}\Omega} \times 100 \% = 10.68 \%$$

**Comments:** The tolerance of the equivalent resistance is even larger than the maximum tolerance of the individual resistors.

### 1.12.2 Power Ratings of Series/Parallel Resistors

Every resistor has a limit to the maximum power level that the resistor can withstand. This maximum power rating, limits the voltage that can be applied across it, and the maximum current that can pass through it. **Generally speaking, the larger their physical size the higher their power rating [see Fig.1-41].** It is always better to select a resistor that is capable of dissipating more power than the calculated power.

When resistors of equal resistance and power ratings are connected in series or parallel, the power of the equivalent resistance will be just some of the power rating of the individual resistors. But when we combine resistors of different resistance and power values, the equivalent power will be less than the sum of the individual powers. However, in some extreme case the power rating of the equivalent resistance may be just equal to the power rating of the smallest resistance connected in series/parallel.

The power dissipated in a parallel combination of dissimilar values is not split evenly between the resistors because the currents are not equal. So, we have to pay closer attention to what we're doing when combining resistors of dissimilar values in parallel. The situation is explain in Fig.1-42(a). Here, two resistors are connected in parallel.  $R_1 (= 1 \text{ k}\Omega)$  is a fixed value resistor, and  $R_2$  is a variable resistor ( $= 10 \text{ k}\Omega \sim 60 \Omega$ ). The power rating of  $R_1$  is 1 W and the power rating of  $R_2$  is 10 W. The supply voltage has been changed in such a way that the power of equivalent resistance is equal

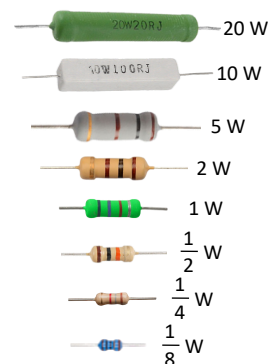


Fig.1-41: Physical size of different resistors of different power

to the sum of the individual powers. That is,  $V_s^2 \times (R_1 || R_2) = 11 \text{ W}$ . For this situation, the power consumed by the individual resistors for different values of  $R_2$  is shown in Fig.1-42(b). As shown in this graphs, the resistors get their rated power only when  $R_1/R_2 = 10$ , which is the reciprocal of the power ratio of the resistors. For other values of  $R_1/R_2$  at least one resistor will get over powered, which is unsafe. Thus, for resistors of different resistance values, the equivalent power will not be the sum of individual powers. Moreover, from these graphs, it is clear that the actual power dissipation will increase if the resistance decreases.

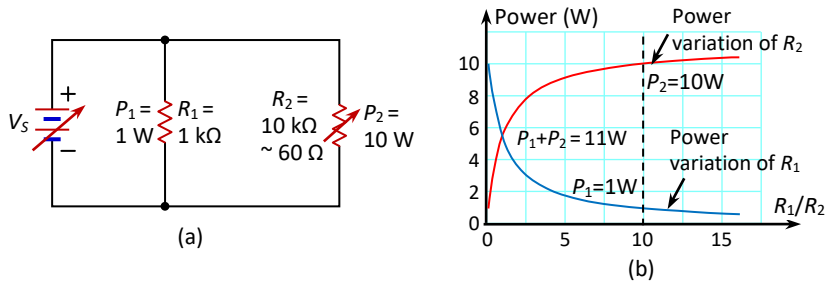


Fig.1-42: Variation of power consumption of two resistors with the variation of their relative value

In case of series connected resistors of different values and power, the equivalent power will be the sum of individual powers, only when the resistors have the same resistance. Here also, for resistors of different values, we have to pay extra care so that one resistor may not be over powered. For series connection, the actual power dissipation will increase if the resistance decreases.

### Example 1-6

Suppose  $R_1 = 10 \Omega$ , 1 W and  $R_2 = 10 \text{ k}\Omega$ , 5 W connected in parallel. Will the power capacity of the combination be the sum of individual powers? Justify your answer.

#### Solution:

The equivalent resistance will be,

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{10 \Omega \times 10 \text{ k}\Omega}{10 \Omega + 10 \text{ k}\Omega} = 9.99 \Omega$$

Now let us assume that the power rating of the combined resistance will be the sum of the individual resistors. That is  $P_{eq} = 1 \text{ W} + 5 \text{ W} = 6 \text{ W}$ . So, the maximum voltage of the equivalent resistance can be calculated as,

$$6 \text{ W} = \frac{V_m^2}{R_{eq}} \text{ or, } V_m = \sqrt{6 \text{ W} \times R_{eq}} = \sqrt{6 \text{ W} \times 9.99 \Omega} = 7.74 \text{ V}$$

So, if 7.74 V is applied across the parallel combination, the power dissipated by the smaller resistor will be,

$$P_1 = \frac{V_m^2}{R_1} = \frac{(7.74 \text{ V})^2}{10 \Omega} = 5.99 \text{ W} \approx 6 \text{ W} !!$$

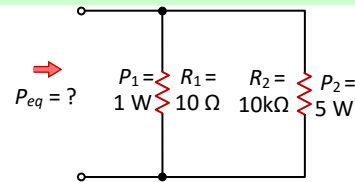


Fig.1-43: Resistor for Example 1.6

That is, the smaller resistor will consume almost all of the power. Therefore, the power capacity of the equivalent resistance will be 1 W.

**Comments:** If the resistors have equal values, then the power capacity will be summed up. With the decrease of resistance of one resistor, the power capacity of the equivalent resistance will gradually get closer to the power capacity of the smaller one.

### 1.13 Inductors

**Inductor** (also called *coil*, *choke*, or *reactor*) is a passive two-terminal electrical component that stores energy in the form of magnetic field when current flows through them. An inductor typically consists of an insulated wire wound into a coil around a core [Fig.1-44]. When current flows through an inductor magnetic field is produced inside the coil as shown in Fig.1-45(a). If the current flowing through the inductor changes with time, magnetic field also varies with time, and this time-varying magnetic field induces an electromotive force or voltage ( $e = -Ndi/dt$ ) across the inductor as described by Faraday's law of electromagnetic induction [Fig.1-45(c)]. According to Lenz's law, the induced voltage has such a polarity (direction) that it opposes the change in current that produces it. As a result, inductors oppose any change in current (AC current) through them by inducing a voltage (in opposite direction). The ability of an inductor to oppose the change in current is measured by a parameter called **inductance** and is denoted by  $L$ . Inductance is defined as the ratio of the induced voltage to the rate of change of current ( $L = e/(di/dt)$ ). In SI system, the unit of inductance is **Henry** (H). Along with capacitor and resistor, inductor is one of the three passive linear circuit elements that are used to build up electrical and electronic circuits.

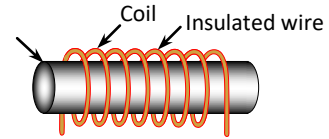


Fig.1-44: Inductor

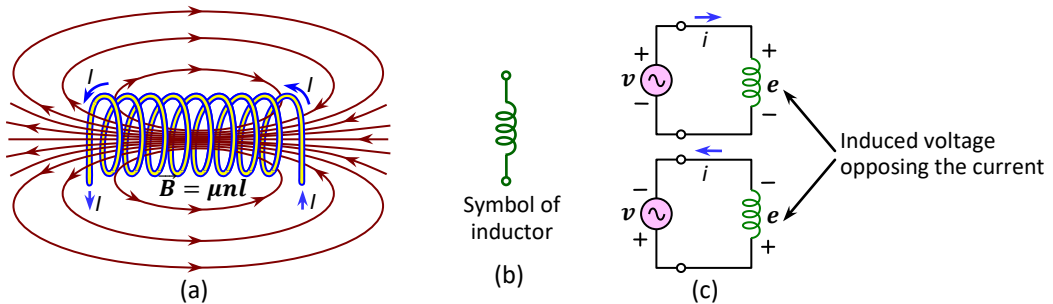


Fig.1-45: Inductor: (a) Coil with magnetic field, (b) Symbol of inductor, and (c) Inductors with induced voltage

Inductors are widely used in alternating current (AC), electronic equipment, particularly in radio equipment. They are used to block AC while allowing DC to pass. Unlike the capacitors, inductors oppose the AC current flowing through them. The opposing parameter of an inductor to AC current is called **inductive reactance** ( $X_L$ ) and is directly proportional to the frequency of the AC current and the value of the inductance. Thus, the reactance is calculated as,

$$X_L = \omega L = 2\pi fL \quad (1-21)$$

There are various types of inductors available in the market with different values, sizes and shapes. To be familiar with them, photographs of some common inductors are given in Fig.1-46.

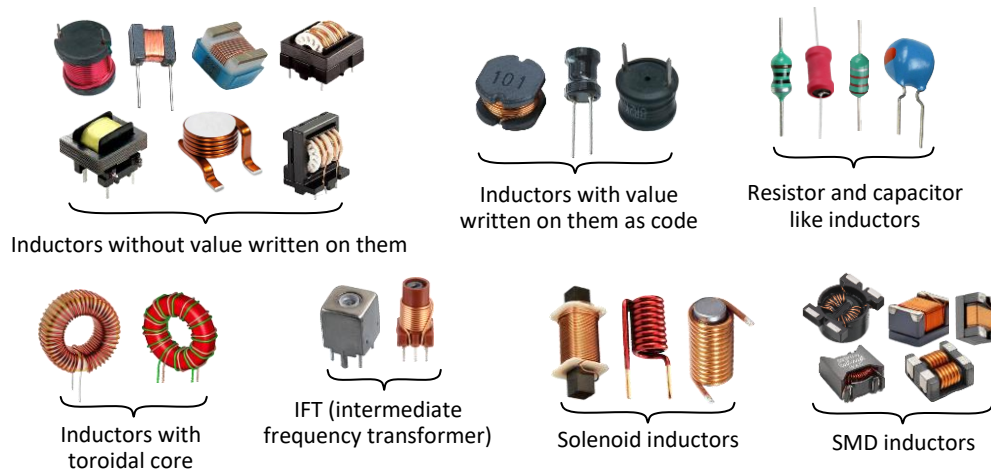


Fig.1-46: Photographs of various inductors available in the market

### 1.14 Types of Inductors

Depending on the number of coils, shape, core materials, current capacity, etc. there are various types of inductors. Some of them are described below.

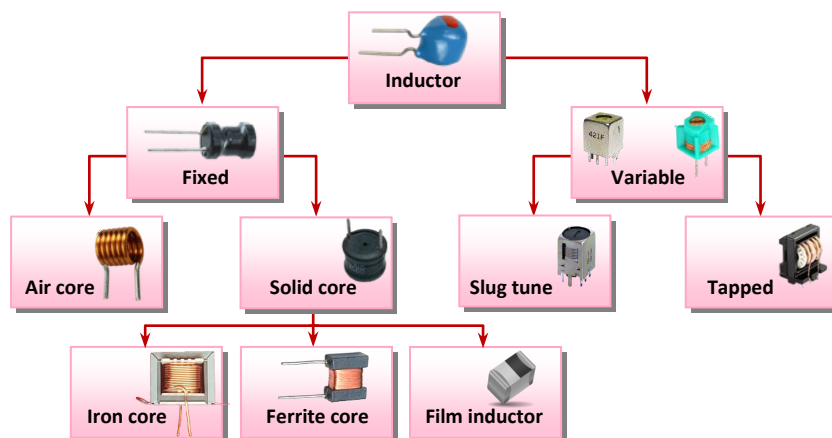


Fig.1-47: Classification of inductors

#### 1.14.1 Air Core Inductors

**Air core** inductors are made winding a coil on either non-magnetic core or no core at all. The use of an air core results in lower inductance, but also reduces the energy losses associated with core inductors. As there is no core losses, air core inductors can be operated at very high frequencies.

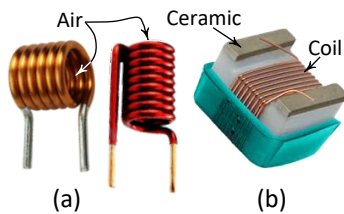


Fig.1-49: (a) Air core inductor  
(b) Ceramic core inductor

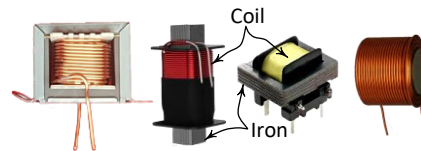


Fig.1-48: Iron core inductors

As a non-magnetic core material, ceramic is the most commonly used. Ceramic has very low thermal co-efficient of expansion, so even for a range of operating temperatures the stability of the inductor's inductance is high. The main function of the ceramic core is to give a form for the coil. In some cases, it will also provide the structure to hold the terminals in place. The main advantage of these inductors are very low core losses, high **quality factor**. These are mainly used in high frequency applications where low inductance values are required.

### 1.14.2 Iron Core Inductor

The value of the inductance of air core inductors depends on the number of turns, length, diameter, thickness of the spiral, etc. The air core inductors have limited inductance values. In order to increase the inductance value of an inductor, an iron core is placed inside it which results in an iron core inductor. These are applicable in audio equipments. When compared with other core inductors these have very limited applications, due to their losses and frequency limitations.

### 1.14.3 Ferrite Core Inductors

Ferrite is a very good ferromagnetic material with very high resistance for electric current. They exhibit magnetic properties with small hysteresis loss. Ferrites are made by sintering a mixture of metal oxides and have the general chemical composition  $MO \cdot Fe_2O_3$ , where M is a divalent metal such as Mn, Mg, Fe, Zn, Ni, Cd, etc.

Two types of ferrites are mainly used- **soft ferrite** and **hard ferrite**. These are classified according to the magnetic coercivity. **Coercivity** is the magnetic field intensity needed to demagnetize the ferromagnetic material from complete saturation state to zero. **Soft ferrite** materials will have the ability to reverse their polarity of magnetization without any particular amount of energy. **Hard ferrite** are also called as permanent magnets. These will keep the polarity of the magnetization even after removing the magnetic field.

Ferrite core helps to improve the performance of the inductor by increasing the permeability of the coil which increases the value of the inductance. The level of the permeability of the ferrite core used within the inductors will depend on the ferrite material. This permeability level ranges from 20 to 15,000 (relative) according to the material of ferrite. Thus, the inductance is very high with ferrite core when compared to the inductor with air core.

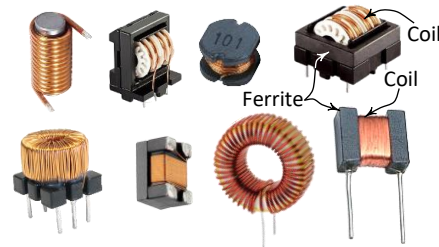


Fig.1-50: Ferrite core inductors

#### 1.14.4 Toroidal Inductor

These inductors are made by winding wires on a ring-shaped or donut-shaped core. Different materials like ferrite, powdered iron, etc. are used to construct the core. Sometimes depending on the applications, more than one coils are used. This inductor has high coupling coefficient between windings and early saturation. Its arrangement gives minimum loss in magnetic flux which helps to avoid coupling magnetic flux with other devices. It has high energy transferring efficiency and high inductance values at low frequency applications. These inductors mainly used in medical devices, switching regulators, air conditioners, refrigerators, telecommunications and musical instruments etc.

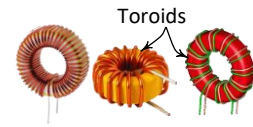


Fig.1-51: Toroidal inductors

#### 1.14.5 Film Inductors

These inductors use a film of conductor on base material. Thus according to the requirement this film is shaped for conductor application. As shown in Fig.1-52, film inductors are mainly constructed in SMD forms. For their very thin size, film inductors are suitable for DC to DC converters that serve as power supplies in smart phones and mobile devices.

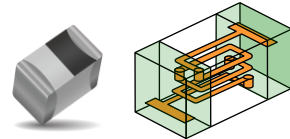


Fig.1-52: Film inductor (SMD)

#### 1.14.6 Variable Inductors

The inductance of a variable inductor can be changed by rotating screw, attached to it. The rotation of screw moves the magnetic core in and out of the inductor windings. By this magnetic core we can adjust the inductance value. Fig.1-53 shows some ferrite core variable inductors. Sometimes variable inductors are made by taking tapes at different turns of the coils.

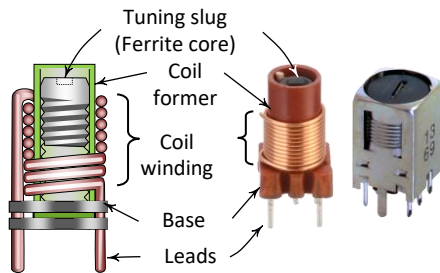


Fig.1-54: Variable inductors

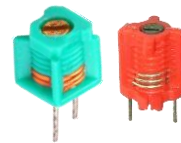


Fig.1-53: Molded inductors

These type of inductors are used in radio and high frequency applications where the tuning is required. These inductors are typically ranged from 10 nH to 100 mH.

#### 1.14.7 Molded Inductors

These inductors are molded by plastic or ceramic insulators. The variable inductors are mainly made in molded structures. These are typically available in bar and cylindrical shapes with wide option of windings.

### 1.14.8 Coupled Inductors

In this arrangements, two coils are constructed in one core to connect them magnetic coupling, just like a transformer. AC current flowing in one inductor induces voltage in second inductor, and gives a mutual inductance,  $M$ . Some photographs of this type of inductors are given in Fig.1-53 to 1.5. Fig.1-55(b) shows its equivalent circuit of a two coils coupled inductor. From this equivalent circuit, we can calculate the input impedance as,



Fig.1-55: Tapped & Coupled inductors: (a) Photographs, and (b) Equivalent circuit

$$Z_{in} = \frac{R_L}{R_L^2 + (\omega L_2)^2} (L_1 L_2 \omega^2 k^2) + j \left\{ \omega L_1 - \frac{\omega L_2}{R_L^2 + (\omega L_2)^2} (L_1 L_2 \omega^2 k^2) \right\} \quad (1-22)$$

Here,  $k$  is the coupling coefficient. For 100 % magnetic coupling  $k = 1$ .

If  $\omega L_2 \gg R_L$ , then:

$$Z_{in} = R_L \frac{L_1 k^2}{L_2} + j \omega L_1 (1 - k^2) \quad (1-23)$$

Again, if  $k = 1$  (ideal case), then:

$$Z_{in} = R_L \frac{L_1}{L_2} \quad (1-24)$$

These inductors can isolate two circuits electrically by transferring impedance from one coil to the other coil. We know that a transformer (step up or step down) isolates the primary (coil) circuit from the secondary (coil) circuit. Thus, a transformer is also one kind of coupled inductor.

### 1.15 Inductor Color Code

The value of inductance is written directly on the device or using color code. Although, there is variation in number of colors, a 4-band color code system is shown in Fig.1-56. 4-band color coding of inductor is same as resistor color code. The numerical values of the colors are always same, e.g., Black = 0, Brown = 1 and so on.

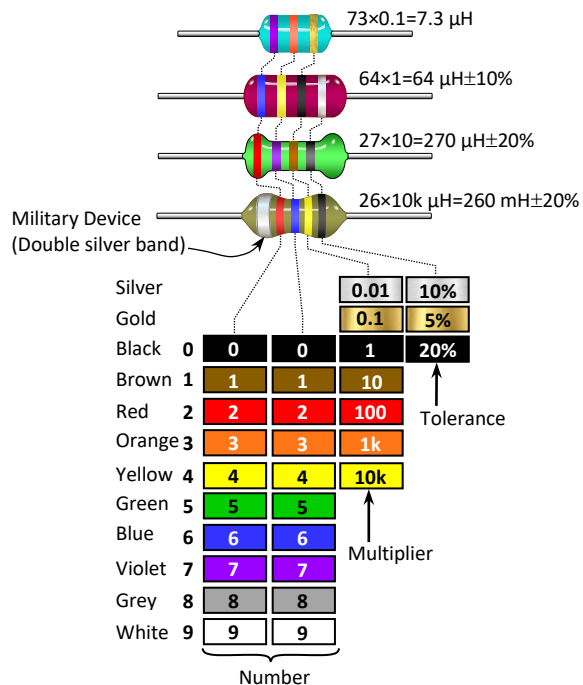


Fig.1-56: Process to read color codes of inductor

Here also, the first two-digit represent a number and the 3<sup>rd</sup> color is the multiplier. When the value formed by the first two color is multiplied by the multiplier the inductance will be found in micro Henry ( $\mu\text{H}$ ). The 4<sup>th</sup> color gives the tolerance that is the range of variation of the values represented by the first three colors. To understand the process, the inductance values of some inductors are determined from their colors and are shown in Fig.1-56.

### Example 1-7

Determine the inductance of the inductor shown in Fig.1-57.

#### Solution:

The color code written on the inductor is: Orange, Blue, Red, and Silver. The value of first two colors is 35. The third color is red which is the multiplier. The value of the multiplier is 100. The fourth color is the tolerance. Silver means 10%. Thus, the value of the inductance will be,  $35 \times 100 \pm 10\%$ .

**Comments:** Actually, the use of color codes in inductors is not as frequent as in resistors.

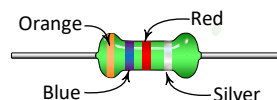


Fig.1-57: Inductor for Example 1

## 1.16 Series and Parallel Connection of Inductors

Like the resistors, inductors are also used in series and parallel combinations. Here also, the series combination increases the inductance and parallel combination decreases it, but for inductor the calculations may be different, because of the mutual coupling of two or more inductors. The theories related to these combinations are discussed below.

### 1.16.1 Inductors in Series

Inductors are connected together in series, when they are daisy chained together sharing a common electrical current (Fig.1-58). In series connection, inductances are simply added because the number of coil turns is effectively increased. The total (or equivalent) inductance  $L_{eq}$  is equal to the sum of all individual inductances added together in series. For  $n$  number of inductors connected in series, the equivalent inductance is calculated as,

$$L_{eq} = L_1 + L_2 + \dots + L_n \quad (1-25)$$

As shown in Fig.1-58, the current flowing through all the inductor is  $I$ . However, the voltage across them will be different and will be directly proportional to their individual inductance.

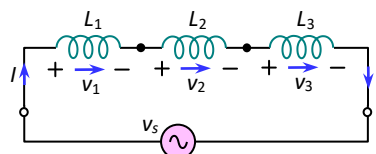


Fig.1-58: Series connected inductors

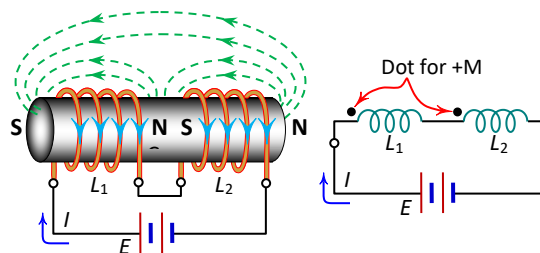


Fig.1-59: Cumulatively coupled series connected inductors: (a) Inductors with similar magnetic field (b) Inductors with dot notations

However, the above equation [Equ.(1-25)] only holds true when there is no mutual inductance or magnetic coupling among the inductors. Two inductors will be mutually coupled if the magnetic field of one inductor affects the other inductor.

When inductors are connected in series in such a way that the magnetic field of one links with the other, that is, there is magnetic coupling between the coils. In this case, the inductance of the individual inductors will be changed due to this mutual effect which is called **mutual inductance** ( $M$ ). The value of this mutual inductance depends on the coupling coefficient ( $k$ ).

Mutual inductance can be either additive or subtractive. If the direction of magnetic flux produced by the current through the coils is in the same direction (as shown in Fig.1-59), then the coils are said to be **Cumulatively Coupled**, but if the direction of flux is in opposite directions (as shown in Fig.1-60) then the coils are said to be **Differentially Coupled**.

When two series connected inductors are cumulatively coupled the resultant inductance will be,

$$L_{eq} = L_1 + L_2 + 2M \quad (1-26)$$

Here  $M$  is called the mutual inductance.

But, when the series connected inductors are oppositely coupled the resultant inductance will be,

$$L_{eq} = L_1 + L_2 - 2M \quad (1-27)$$

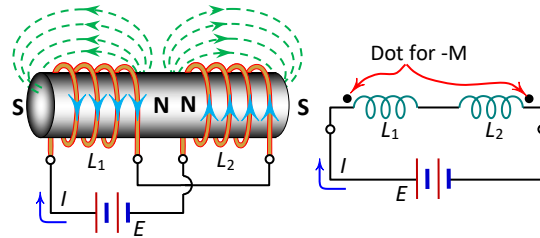


Fig.1-60: Differentially coupled series connected inductors: (a) Inductors with opposite magnetic field (b) Inductors with dot notations

### Example 1-8

Find the equivalent inductance of the following circuit. Assume,  $L_1 = 2.5 \text{ H}$ ,  $L_2 = 3.5 \text{ H}$ ,  $L_3 = 5 \text{ H}$ ,  $M_{12} = 1.5 \text{ H}$ ,  $M_{23} = 1 \text{ H}$ , and  $M_{13} = 0.5 \text{ H}$ .

#### Solution:

The mutual inductance between  $L_1$  and  $L_2$  is additive, other two mutual inductance is subtractive (found from the dot notation). Thus, the effective value of inductances will be,

$$L'_1 = L_1 + M_{12} - M_{13} = 2.5 \text{ H} + 1.5 \text{ H} - 0.5 \text{ H} = 3.5 \text{ H}$$

$$L'_2 = L_2 + M_{12} - M_{23} = 3.5 \text{ H} + 1.5 \text{ H} - 1 \text{ H} = 4.0 \text{ H}$$

$$L'_3 = L_3 - M_{13} - M_{23} = 5 \text{ H} - 0.5 \text{ H} - 1 \text{ H} = 3.5 \text{ H}$$

Using Equ(1-), the equivalent inductance of the combination will be,

$$L_{eq} = L'_1 + L'_2 + L'_3 = 3.5 \text{ H} + 4.0 \text{ H} + 3.5 \text{ H} = 11 \text{ H [Ans.]}$$

**Comments:** The type (additive or subtractive) of mutual inductance is determined from dot notation. The equivalent inductance can also be calculated as  $L_{eq} = L_1 + L_2 + L_3 + 2M_{12} - 2M_{23} - 2M_{13}$

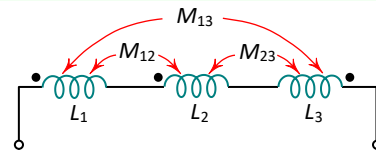


Fig.1-61: Series connected inductors for Example 1-8

### 1.16.2 Inductors in Parallel

Inductors are said to be connected in parallel when one terminal of all inductors are connected to a common point, and all the opposite terminals are connected to another point as shown in Fig.1-62. The voltage drop across all the inductors in parallel will be the same, but current following through the individual inductor will be different.

The equivalent ( $L_{eq}$ ) or total inductance for  $n$  number of inductors in parallel is calculated as,

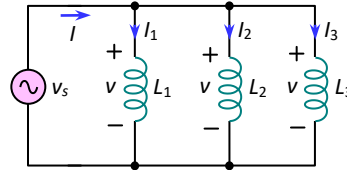


Fig.1-62: Parallel connected inductors

$$\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \frac{1}{L_3} + \dots + \frac{1}{L_n} \quad (1-28)$$

The above equation only holds true when there is no mutual inductance or magnetic coupling between the inductors, (magnetically isolated from each other). Like series connected inductors, when there is coupling between the coils, the total inductance is also affected by the amount of magnetic coupling.

When inductors are connected in parallel in such a way that the magnetic field of one inductor links with the other then a mutual inductance ( $M$ ) is developed. The value of this mutual inductance depends upon coupling coefficient.

Depending of the direction of flux, the mutual inductance can increase or decrease the equivalent inductance. In case of series connected inductance the coupling process is explained using the flux direction. But in case of parallel connections, this is represented using **dot notation**. Mutually coupled parallel coils can be shown as either connected in an aiding or opposing configuration by the use of polarity dots or polarity markers as shown in Fig.1-63 and 1-64.

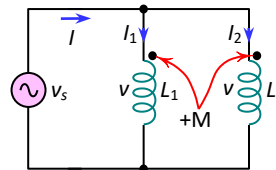


Fig.1-63: Parallel connected inductors

The equivalent inductance,  $L_{eq}$  for two parallel connected inductors with aiding mutual inductance (both dots on upper sides) given as,

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 - 2M} \quad (1-29)$$

Where,  $2M$  represents the influence of coil  $L_1$  on  $L_2$  and likewise coil  $L_2$  on  $L_1$ .

If one of the two coils was reversed with respect to the other, we would then have two parallel inductors with opposing mutual inductance ( $M$ ) as shown in Fig.1-64.

Then the equivalent inductance ( $L_{eq}$ ) for two parallel connected opposing inductors can be calculated as,

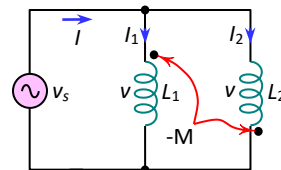


Fig.1-64: Parallel connected inductors

$$L_{eq} = \frac{L_1 L_2 - M^2}{L_1 + L_2 + 2M} \quad (1-30)$$

An important thing to notice that if the two inductances are equal in value (let  $L_1 = L_2 = L$ ) and the magnetic coupling is perfect between them ( $K = 1$ ), the equivalent inductance and also the self-induced emf across the inductors will be zero as the inductors cancel the flux each other. This is because the two currents, ( $i_1$  and  $i_2$ ) flowing through each inductor will be same and the same amount (in magnitude) of flux will be produced by each inductor but in opposite directions. As there is perfect mutual coupling, one flux will completely cancel the flux in the other. Then, the two coils effectively become a short circuit to the flow of current so the equivalent inductance,  $L_{eq}$  becomes

using equ.(1-10) 
$$L_{eq} = \frac{L^2 - M^2}{2L + 2M} = \frac{(L + M)(L - M)}{2(L + M)} = \frac{L - M}{2} \quad (1-31)$$

Again for perfect coupling,  $L = M$ , and hence,  $L_{eq} = 0$ .

### Example 1-9

The values of two mutually coupled inductors are  $L_1 = 5$  H,  $L_2 = 8$  H. If they are connected like i) Fig.1-63 and ii) Fig.1-64 determine the equivalent inductance. Assume,  $M_{12} = 1.5$  H.

#### Solution:

Given,  $L_1 = 5$  H,  $L_2 = 8$  H, and  $M_{12} = 1.5$  H

i) When the inductors are connected as in Fig.1-63, the mutual inductance will be additive. Therefore,

Using Equ.(1-29) 
$$L_{eq} = L_{eq} = \frac{L_1 L_2 - M_{12}^2}{L_1 + L_2 - 2M_{12}}$$

or, 
$$L_{eq} = \frac{5 \text{ H} \times 8 \text{ H} - (1.5 \text{ H})^2}{5 \text{ H} + 8 \text{ H} - 2 \times 1.5 \text{ H}} \approx 3.78 \text{ H [Ans.]}$$

ii) When the inductors are connected as in Fig.1-64, the mutual inductance will be subtractive. Therefore,

Using Equ.(1-30) 
$$L_{eq} = \frac{L_1 L_2 - M_{12}^2}{L_1 + L_2 + 2M_{12}}$$

or, 
$$L_{eq} = \frac{5 \text{ H} \times 8 \text{ H} - (1.5 \text{ H})^2}{5 \text{ H} + 8 \text{ H} + 2 \times 1.5 \text{ H}} \approx 2.36 \text{ H [Ans.]}$$

**Comments:** The additive (dots in the similar terminals) mutual inductance increases the equivalent inductance whereas, subtractive (dots in the opposite terminals) mutual inductance decreases it.



## 1.17 Capacitors and Capacitance

A **capacitor** (originally called condenser) is a passive two-terminal electrical component used to store electrostatic energy (or charge). Though the shapes and sizes of practical capacitors vary widely, but all capacitors contain two electrical conducting plates separated by a dielectric (i.e., insulator) material [Fig.1-65]. The conductors can be thin films of metal, aluminum foil (rolled cylindrically) or disks, etc. Capacitors are widely used as parts of electrical circuits in combination with many common electrical devices. Unlike a resistor, a capacitor does not dissipate energy. Instead, a capacitor stores energy in the form of electric field (in dielectrics) between its plates.

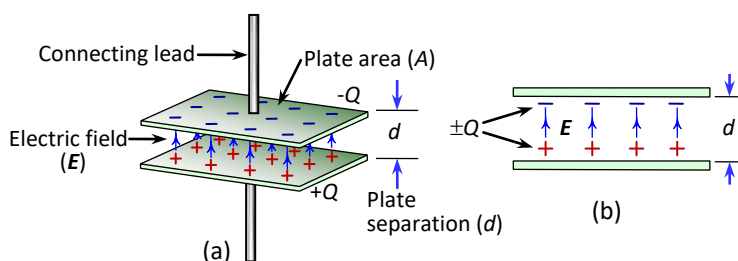


Fig.1-65: Components of a parallel plate capacitor with charge and field: (a) 3-D view, and (b) 2-D view

When potential difference is applied across a capacitor, negative charge accumulates on one plate and positive charge on the other plate as shown in Fig.1-65. Due to these opposite charges between the dielectric film (or air), an electric field is developed, where the energy is stored. The ability of a capacitor to store charge is called **capacitance** and is denoted by  $C$ . The SI unit of capacitance is Farad (F), which is equal to one coulomb per volt (1 C/V). The capacitance of a capacitor will be 1 F, if the capacitor can store 1 Coulomb charge for 1 V potential difference applied to its plates. Farad is a very large unit, so the practical unit of capacitance is microfarad ( $1 \mu\text{F} = 10^{-6} \text{ F}$ ).

In practice, the dielectric between the plates has an electric field strength limit, known as the **breakdown voltage**. This voltage is called the **tolerance voltage** of the capacitor. The maximum voltage up to which the capacitor can withstand (before dielectric breakdown) is called **working voltage** or **tolerance voltage** of the capacitor. In some capacitors (e.g. electrolytic capacitors), the tolerance voltage is written on the capacitor [Fig.1-70 or 93]. The applied voltage to a capacitor must be equal or less than the tolerance voltage.

Capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass. In analog filter networks, they smooth the output of power supplies. In resonant circuits they tune radios to particular frequencies. In electric power transmission systems they stabilize voltage and power flow.

Already we have discussed that a capacitor can store charge. In DC (direct current) circuits capacitors store charges, but in AC circuits capacitors work just like a resistor. The obstacle produced by a capacitor to the AC current is called **capacitive reactance** and is denoted by  $X_C$ . The reactance of a capacitor is inversely proportional to its capacitance and the frequency of the AC current. Mathematically,

$$X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C} \quad (1-32)$$

where,  $f$  is the frequency of AC current and  $C$  is the capacitance of the capacitor.

For AC voltage, the capacitor behaves as a linear device, i.e., if the frequency and the capacitance are constant, the capacitor current will increase linearly with the increase in AC voltage. But for DC circuit, we get the transient response of the capacitors.

The shapes and sizes of practical capacitors vary widely according to their applications. A number of capacitors are available in markets. Some of the common capacitors are shown in Fig.1-67.

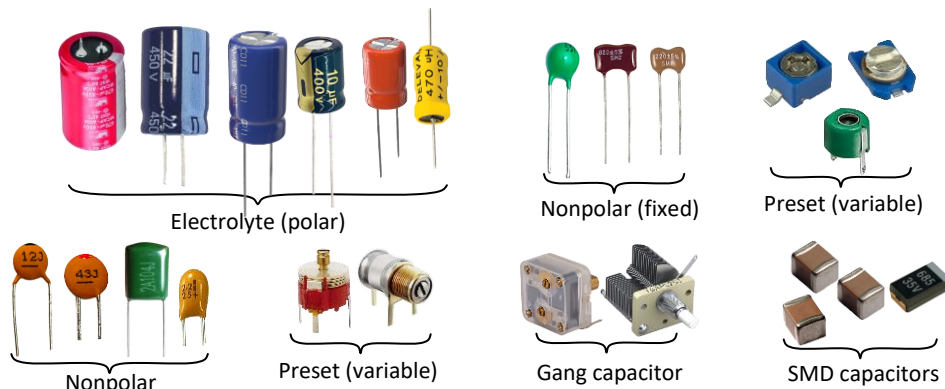


Fig.1-66: Photographs of different types of capacitors

## 1.18 Classifications of Capacitors

The types of capacitors range from a very small delicate trimming capacitors, used in an oscillator or radio circuits, up to large power metal-can type capacitors used in high voltage power factor correction and filtering circuits. Capacitors are classified depending on shapes, constructions, materials, applications, and so on.

According to mechanical structure, capacitors are classified as:

- Fixed Capacitors
- Variable Capacitors

According to polarization, capacitors are classified as:

- Polarized (or polar) Capacitors
- Nonpolarized (or nonpolar) Capacitors

According to the dielectric materials:

- Electrolytic Capacitors
- Ceramic Capacitors
- Film Capacitors
- Tantalum Capacitors
- Paper Capacitors
- Power Film Capacitors

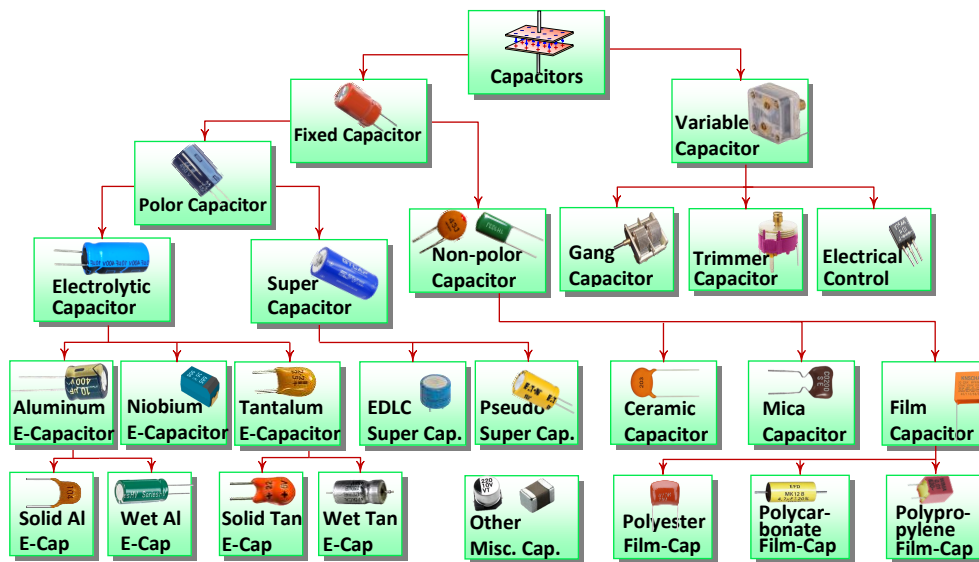


Fig.1-67: Classifications of capacitors

Though not all, Fig.1-68 shows the classifications of major capacitors. A polarized capacitor is an important component of the electronic circuit and often termed as an *electrolytic capacitor*. The electrolytes are used to achieve high capacitive density. The positive and negative terminals are fixed in these capacitors, so, they can be used in a DC circuit, but not in AC circuits.

On the other hand, *nonpolarized capacitors* have no polarity in their terminals. So these capacitors can be used in both DC and AC systems. Nonpolarized capacitors are preferred over polarized capacitors because they are destroyed by reverse voltage. The nonpolar capacitors have very small leakage current and can work in very high frequency.

### 1.18.1 Electrolytic Capacitors

In an *electrolytic capacitor* metallic anode coated with an oxidized layer used as a dielectric. These capacitors are polarized. They are able to offer high capacitance values – typically above  $1\ \mu\text{F}$ . These capacitors are most widely used for low-frequency applications – power supplies decoupling, and audio coupling applications as they have a frequency limit around 100 kHz. Electrolytic capacitors are categorized based on their dielectric.

- Aluminum Electrolytic Capacitors – Aluminum Oxide (Dielectric)
- Niobium Electrolytic Capacitors – Niobium Pentoxide (Dielectric)
- Tantalum Electrolytic Capacitors – Tantalum Pentoxide (Dielectric)

#### Aluminum Electrolytic Capacitors

Aluminum electrolytic capacitors are polarized electrolytic capacitors whose anode electrode is made of a pure aluminum foil with an etched surface. By anodization a very thin aluminum oxide is

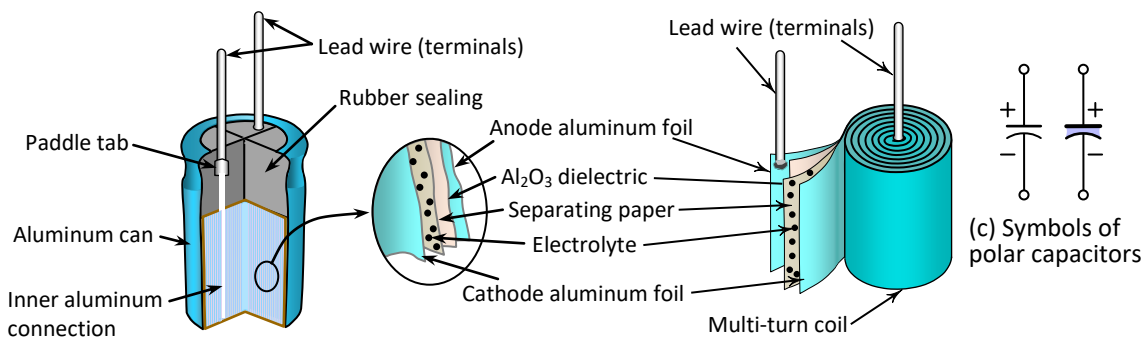


Fig.1-68: Construction of aluminum electrolytic capacitors

made that acts as the dielectric of the capacitor. A non-solid electrolyte covers the rough surface of the oxide layer, serving as the second electrode (cathode) of the capacitor. A second aluminum foil called cathode foil is placed in contact to the electrolyte and serves as the negative terminal of the capacitor. The construction process is illustrated in Fig.1-69. Aluminum electrolytic capacitors are divided into three subfamilies by electrolyte type:

- Non-solid (liquid or wet) aluminum electrolytic capacitors,
- Solid manganese dioxide aluminum electrolytic capacitors, and
- Solid polymer aluminum electrolytic capacitors.



Fig.1-69: Electrolytic capacitors

### Applications of Aluminum Electrolytic Capacitors

- Electrolytic capacitors are used when a large capacitance is required.
- They are used as filtering devices that lower the ripple voltage.
- They are used in audio amplifiers to reduce the electrical noise that is induced by the main supply.
- Electrolytic capacitors are used in smoothing the input and output signals in a DC signal that has a weak AC component.

### 1.18.2 Niobium Electrolytic Capacitors

The construction of niobium electrolytic capacitor is same as aluminum electrolytic capacitor. The main difference is instead of aluminum (Al), niobium (Nb) is used. Here, the anode is made of passivated niobium metal or niobium monoxide, on which an insulating niobium pentoxide layer ( $\text{Nb}_2\text{O}_5$ ) is made that acts as a dielectric. A solid electrolyte on the surface of the oxide layer serves as the capacitor's cathode. The dielectric constant of  $\text{Nb}_2\text{O}_5$  is higher, but the tolerance voltage is low. So, niobium-based capacitors have significantly lower maximum rated voltage compared to tantalum capacitors. Niobium capacitors are most commonly constructed in cylindrical structures and SMD structures.

Niobium capacitors are used for their high reliability, long term operation, safety circuits, with standard ESR requirements. They show high mechanical robustness against shocks and vibrations and stable electrical performance.

### 1.18.3 Tantalum Capacitors

Tantalum capacitors are also electrolytic capacitors and have polarity. They offer a very high capacitance per volume. Tantalum capacitors are constructed using tantalum for at least anode material. Tantalum material tends to naturally form a non-conductive oxide surface layer which functions as the capacitor's dielectric. Because this layer is thinner and lighter than dielectric materials such as paper and plastic, tantalum capacitors can be manufactured as very small components with excellent capacitance to size ratio.

Tantalum metal is ground into a fine powder, and sintered into a pellet at high temperatures. This forms a very porous anode with a high surface area that yields higher capacitance per volume. The anode is then covered with a layer of insulating oxide, which acts as a dielectric. This step is precisely controlled to ensure required voltage tolerances, and capacitance values as the extent of oxide growth determines the dielectric thickness. Electrolyte is added to the anode by means of pyrolysis in the case of solid tantalum capacitors. Solid tantalum capacitors are then dipped into a special solution and baked in an oven to produce a manganese dioxide coat. The process is repeated until a thick coating is produced on all internal and external surfaces of the pellet. Finally, the pellet used in solid tantalum capacitors is dipped into graphite and silver to provide a good cathode connection.

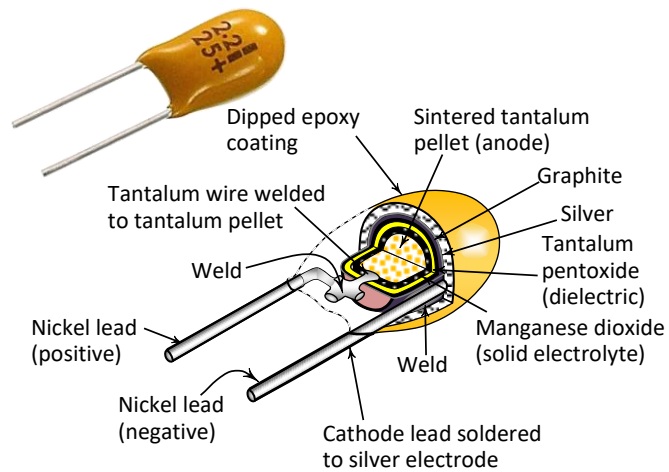


Fig.1-70: Solid tantalum capacitors (dip type)

Fig.1-71 (dipped tantalum) shows a cutaway view of a solid tantalum capacitor. Note that the coatings around the tantalum core (manganese dioxide cathode, carbon, silver paint) are very thin. This results in a higher capacitance value per volume, superior frequency characteristics compared to many other types of capacitors and excellent stability over time.

In contrast to solid tantalum capacitors, wet tantalum capacitors use a semi-liquid electrolyte paste as a second electrode. After the anode is sintered and dielectric layer is grown, it is dipped into a liquid electrolyte within an enclosure. The enclosure and electrolyte together serve as the cathode in wet tantalum capacitors.

Tantalum capacitors are made with capacitance values ranging from 1 nF to 72000  $\mu$ F and they are much smaller in size than other capacitors of the same capacitance. The voltage rating for tantalum capacitors varies from 2 V to 500 V. They have an equivalent series resistance (ESR) ten times smaller than the ESR of aluminum electrolytic capacitors which allow for larger currents to

pass through the capacitor with less heat generated. They are very reliable when handled properly and their shelf life is virtually unlimited.

Applications using tantalum capacitors take advantage of their low leakage current, high capacity and long term stability and reliability. For example, they are used in sample and hold circuits, power supply filtering on computer motherboards and cell phones due to their small size and long term stability. Tantalum capacitors are also available in military versions, which offer better tolerances and a wider operating temperature range. Medical electronics also rely on them because of their high stability. Audio amplifiers sometimes use tantalum capacitors where stability is a critical factor.

While aluminum electrolytic capacitors, which are polarized as well, might survive a short-time reverse voltage, tantalum capacitors are very sensitive to reverse polarization. If a reverse polarity voltage is applied, the dielectric oxide breaks down, sometimes forming a short circuit. This short circuit may later cause thermal runaway and destruction of the capacitor.

Nowadays, these capacitors are mostly comes in the form of surface mounted devices (SMD) [Fig.1-84]. These surface mount tantalum capacitors claim much less space on the printed circuit board and allow for greater packing densities.

#### 1.18.4 Super Capacitors

**Supercapacitors** are electrolytic capacitors with huge charge storing capacity. They are also known as **Electric Double-layer Capacitors** (EDLC) or **ultracapacitors**. Instead of using a conventional dielectric, supercapacitors use two mechanisms to store electrical energy: **double-layer capacitance** and **pseudocapacitance**. Double layer capacitance is electrostatic in origin, while pseudocapacitance is electrochemical. The capacitance of super capacitor can be as high as 12000 F. But the super capacitors have very small operating voltage range. Typical maximum voltage lies between 2.5 V to 2.7 V. Supercapacitors have such properties that they may be used as storage devices in places of batteries. Some important features of these capacitors are described below:

##### Charging time

Supercapacitors can be charged and discharged very quickly. Batteries usually take up to several hours to reach a full-charge-state, while supercapacitors can be brought to the same charge state in less than a couple of minutes.

##### Specific power

The **specific power** of a battery or supercapacitor is defined as the maximum power output per unit mass of the device. Supercapacitors have a specific power 5 to 10 times greater than that of batteries. For example, while Li-ion batteries have a specific power of 1~3 kW/kg, the specific power of a typical supercapacitor is around 10 kW/kg.

##### Life-cycle and safety

Supercapacitors can be charged and discharged millions of times and have a virtually unlimited cycle life, while batteries have only cycle life of about 500. This makes supercapacitors very useful in applications where frequent storage and release of energy is required. Supercapacitors are safer than ordinary batteries.

##### Disadvantages

The main disadvantage of super capacitor is a relatively lower specific energy. The **specific energy** is defined as the amount of energy stored per unit mass of the device. While Li-ion batteries

commonly used in cell phones have a specific energy of 100~200 Wh/kg, supercapacitors may have only 5 Wh/kg.

Another disadvantage is a linear discharge voltage. For example, a battery rated at 2.7 V, when at 50 % charge would still output a voltage close to 2.7 V, while a supercapacitor rated at 2.7 V at 50 % charge would output exactly half of its maximum charge voltage, 1.35 V.

Cost is the third major disadvantage of currently available supercapacitors. The cost per Wh of supercapacitors is more than 20 times higher than that of Li-ion batteries.

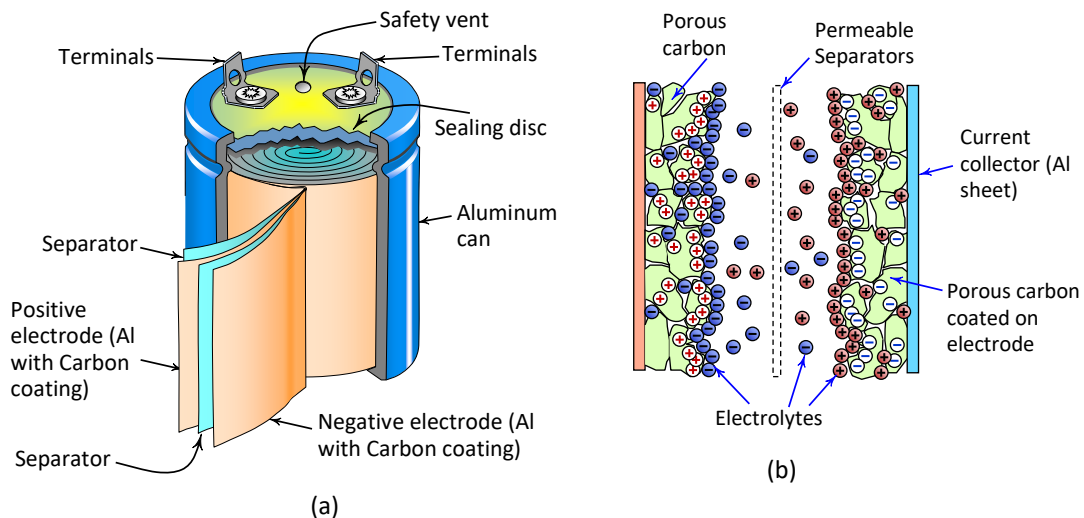


Fig.1-71: (a) Construction of a super capacitor, and (b) Charge stored on the pores of carbon granules

### Construction and properties of supercapacitors

The construction of supercapacitor is similar to the construction of electrolytic capacitors in that they consist of two foil electrodes, an electrolyte and a foil separator [shown in Fig.1-72]. The separator is sandwiched between the electrodes and the foil is rolled or folded into a shape, usually cylindrical or rectangular. This folded form is placed into a housing, impregnated with electrolyte and hermetically sealed. The electrolyte used in the construction of supercapacitors as well as the electrodes, are different from those used in ordinary electrolytic capacitors.

In order to store electrical charge, a supercapacitor uses porous materials as separators in order to store ions in those pores at an atomic level. The most commonly used material in modern supercapacitors is activated carbon. The fact that carbon is not a good insulator results in a maximum operating voltage limited to under 3 V.

One of the most exciting materials used in supercapacitor research is graphene. Graphene is a substance consisted of pure carbon, arranged

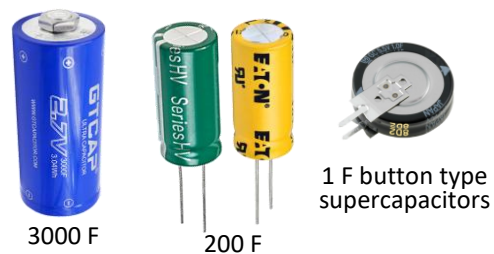


Fig.1-72: Photographs of various super capacitors

in a planar sheet only one atom thick. It is extremely porous, acting as an ion sponge. Energy densities achievable using graphene in supercapacitors are comparable to energy densities found in batteries. However, even though prototypes of graphene supercapacitors have been made as a proof of concept, graphene is difficult and expensive to produce in industrial quantities, which postpones the use of this technology.

### Applications for supercapacitors

Due to high life-cycle and fast charging/discharging time, supercapacitors are used in some systems instead of batteries. One interesting application is the storage of energy in KERS (Kinetic Energy Recovery System), or dynamic braking systems in automobile industry. Another example is low-power applications where a high capacity is not essential, but a high life cycle or quick recharging is important. Such applications are photographic flash, static memories (SRAM) which need a low power constant voltage source to retain information and so on.

#### 1.18.5 Ceramic Capacitors

A ceramic capacitor is considered to be one of the most commonly used capacitors. Ceramic is used in this capacitor as dielectric [Fig.1-74]. Ceramic capacitors are non-polar device so they can be used in any direction in the circuit. Values of ceramic capacitors range from a few picofarads to

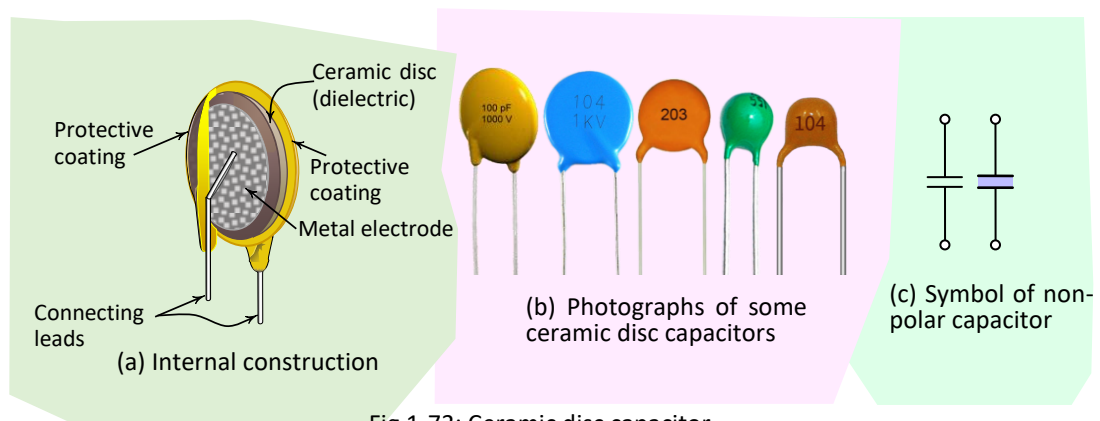


Fig.1-73: Ceramic disc capacitor

around 0.1 microfarads. Ceramic capacitors are cheap and reliable and their loss factor is particularly low. Depending on the physical connection of the capacitor on the circuit board, ceramic capacitors are classified into three groups:

- Leaded disc ceramic capacitors
- Surface mount multi-layered ceramic capacitors
- Microwave bare lead-less disc ceramic capacitors

Depending on the temperature range, temperature drift, and tolerance, ceramic capacitors are classified into the following classes:

- **Class 1 ceramic capacitors:** These capacitors are considered to be the most stable capacitors with linear characteristics.

- **Class 2 ceramic capacitors:** These capacitors perform better for volumetric efficiency but their accuracy and stability are at stake. They find applications in coupling and decoupling circuits.
- **Class 3 ceramic capacitors:** These capacitors have high volumetric efficiency with low accuracy and low dissipation factor. They are used in decoupling.

### 1.18.6 Mica Capacitor

**Mica capacitors** are made by sandwiching mica sheets coated with metal on both sides that work as the plates of the capacitor [Fig.1-75]. This assembly is then encased in epoxy in order to protect it from the environment. Sometimes, the mica dielectric is silvered to produce the electrodes or plates of the capacitor. This variety is called **silver mica capacitor**. They offer very high levels of stability, high precision, low loss, and accuracy. Though are not widely used these days, they were frequently used for high frequency RF applications. They are available in small values and limited to maximum values of approximately 1000 pF.

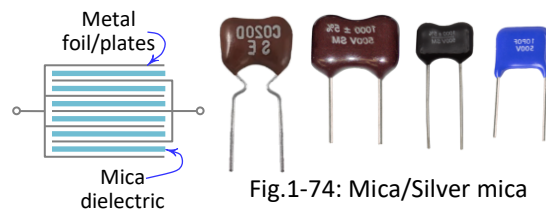


Fig.1-74: Mica/Silver mica capacitors

### 1.18.7 Film Capacitors

Film capacitors are also known as **polymer film**, **plastic film**, or **film dielectric** capacitors. The advantages of film capacitors are- they are inexpensive and come with limitless **shelf life**. The typical voltage range of these capacitors is from 50 V to 2 kV.

Film capacitors are made out of two pieces of plastic films covered with metallic electrodes, wound into a cylindrical/rectangular shaped winding, with terminals attached, and then encapsulated [Fig.1-76]. The dielectric

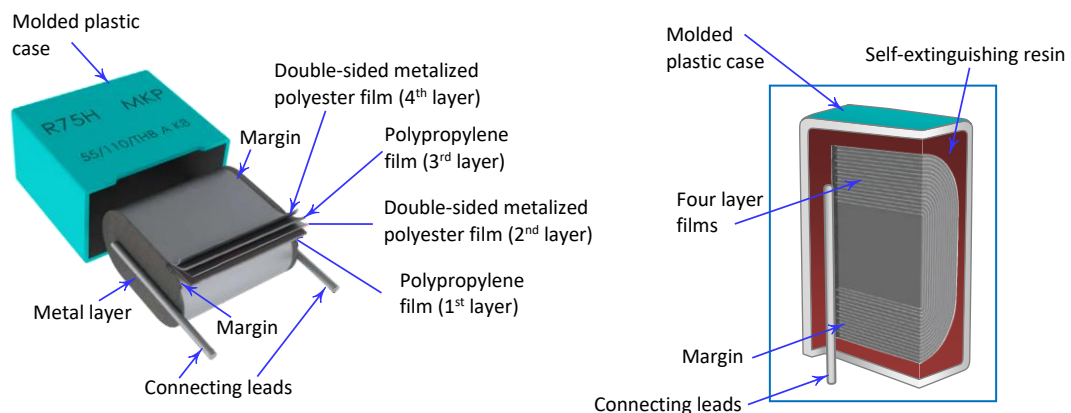


Fig.1-6: Construction of film capacitor

films, depending on the desired dielectric strength, are made in a special process to be extremely thin. The electrodes of film capacitors may be metallized aluminum or zinc applied directly to the surface of the plastic film, or separate metallic foils. These conductive layers or sheets are wound into a cylinder shaped winding, usually flattened to reduce mounting space requirements on a

printed circuit board [Fig.1-77], or layered as multiple single layers stacked together, to form a capacitor like body [Fig.1-84].

In general, film capacitors are not polarized, so the two terminals are interchangeable. There are two different types of plastic film capacitors, made with two different electrode configurations:

- Film/foil capacitors or metal foil capacitors and
- Metalized film capacitors

**Film/foil capacitors** or **metal foil capacitors** are made with two plastic films as the dielectric, with usually aluminum as the electrodes. Advantages of this construction are- easy electrical connection to the metal foil electrodes, and its ability to handle high current surges.

**Metalized film capacitors** are made of two metalized films with plastic film as the dielectric. A very thin ( $\sim 0.03 \mu\text{m}$ ) vacuum-deposited aluminum metallization is applied to one or both sides to serve as electrodes.

The structure of film capacitor results in very low ohmic loss and a very low parasitic inductance, which makes them especially suitable for applications with very high surge currents (snubbers) and for AC power applications, or for high frequency applications.

Another feature of film capacitors is the possibility of choosing different film materials for the dielectric layer to select desirable electrical characteristics, such as stability, temperature range, or ability to withstand very high voltages. So depending on the dielectric materials, there are various types of film capacitors. Some of them are discussed here.

### Polystyrene Film Capacitor

Polystyrene capacitors are a relatively cheap capacitor but offer a good tolerance. They are tubular or rectangular in shape. To make them, plate and dielectric sandwich is rolled together which adds inductance, limiting their frequency response to a few hundred kHz. They are generally available as leaded components.

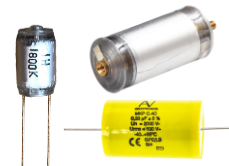


Fig.1-75: Polystyrene film

### Polyester Film Capacitor

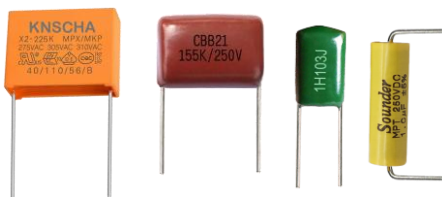


Fig.1-77: Polyester film capacitor

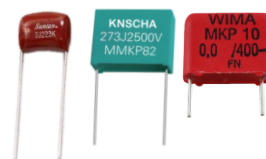


Fig.1-77: Metalized polypropylene film capacitor

Polyester film capacitors are used where cost is a consideration as they do not offer a high tolerance. Many polyester film capacitors have a tolerance of 5 % or 10 %, which is adequate for many applications. They are also available as leaded components [Fig.1-78].

### Metalized Polyester Film Capacitor

This type of capacitor is essentially a form of polyester film capacitor where the polyester films themselves are metalized to form the conducting plates of the capacitor. This process results thin electrodes, and hence, the overall capacitor can be contained within a relatively small package. This variety of capacitor is generally only available as leaded components [Fig.1-79].

### Polycarbonate Capacitor

The polycarbonate capacitors have been used in applications where reliability and performance are critical. It enables us to make high tolerance capacitors. These capacitors will hold their capacitance value over time. In addition, they have a low dissipation factor, and they remain stable over a wide temperature range, many being specified for  $-55^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ . However, the manufacture of polycarbonate dielectric has been very limited [Fig.1-80].



Fig.1-78: Polycarbonate capacitors

### Polypropylene Capacitor

The polypropylene capacitor is sometimes used when a higher tolerance type of capacitor is necessary than polyester capacitors offer. As the name implies, this capacitor uses a polypropylene film for the dielectric. One of the advantages of the capacitor is that there is very little change of capacitance with time and voltage applied. This type of capacitor is used for low frequencies to 100 kHz. They are generally available as leaded electronics components [Fig.1-81].

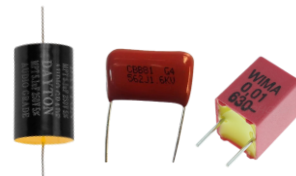


Fig.1-79: Polypropylene capacitors

### Power Film Capacitors

These capacitors are used for high power applications. Construction process and materials, used in power film capacitors, are similar to that of ordinary film capacitors but the thickness of the dielectric material is higher. Polypropylene film is more commonly used as a dielectric in power capacitor. Power film capacitors are designed for different high power applications-like induction heating, plasma generation, medical equipment, magnetizers, wireless electric vehicle chargers, auxiliary power supplies, resonant circuits, etc. These capacitors are designed to work over a wide frequency range 5 kHz to 1 MHz. Other salient features of these capacitors are- current ratings 200 to 1250  $\text{A}_{\text{rms}}$ , voltage ratings 400 V to 1200  $\text{V}_{\text{rms}}$  and power ratings 150 to 1000 kVAR [Fig.1-82].



Fig.1-80: Power film capacitors

### Paper Capacitor

**Paper capacitor** is a fixed capacitor in which paper is used as the dielectric material. The amount of electric charge stored by the paper capacitor is fixed. It consists of two metallic plates and paper is placed between these plates, which works as a dielectric material.

### Applications of Paper Capacitors

- These capacitors are used in noise filtering, coupling, and decoupling systems.

- They are also used for blocking the DC signals so that AC signals are passed through.
- Sensors such as humidity sensors, fuel level sensors.
- Paper capacitors are used in audio systems of cars as they provide extra power to the amplifiers.

### Glass Capacitors

These capacitors are made using *glass as the dielectric*. Glass capacitors are generally expensive. Although expensive, these capacitors offer extremely low loss, high RF current capability, no piezoelectric noise, and some other good features. Typically a glass capacitor will have a relatively low capacitance value ranging from a fraction of a picofarad up to 1000 picofarads.

In particular, glass capacitors are applicable for very high-performance RF applications. Glass capacitors have a low-temperature coefficient. Some forms of capacitor exhibit hysteresis in their temperature characteristic but glass capacitors follow the same characteristic with temperature rising and falling. Glass capacitors do not exhibit aging effect and retain their original value over a long period of time.



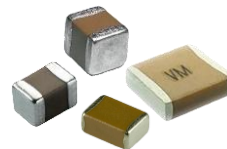
Fig.1-81: Glass capacitors

### SMD Capacitors

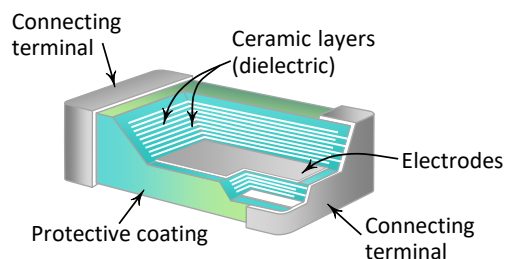
At present, the most frequently used capacitors are SMD capacitors as they are leadless, small size and simple to arrange on a printed circuit board (PCB). The performance of these capacitors is very good, particularly at radio frequency. The designing of this capacitor can be done using metallic plates separated by the dielectric material as shown in Fig.1-84. SMD capacitors are classified into different types based on the dielectric material used. Some very common types are:

- Multilayer Ceramic Capacitor
- Tantalum Capacitor
- Electrolytic Capacitor

In multilayer ceramic capacitor, ceramic is used as a dielectric material [Fig.1-84]. In ceramic capacitors, different ceramic dioxides are used such as barium strontium, barium titanate & titanium dioxide, etc. as two conducting films. Generally, the electrodes of these capacitors are covered with silver so that it gives a premium soldering property.



(a) Photographs of SMD capacitor



(b) Multi layer ceramic SMD capacitor

Fig.1-82: SMD capacitors

As discussed before, tantalum capacitors are used widely to give high values of capacitance. So they are also manufactured as SMD forms.

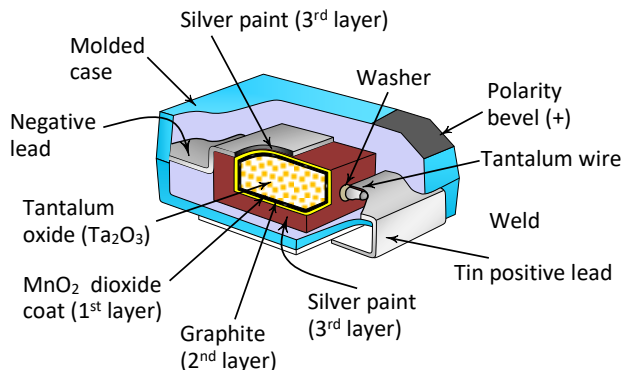


Fig.1-85 (a): Construction of solid tantalum SMD capacitor

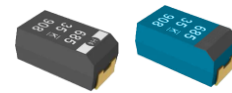


Fig.1-85 (b): Photographs of solid tantalum SMD capacitors



Fig.1-85 (c): Photographs of electrolytic SMD capacitors

Electrolytic capacitors are also used in SMD designs due to the high value of capacitance and low cost. These capacitors are frequently marked with voltage and its value. In this type, two kinds of methods are used. The first method is to include  $\mu\text{F}$  values whereas another method is to utilize code. In the first method, when the capacitor is marked with 33,6 V then the value of the capacitance is 33  $\mu\text{F}$  and the tolerance voltage level is 6 V.

### 1.18.8 Variable Capacitors

The capacitors, whose capacitance can be changed within a certain range, are called **variable capacitors**. The capacitance can be varied either mechanically or electronically. Here, only the mechanical variable capacitors are described. Basically, there are two most common types of (mechanical) variable capacitors: i) **Tuning capacitors** (also called **rotor-stator capacitors**, or gang capacitors) and ii) **Trimmers**.

#### Gang or Tuning Capacitors

The rotor-stator type capacitors are mainly used for tuning of radio receiver or TV. For this reason they are called **tuning capacitors**.

The rotor-stator type capacitor comprises of two metallic plate sets [Fig.1-88]. The moving plates are attached conjointly on the shaft and make the rotor, whereas the fixed plates are linked together and constitute the stator. Capacitance is altered by turning out the shaft in a way that the rotor plates and stator plates overlap with one another while air acts as a dielectric.

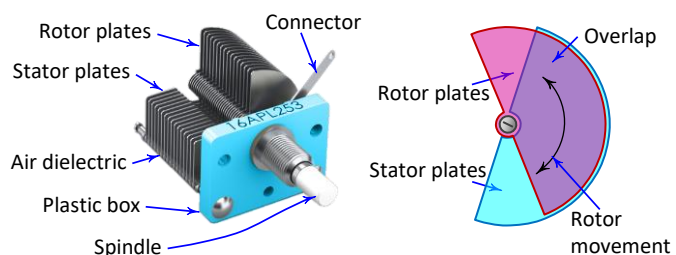


Fig.1-86: Construction of gang/tuning/rotor-stator capacitor

#### Trimmer Capacitor

A **trimmer capacitor** is a variable capacitor used for initial calibration and recalibration of equipment. Trimmer capacitors are almost always mounted directly on the PCB (Printed Circuit

Board), so the user does not have access to them, and set during manufacturing using a small screwdriver. The trimmer capacitors may provide capacitance in the range of 0.5 pF to 120 pF with tolerance voltage upto  $\approx 300$  V.

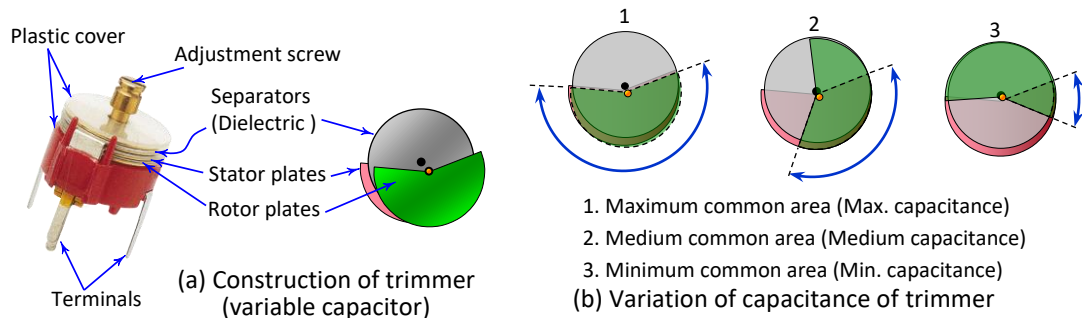


Fig.1-87: Variable (trimmer) capacitor

There are two types of trimmer capacitors: i) Air trimmer capacitor, and ii) Dielectric trimmer capacitors. The first one uses air as dielectric and provides very large tolerance voltage. On the other hand the second one uses different materials (glass, quartz, polytetrafluoro-ethylene etc.) as the dielectric and hence provides lower tolerance voltage. Both types use rotating action to change the capacitance. Trimmer capacitors can be made of semi-circular metal plates [Fig.1-89]. One is fixed, while the other can be rotated using a screwdriver. The user changes the capacitance by rotating the shaft and increasing or decreasing the overlap between the two plates. Gear mechanisms may be used to improve the precision by allowing multiple turns between the lowest and highest settings.

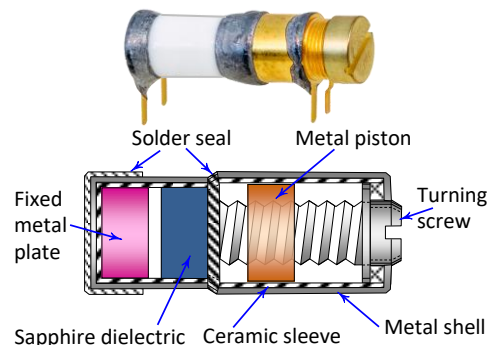


Fig.1-88: Cylinder type variable capacitor

Another way to make a trimmer capacitor is shown in Fig.1-90. Inside a non-conducting cylinder a fixed metal plate, a dielectric and a moveable plate are placed. By rotating a screw, the distance



Fig.1-89: Photographs of various variable capacitors

between the two plates can be varied which results in a change of capacitance. This construction provides very precision adjustment of capacitance. Photographs of different variable capacitors, commonly used in electronic equipment, are shown in Fig.1-91.

### Equivalent Model of a Capacitor

Fig.1-92 shows the most common electrical model of a capacitor. It is a fairly good model to understand how capacitors will behave under most common conditions. Actually real capacitors are a bit more complicated. The model given here does not show the temperature and frequency dependence of the various elements. Also, the model does not show all the parasitic elements that may be important in RF and microwave applications. Here,  $C$  is the primary capacitance, that is, the capacitance printed on the capacitor.  $R_p$  is the leakage resistance.  $ESR$  (represented as  $R_s$ ) is the equivalent series resistance. ESR of a capacitor is generally made up of three components, the **lead and end-termination resistances**, the **electrode resistance**, and **dielectric loss**. The model implies that ESR is constant with frequency. Actually, the skin effect can be a factor in high frequency applications, while the dielectric loss is frequency and temperature dependent for most materials.  $ESL$  is the equivalent series inductance (represented as  $L_s$ ), mostly of interest to RF and microwave applications.  $R_{da}$  and  $C_{da}$  combinedly represent the dielectric absorption.

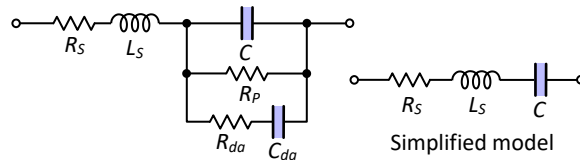


Fig.1-90: Equivalent model of a capacitor

A more realistic equivalent circuit is an infinite series of  $R_{da} - C_{da}$  networks, all in parallel. In each succeeding network, the  $R_{da}$  would get larger and the  $C_{da}$  smaller. The sum of the  $C_{da}$  capacitors is substantial, somewhere on the order of magnitude of the primary capacitance.

### Reading Capacitor Values

Different types of capacitors have different ways of representing the capacitance values and the maximum operating voltage (or tolerance voltage). Some most commonly used methods are described here.

Electrolytic capacitors, non-polar capacitors, large AC paper capacitors have capacitance, and voltage tolerance values written on its body in the form of alphanumeric characters. For example, look at the electrolytic capacitor in Fig.1-93. Here, the capacitance of the 1<sup>st</sup> capacitor is  $10\ \mu\text{F}$  and the tolerance voltage is 400 V. Similarly, the capacitance of the 2<sup>nd</sup> capacitor is  $470\ \mu\text{F}$  and the tolerance voltage is 450 V.

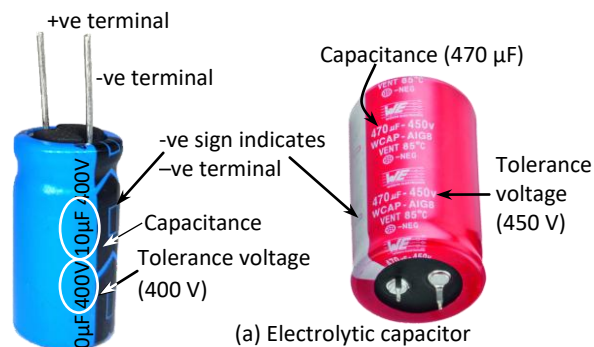


Fig.1-91: Values written on electrolytic capacitors

In non-polar capacitors (like ceramic, miller, tantalum etc.) the value of the capacitance is either written as a code number or using colors (Fig.1-94). In coding system, generally the code consists of 2 or 3 numbers and an optional tolerance letter code to identify the tolerance. In case of two-number code, only the value of the capacitor is given in picofarads. For example, a code '47' = 47 pF and a code 100 = 100 pF etc. But in case of a three-digit code, the first 2 digits give the

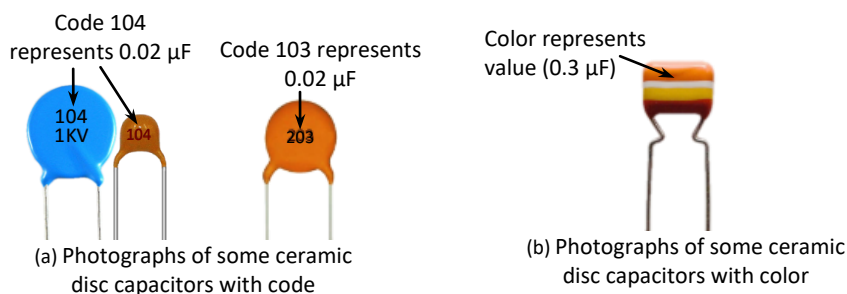


Fig.1-92: Numeric code and color code on capacitors

value and the 3<sup>rd</sup> digit is the multiplier that is used as a power of 10 and the value will be in picofarad (pF). For example, the code number written on a ceramic capacitor is 203 [Fig.1-94(a)]. Its value will be  $20 \times 10^3 \text{ pF} = 20000 \text{ pF} = 0.02 \text{ μF}$  or 20 nF. Three-digit codes are often accompanied by an additional tolerance (variation of capacitance) letter as given below.

### Example 1-10

Determine the value of capacitance of the capacitor shown in Fig.1-66. Calculate the value of reactance ( $X_C$ ) if the capacitor is used for 50 Hz AC voltage.

#### Solution:

The code number written on the capacitor is 104. So, the value of capacitance will be  $10 \times 10^4 \text{ pF} = 100000 \text{ pF} = 0.1 \text{ μF}$  or 100 nF. The reactance can be calculated using Equ.(1-2).

$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi(50 \text{ Hz})(0.1 \times 10^{-6})} \approx 32 \text{ k}\Omega \text{ [Ans.]}$$

**Comments:** In Bangladesh, the frequency of AC supply voltage is 50 Hz. Hence, here we have considered  $f = 50 \text{ Hz}$ .



Fig.1-93: Ceramic capacitor for Example 1-2

### Capacitor Tolerance Letter Codes

| Letter    | Capacitance Range                 | B   | C    | D   | F | G | J | K  | M  | Z      |
|-----------|-----------------------------------|-----|------|-----|---|---|---|----|----|--------|
| Tolerance | $C < 10 \text{ pF} \pm \text{pF}$ | 0.1 | 0.25 | 0.5 | 1 | 2 |   |    |    |        |
|           | $C > 10 \text{ pF} \pm \%$        |     |      | 0.5 | 1 | 2 | 5 | 10 | 20 | +80-20 |

When the value of the capacitance, written on the body of the capacitor, is of a decimal value problems arise with the marking of the “Decimal Point” as it could easily not be noticed resulting in misreading of the actual capacitance value. Therefore, letters such as *p* (pico) or *n* (nano) are used in place of the decimal point to identify its position and the weight of the number. For example, a capacitor can be labeled as,  $n47 = 0.47 \text{ nF}$ ,  $4n7 = 4.7 \text{ nF}$  or  $47n = 47 \text{ nF}$  and so on. Also, sometimes capacitors are marked with the capital letter K to signify a value of one thousand picofarads, so for example, a capacitor with the markings of 100 K would be  $100 \times 1000 \text{ pF}$  or  $100 \text{ nF}$ .

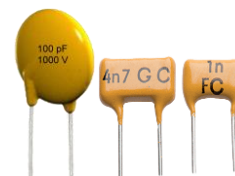


Fig.1-94: Values written on non-polar capacitors

To reduce the confusion in reading letters, numbers, and decimal points, an *International Color Coding Scheme* is used to identify capacitor values and tolerances. It consists of color-bands (in spectral order) known commonly as the *Capacitor Color Code* system. The color coding system for capacitors is illustrated below:

|        | A<br>Digit  | B<br>Digit | D<br>Digit | T>10pf   | T<10pf   | T <sub>c</sub> | V    |
|--------|-------------|------------|------------|----------|----------|----------------|------|
| Black  | 0           | 0          | 1          | ± 20%    | ± 2.0pf  | -33x*          |      |
| Brown  | 1           | 1          | 10         | ± 1%     | ± 0.1 pf | -75x*          | 250V |
| Red    | 2           | 2          | 100        | ± 2%     | ± 0.25pf | -150x*         |      |
| Orange | 3           | 3          | 1k         | ± 3%     |          | -220x*         | 400V |
| Yellow | 4           | 4          | 10k        | ± 4%     |          | -330x*         | 100V |
| Green  | 5           | 5          | 100k       | ± 5%     | ± 0.5pf  | -470x*         | 630V |
| Blue   | 6           | 6          | 1M         |          |          | -750x*         |      |
| Violet | 7           | 7          |            |          |          |                |      |
| Grey   | 8           | 8          | 0.01       | +80%-20% |          |                |      |
| White  | 9           | 9          | 0.1        | ± 10%    | ± 1.0pf  |                |      |
| Gold   | Values (pF) |            |            | 0.1      | ± 5%     |                |      |
| Silver |             |            |            | 0.01     | ± 10%    |                |      |

Multiplier      Tolerance

Temp. Coefficient  
\* =  $10^{-6}$

Max. Voltage

Fig.1-95: Color coding system for capacitors

The capacitor color code system was used for many years on non-polarized capacitors. Nowadays, small capacitors such as film or disk types the colors have been replaced by a letter or number coded system. There are some capacitor types, whose tolerance voltage is represented using colors. The maximum operating voltage represented by the color is different for different types of capacitors [Fig.1-97].

| Types of Capacitor for Tolerance Volta |                                  |
|----------------------------------------|----------------------------------|
| Type J                                 | Dipped Tantalum Capacitors       |
| Type K                                 | Mica Capacitors                  |
| Type L                                 | Polyester/Polystyrene Capacitors |
| Type M                                 | Electrolytic 4-Band Capacitors   |
| Type N                                 | Electrolytic 3-Band Capacitors   |

(a)

Fig.1-96: Color represents tolerance voltage for different types of capacitors

| Color Band | Tolerance Voltage |        |        |        |        |
|------------|-------------------|--------|--------|--------|--------|
|            | Type J            | Type K | Type L | Type M | Type N |
| Black      | 4                 | 100    |        | 10     | 10     |
| Brown      | 6                 | 200    | 100    | 1.6    |        |
| Red        | 10                | 300    | 250    | 4      | 35     |
| Orange     | 15                | 400    |        | 40     |        |
| Yellow     | 20                | 500    | 400    | 6.3    | 6      |
| Green      | 25                | 600    |        | 16     | 15     |
| Blue       | 35                | 700    | 630    |        | 20     |
| Violet     | 50                | 800    |        |        |        |
| Gray       |                   | 900    |        | 25     | 25     |
| White      | 3                 | 1000   |        | 2.5    | 3      |
| Golden     |                   | 2000   |        |        |        |
| Silver     |                   |        |        |        |        |

(b)

### Example 1-11

For the polyester film capacitor of Fig.1-98, determine the value of the capacitance.

#### Solution:

The capacitance of the capacitor is written by a five-band color code. The first two colors indicate the value in pF. Here, the first two colors are: Brown = 1, and Black = 0. Thus, the value represented by these colors is 10 pF. The third color is yellow that is the multiplier. From Fig.1-96, we find that the multiplier value for yellow color is 10 k. The fourth color is red that is the tolerance. From Fig.1-96, we find that the tolerance value for red color is 2%. The last color is orange that indicates the voltage rating of the capacitor is 250 V. Putting all together, the value of capacitance of this capacitor will be:

$$C = 10 \text{ pF} \times 10 \text{ k} \pm 2\%, 250 \text{ V}$$

$$C = 100 \text{ nF} \pm 2\%, 250 \text{ V [Ans.]}$$

**Comments:** For three to five-band color code,  $T_C$  will not be used.  $T_C$  will be used only for six-band color codes.

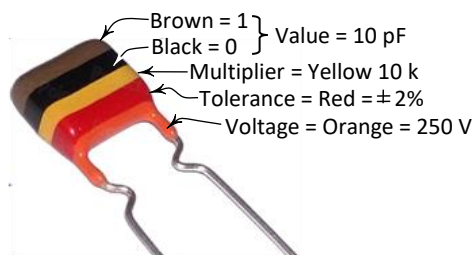


Fig.1-97: Polyester film capacitor with color code

### 1.19 Series and Parallel Connection of Capacitors

Like resistors, we can combine capacitors in series or parallel. But the value of the equivalent capacitance will be just opposite to resistance. That is, if capacitors are connected in series, the equivalent capacitance will decrease. This happens due to the division of the applied voltage across the capacitors. For example, if we connect three capacitors in series and a current of 1 mA flows

through that series path, then all the capacitors will accumulate 1 mC of charge every second. If the capacitors are of equal values, the voltage of individual capacitor will increase at the same rate. Thus, the voltage across the combination will increase 3 times faster than it would have been increased for a single capacitor. We know for a certain amount of charge, the voltage will increase faster in a lower value capacitor as  $V = Q/C$ . Therefore, the series combination will be equivalent to a single capacitor with a capacitance of one-third the capacitance of a single capacitor.

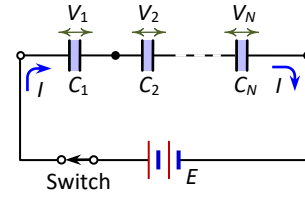


Fig.1-98: Series connected capacitors:  
Same current different voltage

In fact, by placing the capacitors in series, we've effectively spaced the plates farther apart because the spacing between the plates of the two capacitors adds together. So, the reciprocal methods are valid for adding capacitors in series. Mathematically,

$$\frac{1}{C_{eq}} = \left( \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N} \right) \quad (1-33)$$

Though the capacitance decreases, when we put capacitors in series the tolerance voltages add up, just like batteries. However, in series combination of capacitors of different values, the voltage will not simply add up. So, we have to pay extra care in determining the tolerance voltage.

### Example 1-12

Suppose three capacitors have the ratings as  $C_1$ : 10  $\mu\text{F}$ , 16 V;  $C_2$ : 33  $\mu\text{F}$ , 20 V;  $C_3$ : 47  $\mu\text{F}$ , 35 V. Determine the equivalent capacitance and tolerance voltage when the capacitors are connected in series.

### Solution:

The equivalent capacitance of the series connected capacitors can be calculated using Equ.(1-33).

$$\frac{1}{C_{eq}} = \frac{1}{10 \mu\text{F}} + \frac{1}{33 \mu\text{F}} + \frac{1}{47 \mu\text{F}} = 10 \times 10^4 + 3.03 \times 10^4 + 2.13 \times 10^4 = 15.16 \times 10^4$$

$$\therefore C_{eq} = \frac{1}{15.16 \times 10^4} \approx 6.6 \mu\text{F}$$

When capacitors are connected in series, the charging current ( $I_C$ ) flowing through the capacitors will be the same. Therefore, each capacitor will store the same amount of charge,  $Q$ , but the voltage across the capacitor will be inversely proportional to its capacitance. Here, the voltage of the smallest capacitor ( $C_1$ ) will be the maximum. But the tolerance voltage of this capacitor is 16 V. So, the maximum amount of charge  $C_1$  can store will be,

$$Q = C_1 V_{C1} = 10 \mu\text{F} \times 16 \text{ V} = 160 \mu\text{C}$$

For the series connection, all other capacitors will store the same amount of charge (160  $\mu\text{C}$ ). Therefore, the voltage across them will be

$$V_{C2} = \frac{Q}{C_2} = \frac{160 \mu\text{C}}{33 \mu\text{F}} \approx 4.45 \text{ V}$$

and,

$$V_{C3} = \frac{Q}{C_3} = \frac{160 \mu\text{C}}{47 \mu\text{F}} \approx 3.4 \text{ V}$$

Therefore, the total voltage across the three capacitors will be,

$$V_T = V_{C1} + V_{C2} + V_{C3} = 16 \text{ V} + 4.45 \text{ V} + 3.4 \text{ V} = 23.85 \text{ V}$$

This is the maximum voltage that can be applied across the combination. So the tolerance voltage of the series combination will be 23.85 V [Ans.]

**Comments:** Tolerance voltage increases when capacitors are connected in series. If the capacitances are same, the equivalent tolerance voltage will be just sum of the individual tolerances, but not for different capacitance values.

Adding capacitors in parallel is like adding resistors in series. Putting them in parallel effectively increases the size of the plate-area without increasing the distance between them. The equivalent capacitance of parallel connected capacitors will be,

$$C_{eq} = C_1 + C_2 \dots + C_N \quad (1-34)$$

Unlike the series combination, the tolerance voltage will not increase, rather it will remain same. But if capacitors of different voltage ratings (tolerance) are connected in parallel, the equivalent voltage rating will be the smallest voltage rating of the combination [see Example 1-13].

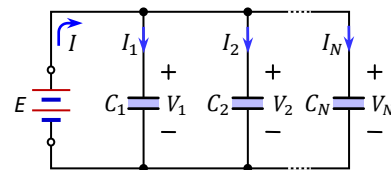


Fig.1-99: Parallel connected capacitors: Same voltage but different current

### Example 1-13

Repeat Example for parallel combination of the capacitors.

#### Solution:

For parallel connected capacitors, we know,

$$C_{eq} = C_1 + C_2 \dots + C_N$$

Here,  $C_1 = 10 \mu\text{F}$ , 16 V;  $C_2 = 33 \mu\text{F}$ ; 20 V,  $C_3 = 47 \mu\text{F}$ , 35 V.

$$\therefore C_{eq} = 10 \mu\text{F} + 33 \mu\text{F} + 47 \mu\text{F} = 90 \mu\text{F}$$

When capacitors are connected in parallel, the charging current ( $I_C$ ) will be different but the voltage across them will be same. If we try to apply 35 V (the maximum tolerance of the combination),  $C_1$  will explode, as it can withstand only 16 V. Thus the maximum voltage capacity of the combination will 16 V, i.e., the minimum tolerance of the combination.

**Comments:** In parallel combination of capacitors, the tolerance voltage will be the minimum tolerance of the individual capacitors!

## 1.20 Use of Multimeter

There are various types of instruments used to measure voltages and currents in circuits. Among them the oscilloscope is the most versatile one. An oscilloscope displays the levels of voltage versus time at one or more points in a circuit. Logic probes and logic analyzers are special purpose instruments used for trouble shooting digital circuits. As these instruments are costly, low-cost instruments like multimeters are commonly used in laboratories.

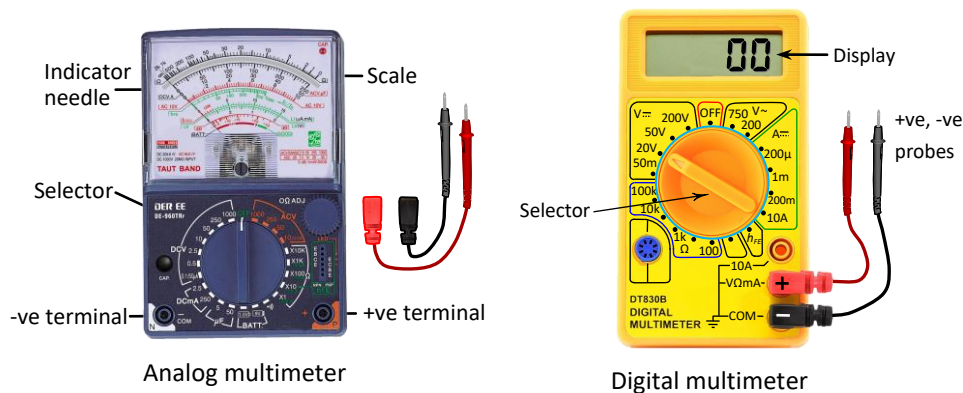


Fig.1-100: Analog and digital multimeters

A **multimeter** is an electronic (or electrical) instrument that is used to measure voltage, current, resistance etc. Actually a multimeter is a combination of **voltmeter**, **ammeter**, and **ohmmeter** and so on. Being a versatile instrument, it is most commonly used for measuring different components (resistors, inductors, capacitors, diodes, transistors etc.) and parameters (voltage, current, power etc.) of electronic (and electrical) circuits. It is also used for continuity testing purpose in servicing electrical and electronic circuits. There are mainly two types of multimeters- **Analog multimeter**, and **Digital multimeter**.

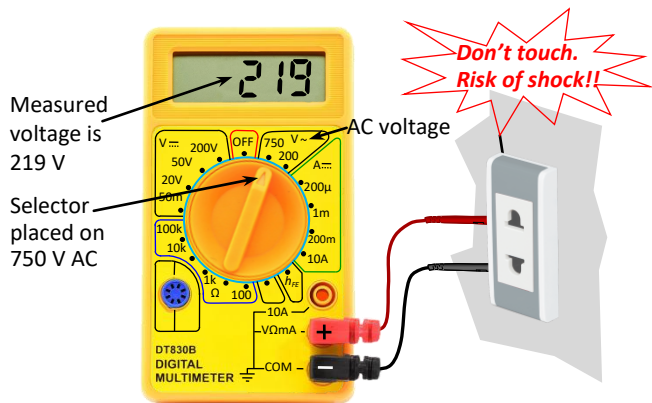


Fig.1-101: Digital multimeter measuring line voltage

The photographs of an analog and a digital multimeter are shown in Fig.1-101.

The main difference between them is: the analog multimeter has an indicating needle that moves on a scale to indicate the value of the measured parameter. On the other hand, the digital multimeter has a display unit, that directly shows the value (numerically) of the measured parameter. The analog multimeter consist of a **d'Arsonval galvanometer** and other components. A galvanometer can measure only milliamperes but can be converted into a voltage meter and Ohm meter. As the analog meter can mainly measure voltage, Ohm, and milliamperes they are widely

known as **VOM**. On the other hand digital multimeters consist of digital circuits and LCD/LED type display. In addition to measure current, voltage and, resistance, digital meters can measure diode's barrier voltage, transistor's current gain, frequency and so on. Both multimeters have a selector that has to be set to appropriate position (type of parameter and values written on the dial) before using. We have to be careful to use these meters. The selector switch has to be set to the lowest range of the meter but sufficient enough to measure the values. The lower ranges give better accuracy of the measurement. For example, if we want to measure line voltage (220 V AC) using a digital multimeter, the selector has to be set in the voltage range 750 V AC not in 200 V AC. But if we want to measure the line voltage in U.S.A. [110 V AC] we have to set the selector in 200 V AC. Fig.1-102 shows how to measure the supply voltage in our house using a digital multimeter. As there is risk of shock so we must not touch the metallic portion of the meter probes.

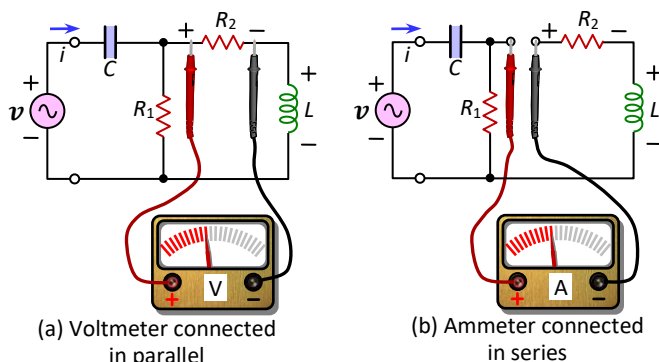


Fig.1-102: Meter connection methods: (a) Voltage measurement, and (b) Current measurement.

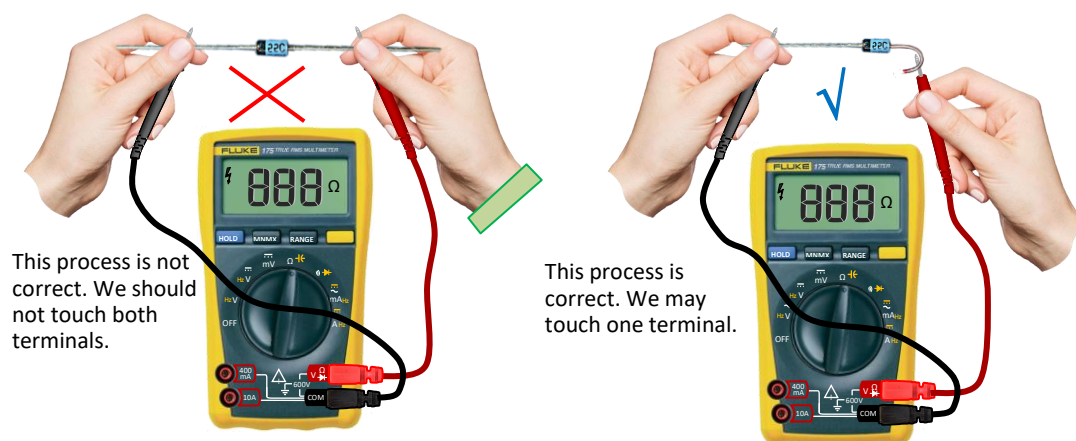


Fig.1-103: Measuring/testing method of components using multimeter

Measuring process of different parameters and elements in circuits are shown in Fig.1-103. To measure voltage the voltmeter has to be connected across two points or across the components [Fig.1-103(a)]. Whereas, to measure current the meter has to be connected in series, that is, the current measuring point has to be opened and then connect the ammeter [Fig.1-103(b)]. Moreover, to measure resistance, capacitance, inductance etc. the meter has to be connected in parallel to those components [as shown in Fig.1-104]. As human body has a finite resistance, we should not hold the components' both terminals by hands (holding one terminal is OK) while

measuring these components. If we hold both terminals, our body resistance will work in parallel to the components, so the reading may not be accurate. The situation is illustrated in Fig.1-104.

## 1.21 Different Types of Signals

In the field of electronics and telecommunications, **signal** is defined as any time-varying voltage, current, or electromagnetic wave that carries information. In other words, a signal is a function that conveys information about a phenomenon. In that context, any quantity that varies in magnitude over space or time can be used as a signal to share messages between observers. The most common examples of signals are audio, video, speech, image, sonar, and radar signals. A signal may also be defined as any observable change in a quantity over space or time (a time series), even if it does not carry information. Signals may have various shapes in waveforms. Some most common signals are described here.

### 1.21.1 Sinusoidal signals

Sinusoidal signals are the most popular signals as most of the natural periodic parameters (like sound, light, vibrations, swings etc.) follow this sinusoidal shape. If someone says something like "take a 10.0 V signal at 1 kHz," they mean a sine wave. The AC voltage available in the supply line is sinusoidal (or sine wave). Mathematically, this signal can be represented as,

$$V = A \sin(\omega t) = A \sin(2\pi f t) \quad (1-35)$$

where,  $A$  is called the amplitude,  $\omega$  is the angular frequency in radian/s, and  $f$  is the frequency in Hertz (cycles per second). A sine wave looks like the wave shown in Fig.1-105. Sometimes it is important to know the value of the signal at some arbitrary time, say,  $t = 0$ , in which case we may see a phase  $\phi$  in the expression:

$$V = A \sin(\omega t + \phi) = A \sin(2\pi f t + \phi) \quad (1-36)$$

The great merit of sine waves is the fact that they are the solutions to certain linear differential equations used to describe many phenomena in nature as well as the properties of linear circuits. A linear circuit has the property that its output when driven by the sum of two input signals, equals the sum of its individual outputs when driven by each input signal in turn. That is, if  $\vartheta(a)$  represents the output when driven by signal  $a$ , then a circuit is linear if  $\vartheta(a + b) = \vartheta(a) + \vartheta(b)$ . A linear circuit driven by a sine wave always responds with a sine wave, although in general, the phase and amplitude are changed. No other periodic signal can make this output. It is standard practice, in fact, to describe the behavior of a circuit by its frequency response, by which we mean the way the circuit alters the amplitude of an applied sine wave as a function of frequency. A stereo amplifier, for instance, should be characterized by a "flat" frequency response over the range 20 Hz to 20 kHz, at least. The sine wave frequencies we usually deal with range from a few hertz to a few tens of megahertz. Lower frequencies, down to 0.0001 Hz or lower, can be

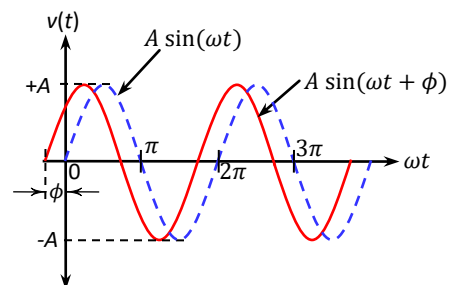


Fig.1-104: Sinusoidal signals

generated with carefully built circuits, if needed. Higher frequencies, up to say 2000 MHz (2 GHz) and above, can be generated by using special transmission-line techniques.

### Signal Amplitudes and Decibels

In addition to its amplitude, there are several other ways to characterize the magnitude of a sine wave or any other signal. We sometimes see it specified by peak-to-peak value (pp value), which is just twice the amplitude. The other method is to give the root-mean-square (RMS) value, which is calculated as  $V_{rms} = A/\sqrt{2}$  for sine wave only, that is,  $A/V_{rms} = \sqrt{2}$ . The ratio of amplitude to RMS value will be different for different waveforms. The nominal voltage across the terminals of a wall socket (in Bangladesh) is 220 V (RMS), 50 Hz. Thus, the amplitude is  $A = 220 \times \sqrt{2} \text{ V} \approx 311 \text{ V}$ .

#### Example 1-14

An AC voltage is represented as  $v_s = 24 \sin(100t + 20^\circ)$ .

Determine the amplitude, peak-to-peak value, frequency, and phase angle. Draw the graph for this AC voltage.

#### Solution:

The general equation of an AC voltage is  $v(t) = V_m \sin(\omega t + \phi)$

where,  $v(t)$  is the instantaneous value of the AC voltage,  $V_m$  is the peak value or amplitude,  $\omega$  is the angular frequency. Comparing this equation with the given equation, we get,

The amplitude  $V_m = 24 \text{ V}$  [Ans.], the phase angle  $\phi = 20^\circ$  [Ans.], and angular frequency  $\omega = 100$  [Ans.]. But we know,  $\omega = 2\pi f$

$$\therefore f = \frac{\omega}{2\pi} = \frac{100}{2\pi} \approx 16 \text{ Hz [Ans.]}$$

**Comments:** At  $t = 0$ , the instantaneous value of this AC voltage will be  $v_{s(t=0)} = 24 \sin(20^\circ) \approx 8.21$ . This voltage is due to the leading phase angle  $\phi$ .

### 1.21.2 Other signals

#### Ramp

The **ramp** is a signal that looks like the one shown in Fig.1-106. It is simply a voltage, rising (or falling) at a constant rate. But, it can't go on forever. It is sometimes approximated by a **finite ramp** [Fig.1-106(b)] or by a **periodic ramp** also known as a saw-tooth wave [Fig.1-106(c) and (d)].

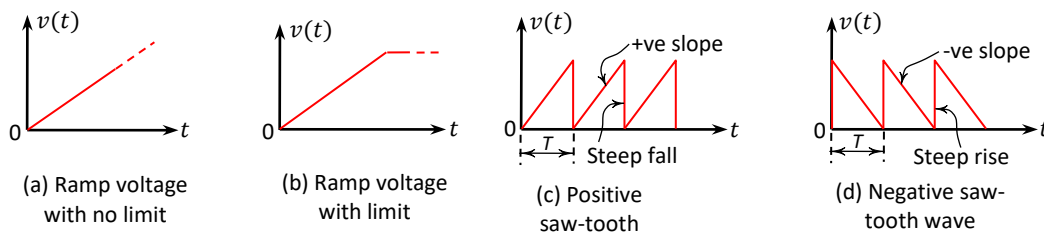


Fig.1-105: Different types of ramps

### Triangular wave

The **triangular wave** is a periodic or symmetrical ramps [Fig.1-107(a)]. A triangular wave or triangle wave is a non-sinusoidal waveform named for its triangular shape. It is a periodic, piecewise linear, continuous real function.

### Square wave

A **square wave** is a signal that has only two states- HIGH and LOW. The variation of the states is shown in Fig.1-107(b). Like the sine wave, it is characterized by amplitude and frequency (and perhaps phase). A linear circuit driven by a square wave rarely produces a square wave as the output. For a square wave, the peak amplitude and the RMS amplitude are the same.

The edges of a square wave are not perfectly square. In typical electronic circuits the rise time,  $t_r$ , ranges from a few nanoseconds to a few microseconds. The rise time is conventionally defined as the time required for the signal to go from 10 % to 90 % of its final value, as shown in Fig.1-107(c).

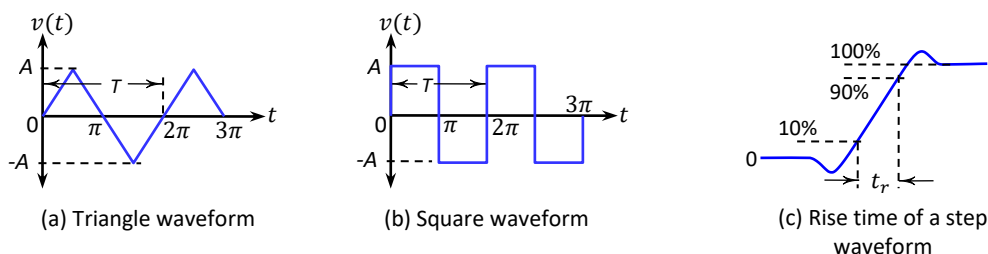


Fig.1-106: Triangular and square waveforms

### Pulses

A **pulse** is a signal that is single square wave as shown in Fig.1-108(a) and (b). It is defined by amplitude and **pulse width**. We can generate a train of periodic (equally spaced) pulses, in which case we can talk about the frequency, or pulse repetition rate, and the **duty cycle**. Duty cycle is defined as the ratio of pulse width to repetition period (duty cycle ranges from zero to 100 %). Pulses can have positive or negative polarity, in addition, they can be positive-going or negative-going. For instance, the pulse in Fig.1-108(b) is a negative-going pulse of positive polarity.

### Steps and Spikes

Steps and spikes are signals that are talked about a lot but are not so often used. They provide a nice way of describing what happens in a circuit. If we could draw them, they would look something like the example in Fig.1-108(c) and (d). The step function is part of a square wave; the spike is simply a jump of vanishingly duration.

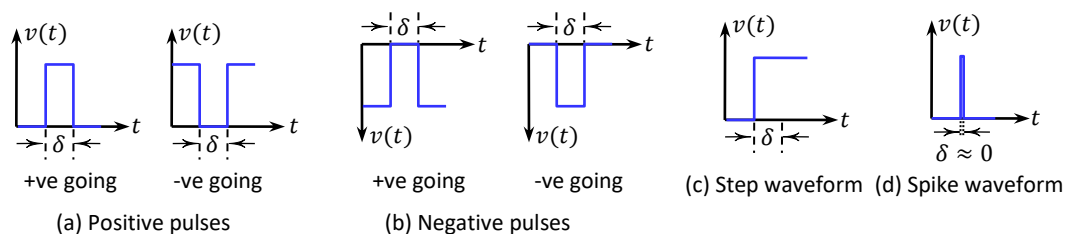


Fig.1-107: Different pulses

## 1.22 Breadboard

**Breadboards** are one of the mostly used devices in electronics laboratory. Breadboards are used for making temporary circuits and prototyping, and they require absolutely no soldering. Another common use of breadboards is testing out new parts, such as Integrated Circuits (ICs).

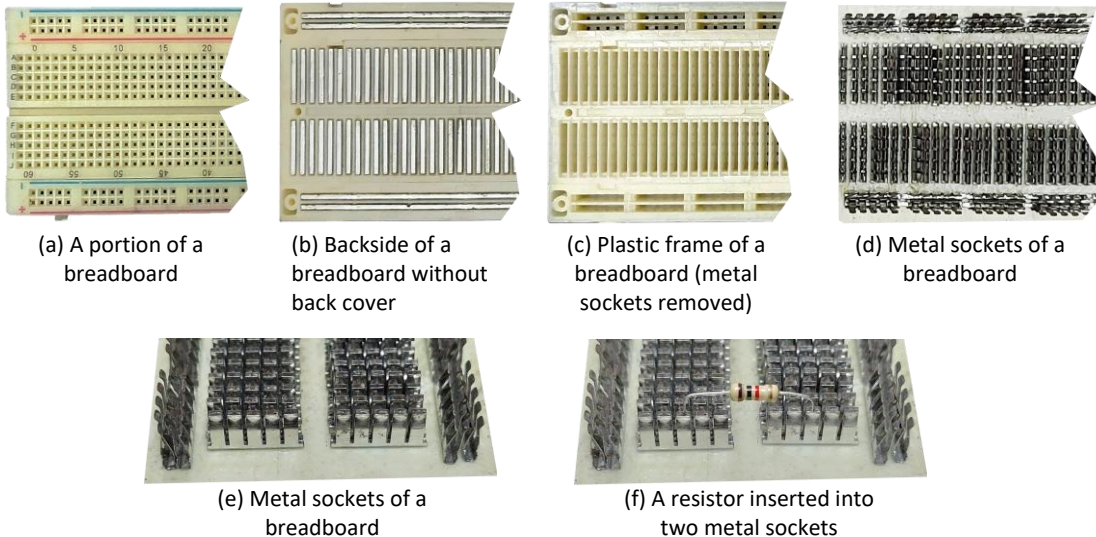


Fig.1-108: Solderless breadboards

A Breadboard is simply a board for prototyping or building circuits on it. It allows us to place components and connections on the board to make circuits without soldering. The holes in the breadboard take care of the connections by physically holding the parts or wires where we put them and electrically connecting them inside the board. The ease and speed of use are great for learning and quick prototyping of simple circuits. More complex circuits and high frequency circuits are less suited to breadboarding. Breadboard circuits are also not ideal for long term use like circuits built on perfboard (or protoboard) or PCB (printed circuit board). To understand the function of breadboard, the internal construction and component holding process is illustrated in Fig.1-109.

## 1.23 Noise in Electronic Circuits

The unwanted signals in any circuit which are not derived or related to the input signals are called **noises**. Most of the noises are random in nature and are combined with the circuit signal to affect the overall performance of a system [Fig.1-110]. However, there are some periodic noises too. Actually, noise is unwanted signal whether periodic or random in nature. Random noise cannot be predicted and represented by any equation.

In the context of generating source, noise may be divided into two types:

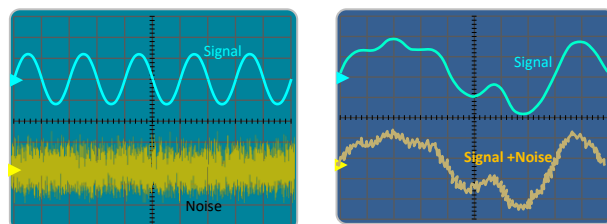


Fig.1-109: Waveforms of signal and noise

- i) External Noise and
- ii) Internal Noise

### 1.23.1 External Noise

The **external noise** is induced with the signal when it travels through the medium or free space. The main types of external noises are: **man-made noise**, **atmospheric noise** and **space noise**.

#### Man-made Noise

It is often produced by spark-producing mechanisms such as engine ignition systems, fluorescent lights, and commutators in electric motors. This noise is actually radiated from its generating sources through the atmosphere. Another common source of man-made noise is contained in the power lines that supply the energy for most electronic systems. AC power lines contain surges of voltage caused by the switching ON and OFF of electrical motors, harmonics, produced by the non-linear electronics load connected to the power line. All of these produce noises that appear to our circuit through the DC power supply. Man-made noise occurs randomly at frequencies up to around 500 MHz. This is also called **industrial noise**.

#### Atmospheric Noise

Atmospheric noise is caused by naturally occurring disturbances within the earth's atmosphere. Lightning discharges is the most prominent contributor of atmospheric noise. The frequency content is spread over the entire radio spectrum, but its intensity is inversely related to frequency. It is therefore most troublesome at the lower frequencies. This noise is insignificant for frequencies above about 20 MHz.

#### Space Noise

This noise arrives from outer space, i.e., above the atmosphere. So, this noise is also called **extraterrestrial noise**. The noise originating from the sun is termed **solar noise**. Other stars also generate the space noise which is termed as **cosmic noise**. Since they are much farther away than the sun, their effects are very small. Space noise occurs at frequencies from about 8 MHz up to 1.5 GHz.

### 1.23.2 Internal Noise

On the other hand, the internal noises are generated in the circuits. There produce various types of noises in circuits. The following are some most common types of internal noises:

- Thermal noise
- Shot noise
- Flicker noise
- Transit-time noise

#### Thermal noise

Due to the temperature electrons and ions of any resistor or conductor vibrate randomly about their normal positions and produce continuous collisions. This is the source of resistance in a conductor or resistor. This random of movements of electrons give rise to a voltage which is called **thermal noise**. Thus, noise generated in any resistance due to random motion of electrons is called thermal noise or **white noise** or **Johnson noise**. The power of thermal

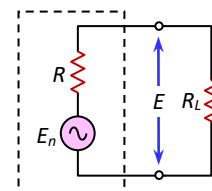


Fig.1-110: Noise source equivalent circuit

noise is proportional to temperature. Noise power is also proportional to the bandwidth over which the noise is considered. Mathematically,

$$P_n \propto TB = KTB \quad (1-37)$$

where,  $P_n$  is the noise power,  $K$  is Boltzmann constant,  $T$  is absolute temperature, and  $B$  is bandwidth.

Fig.1-111 shows an equivalent circuit of a resistor with noise voltage. Here,  $R$  is a noise free resistor,  $E_n$  is RMS value of noise voltage and  $R_L$  is the load resistance. Therefore, the noise power at the load will be  $P_n = E^2/R_L$ . For maximum power transfer,  $R_L$  will be equal to  $R$ . So,  $E = E_n/2$ . Thus,

$$P_n = E_n^2/4R \quad (1-38)$$

$$\text{or,} \quad E_n^2 = 4RP_n \text{ or, } E_n = \sqrt{4RP_n} \quad (1-39)$$

$$\text{Using Equ.(1-37),} \quad E_n = \sqrt{4RKT B} \quad (1-40)$$

Equ.(1-40) gives the RMS value of the noise voltage of a resistor with resistance  $R$ .

#### Example 1-15

An amplifier has a bandwidth of 5 MHz with 4.7 kΩ of input resistance. Determine the RMS value of the thermal noise at room temperature,  $T = 25^\circ\text{C}$ .

#### Solution:

Here given: Bandwidth  $B = 5$  MHz, Resistance  $R = 4.7$  kΩ, and temperature  $T = 25^\circ\text{C} = 298$  K. Thus, using Equ.(1-40),

$$E_n = \sqrt{4(4.7 \times 10^3)(1.38 \times 10^{-23})(298)(5 \times 10^6)} \\ \approx 19.66 \mu\text{V}$$

**Comments:** The effect of this noise will remain in the entire bandwidth (5 MHz). This represents the limit for the smallest voltage we can resolve across this resistor in this bandwidth.

#### Shot Noise

The most common type of noise is referred to as **shot noise**. Shot noise, in electronic circuits, consists of random fluctuations of the electric current which originate due to the fact that current actually consists of a flow of discrete charges (electrons). Because the electron has such a tiny charge, however, shot noise is of relative insignificance in many (but not all) cases of electrical conduction. For instance, 1 A of current consists of about  $6.24 \times 10^{18}$  electrons flowing per second equivalently  $6.24 \times 10^9$  electrons per nano-second. But, if we measure the number of electrons flowing per nanosecond it will fluctuate due to its discrete and random nature [as shown in Fig.1-112]. Thus, with very small currents, particularly when measuring on short time scales (thus wider bandwidths), shot noise can be significant.

Shot noise is particularly noticeable in semiconductor devices, such as tunnel junctions, Schottky barrier diodes and PN junctions. Unlike thermal noise, this noise is dependent on the current and independent of operating temperature. It is not white noise in the sense that it tends to diminish at higher frequencies. At high frequencies and low temperatures shot noise may become the dominant source of noise.

Shot noise normally occurs when there is a potential barrier (voltage differential). PN junction diode is an example that has potential barrier. When the electrons and holes cross the barrier, shot noise is produced. For example, a diode, a transistor, and vacuum tube will all produce shot noise. On the other hand, a resistor normally does not produce shot noise, since, there is no potential barrier built within a resistor. Current flowing through a resistor will not exhibit any fluctuations. However, current flowing through a diode produces small fluctuations. This is due to electrons (in turn, the charge) arriving in quanta, one electron at a time. The current flow is not continuous, but limited by the quantum of the electron charges.

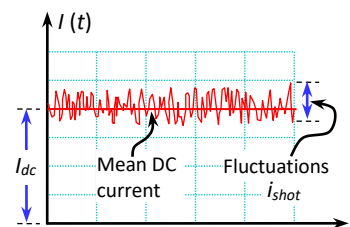


Fig.1-111: Shot noise in DC current

### Transit-time Noise

**Transit-time** is defined as the duration of time that it takes for a current carrier such as a hole or electron to move from input to the output. Transit-time noise is similar to shot noise in that it affects systems more as they get smaller due to the quantized nature of charge. Transit-time noise results when the time period of a signal is comparable as the time an electron takes to travel from input to output.

In PN junction diodes, if the time taken by the electrons to travel from cathode to anode, or in transistor from emitter to collector, becomes comparable to period of the signal being processed, transit-time effect takes place. So the shape of the signal is changed as shown in Fig.1-113. Transit-time noise is directly proportional to the frequency of the signal.

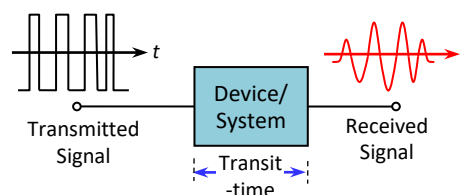


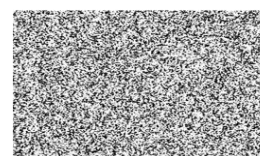
Fig.1-112: Effect of transit-time noise

### Flicker Noise

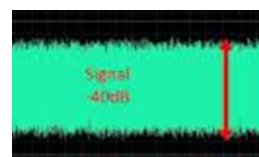
Flicker noise is found in all active electronic components as well as some of the passive devices. Like shot noise, flicker noise is associated with a DC current flow. A characteristic property of this noise is that its magnitude decreases with frequency, and therefore, is sometimes referred to as  $1/f$  noise.

#### 1.23.3 White Noise

**White noise** is a signal that has its energy evenly distributed over the entire frequency spectrum, within the frequency range of interest. White noise, therefore, is a combination of various noise frequencies. For example, in acoustics, white



(a)



(b)

Fig.1-113: White noise in (a) Image signal, and (b) Sound signal

noise has the same amplitude, or intensity, throughout the entire audible frequency range 20 Hz to 20 kHz, that is heard as a *hiss* sound from the speaker. White noise is so named because it's analogous to white light, which is a mixture of all visible wavelengths of light. A pure white noise signal may only be produced by a generator called a white noise generator and is often used as a sound source in sound synthesis. However, many sounds in nature and industry resemble white noise, such as steady rainfall, wind, storm etc. Although in electronics, any noise is harmful and unwanted, white noise is sometimes used for better night-sleep by masking environmental or outside noises that disturbs sound sleeping. Fig.1-114 shows the white noise in image signal and in sound signal.

### 1.23.4 Interference

Usually, interference comes from another electronic system such as an interfering radio sources. The name given to any predictable, periodic signal that occurs in an electronic circuit in addition to the signal the circuit is designed to process. This is distinguished from a noise signal by the fact that it occupies a relatively small frequency range, and because it is predictable it can often be filtered out. The source of interference can be both internal or external.

## 1.24 SNR and Noise Figure

**Signal-to-noise ratio** (SNR or S/N) is a measure used in science and engineering to compare the level of a desired signal to the level of background noise. The ratio is typically expressed as a single numeric value in decibels (dB). Mathematically,

$$SNR = \frac{P_s}{P_n} \quad \text{or,} \quad SNR (dB) = 10 \log \left( \frac{P_s}{P_n} \right) \quad (1-41)$$

where,  $P_s$  is the power of useful signal and  $P_n$  is the power of noise signal.

The ratio can be zero, a positive number or a negative number. A signal-to-noise ratio over 0 dB indicates that the signal level is greater than the noise level. **The higher the ratio, the better the signal quality [H\_E\_atn].** For example, a Wi-Fi signal with S/N of 40 dB will deliver better network services than a signal with S/N of 20 dB.

**Noise factor** (F) is a figure of merit that indicates degradation of the signal-to-noise ratio (SNR) caused by the components of a system. This figure of merit is used to evaluate the performance mainly of an amplifier or a radio receiver. Lower values noise factor indicate better performance. The noise factor is defined as the ratio of the input SNR to the output SNR. Mathematically,

$$F = \frac{SNR_i}{SNR_o} = \frac{P_{si}/P_{ni}}{P_{so}/P_{no}} \quad (1-42)$$

where,  $P_{si}$  is the power of input signal and  $P_{ni}$  is the power of input noise signal and  $P_{so}$ ,  $P_{no}$  are corresponding values at the output.

**Noise figure** (NF) is a similar parameter by which the noise performance of a radio receiver, amplifier, mixer or other circuit can be specified. The noise figure uses a logarithmic scale and is simply the noise factor expressed in decibels. **The lower the value of the noise figure, the better the performance of the system [H\_E\_atn].**

$$NF = 10 \log \left( \frac{P_{si}/P_{ni}}{P_{so}/P_{no}} \right) \quad (1-43)$$

**Example 1-16**

The input signal power of an amplifier is  $200 \mu\text{W}$  and that of noise power is  $2.5 \mu\text{W}$ . If the signal power at the output is  $3 \text{ W}$ , and the noise power is  $200 \text{ mW}$ , calculate the noise factor and noise figure of the amplifier.

**Solution:**

Here given: Input signal power,  $P_{si} = 200 \mu\text{W}$ , input noise power,  $P_{ni} = 2.5 \mu\text{W}$ .

Output signal power,  $P_{so} = 3 \text{ W}$ , output noise power,  $P_{no} = 200 \text{ mW}$ .

Signal to noise ratio at input:  $SNR_i = \frac{P_{si}}{P_{ni}} = \frac{200 \mu\text{W}}{2.5 \mu\text{W}} = 80$

Signal to noise ratio at output:  $SNR_o = \frac{P_{so}}{P_{no}} = \frac{3 \text{ W}}{200 \text{ mW}} = 15$

Noise factor:  $F = \frac{SNR_i}{SNR_o} = \frac{80}{15} \approx 5.33 \text{ [Ans.]}$

Noise figure:  $NF = 10 \log(5.33) \approx 7.27 \text{ dB [Ans.]}$

**Comments:** The value of  $NF$  is comparative small. It means that, in terms of noise, the amplifier's quality is good.

## 1.25 Electro-Mechanical Switches (Relays)

**Relays** are basically electromechanical switches used to turn ON or OFF a circuit with the help of electromagnets. Relays are one of the most important electromechanical devices highly used in

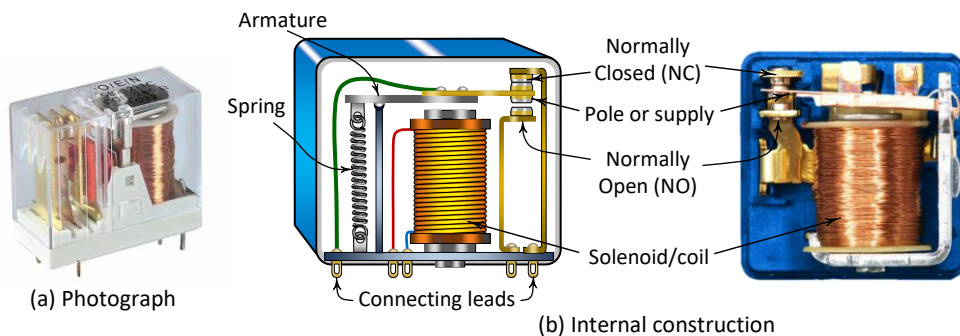


Fig. 1-114: Electromagnetic relays: (a) Photograph, (b) Internal components

industrial applications, specifically in automobiles. A relay can be used to switch ON or OFF electrical circuits operating at high AC voltage, using a low DC control voltage. It provides electrical isolation between the controlling circuit (DC circuit) and the controlled circuit (AC circuit). Relays

were used as constructive parts of some early calculators and computers. US scientist Joseph Henry (1797–1878) invented the relays in 1835.

Relays have basically three main parts. These are: (i) **Electromagnetic coil** (solenoid), (ii) **Contacts-Normally Open (NO) and Normally Closed (NC)**, and (iii) A **movable Contact or Pole** fitted with a spring [as shown in Fig.1-115].

An electromagnetic coil is made up of a copper wire wound on a ferromagnetic core in the shape of a solenoid. When voltage is applied to this electromagnetic coil, current flows through the coil and a magnetic flux is produced in the core. Thus, the core works a bar magnet.

There are two types of contacts in the electromagnetic relay. These are: Normally Open (NO), and Normally Closed (NC) contacts. Normally open contact is connected to the pole when the coil is energized. Normally closed contact remains connected to the pole, when the coil is not energized. Generally the supply voltage is connected to the pole, which is supplied through the NO or NC terminals.

The NO and NC contacts are also called throws and the moveable contact (attached to armature) is called the pole. Mostly used relays have one pole and two throws. So these types of relays are called **SPDT** (Single Pole Double Throw) relays. On the other hand, an SPST (Single Pole Single Throw) relay has only one pole and one throw (ON). There are some other types of relays. The schematic symbols of different types of relays are shown in Fig.1-116 and Fig.1-117 shows the pin-connection of a common relay.

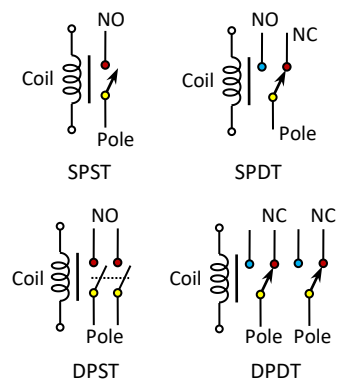


Fig.1-115: Symbols of different types of relays

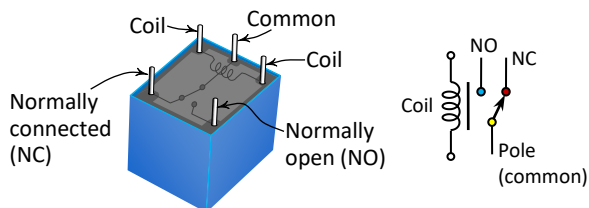


Fig.1-116: Pin-configuration of a common relay

### 1.25.1 Operation of Electromagnetic Relays

Fig.1-118 2.25 shows the working principle of a relay. The line voltage (phase) is connected to the pole (pin 1), a bulb is connected to NO (pin 2) terminal and a fan is connected to NC terminal (pin 3) of the relay. A DC voltage (batteries) is connected to the solenoid of the relay through a switch. When the batter switch is OFF [Fig.1-118(a)] the relay is not energized. So the AC voltage will be connected to the fan through NO (pin 3) terminal. So the fan will run but the bulb will remain OFF. On the other hand, when the DC voltage is applied to the coil, the coil is energized and a magnetic field is set up in the core, which attracts the armature downward. So the pole will be connected to NO (pin 2) contact. Thus, the bulb will be ON, but the fan will be OFF. In this way the power of the loads are controlled by energizing and de-energizing the relay coil. Look at the figure, the DC circuit and the AC circuits are completely isolated.

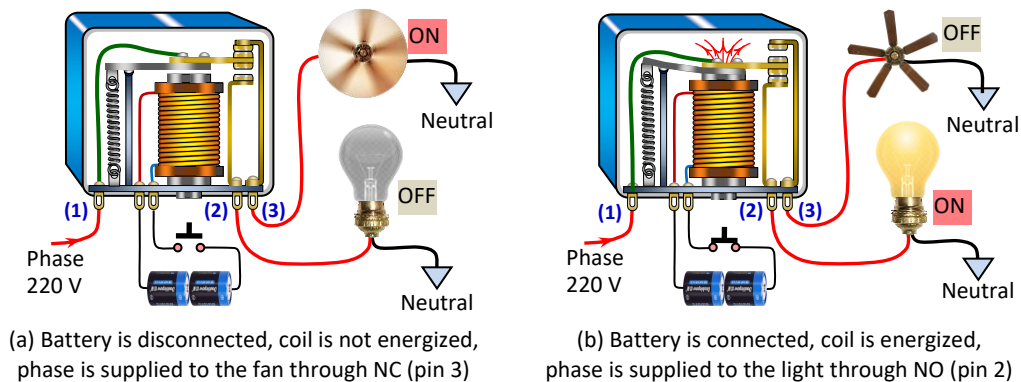


Fig.1-117: Controlling loads using relay

## 1.26 Wire Gauge and Ampacity

Wire gauge is a measurement of wire diameter. This determines the amount of electric current the wire can carry safely, as well as its electrical resistance and weight. Two standards are mainly used to represent the wire size and ampacity (carry capacity to current). These are **British Standard Wire Gauge (SWG)** dropping the British part) and **American Wire Gauge (AWG)**. These two are actually not equivalent in size. In general, an SWG wire would have larger diameter (though not for all) than an AWG wire of the same gauge. AWG is widely used in electrical wires while SWG is not so much. AWG is a logarithmic stepped standardized wire gauge system used since 1857. In AWG, there are 40 gauge numbers (1, 2 upto 40). The higher the AWG numbers (like 14, 16, 20, 40), the thinner the wire. The biggest 4/0 wire is the thickest and can carry the maximum amps (230 A). The thicker wire sizes are represented with multiple zeros as **number of zeros** or **number of zeros/0** as shown in Table 2. Different AWG wires have different diameters, cross-sections, and current capacity to flow through them.

AWG wire gauge chart for a standard copper wire is shown in Table 1-2 (**full chart is given in Appendix-H\_C\_atn**). We can read the diameter (in inches or mm), cross-section (in  $\text{mm}^2$ ), and ampacity, ie, the amount of amps a wire can handle. From this chart it can be observed that if the gauge number of a wire is increased by 3, the resistance ( $\Omega/\text{km}$ ) becomes double and if it is increased by 10 the resistance ( $\text{m}\Omega/\text{ft}$ ) will also be double. To understand it, look at the AWG chart in Table 1-2. The resistance of 1 AWG wire is  $0.1239 \text{ m}\Omega/\text{ft}$  and the resistance of 4 AWG (increased by 3) is  $0.2485 \text{ m}\Omega/\text{ft}$ .

In addition to the ampacity, the AWG Table, gives the maximum frequency. This data is useful for high frequency AC applications. When high frequency AC is conducted by a wire there is a tendency for the current to flow along the outside of the wire- called **skin-effect**. Skin-effect increases the effective resistance. The frequency listed in the table shows the frequency at

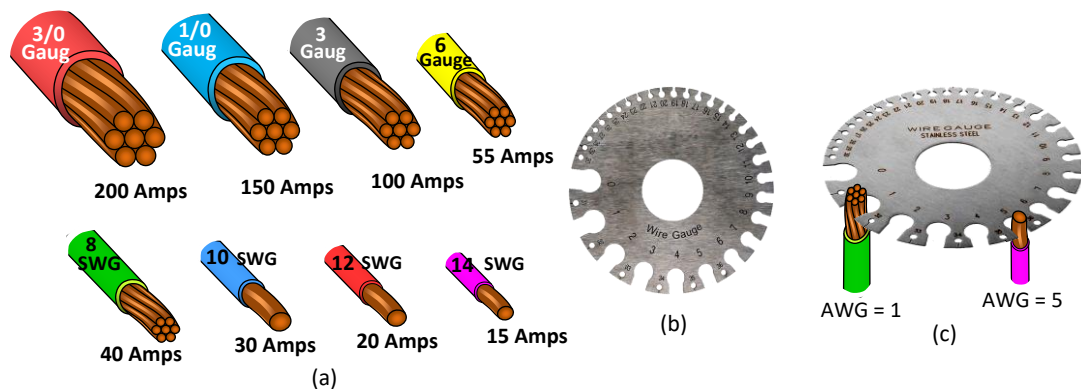


Fig.1-118: (a) AWG of different wire (b) AWG measuring disc, and (c) AWG measuring process

which the calculated skin depth is equal to the radius of the wire, and is an indication that above this frequency we must consider the skin effect when calculating the wire resistance.

Table 1-2: AWG and Their Related Parameters

| AWG       | Diameter (inches) | Diameter (mm) | Area (mm <sup>2</sup> ) | Resistance (mΩ/ft) | Resistance (Ω/km) | Max. Current (A) | Max. Frequency for 100% skin depth |
|-----------|-------------------|---------------|-------------------------|--------------------|-------------------|------------------|------------------------------------|
| 0000(4/0) | 0.46              | 11.684        | 107                     | 0.049              | 0.16072           | 302              | 125                                |
| 000(3/0)  | 0.4096            | 10.40384      | 85                      | 0.0618             | 0.202704          | 239              | 160                                |
| 00 (2/0)  | 0.3648            | 9.26592       | 67.4                    | 0.0779             | 0.255512          | 190              | 200                                |
| 0 (1/0)   | 0.3249            | 8.25246       | 53.5                    | 0.0983             | 0.322424          | 150              | 250                                |
| 1         | 0.2893            | 7.348         | 42.409                  | 0.1239             | 0.406392          | 119              | 325                                |
| 2         | 0.2576            | 6.544         | 33.632                  | 0.1563             | 0.512664          | 94               | 410                                |
| 3         | 0.2294            | 5.827         | 26.671                  | 0.197              | 0.64616           | 75               | 500                                |
| 4         | 0.2043            | 5.189         | 21.151                  | 0.2485             | 0.81508           | 60               | 650                                |
| 5         | 0.1819            | 4.621         | 16.774                  | 0.3133             | 1.027624          | 47               | 810                                |
| 6         | 0.1620            | 4.115         | 13.302                  | 0.3951             | 1.295928          | 37               | 1100                               |
| 7         | 0.1443            | 3.665         | 10.549                  | 0.4982             | 1.634096          | 30               | 1300                               |
| 8         | 0.1285            | 3.264         | 8.366                   | 0.6282             | 2.060496          | 24               | 1650                               |
| 9         | 0.1144            | 2.906         | 6.634                   | 0.7921             | 2.598088          | 19               | 2050                               |
| 10        | 0.1019            | 2.588         | 5.261                   | 0.9989             | 3.276392          | 15               | 2600                               |

The gauge number can be measured using different instrument, the most easy way is to use gauge disks [Fig.1-119(b),(c)]. We have to place the wire into different grooves until we find the best fit.

The fit should be snug but not too tight. The AWG value of the wire will be the size of the smallest gauge groove through which it barely slides into.

For a given AWG, the diameter of the wire can be calculated using the following Equ.(1-44)

$$D_n = 0.127 \text{ mm} \times 92^{\frac{36-n}{39}} \quad (1-44)$$

For example, the diameter of a 30 AWG will be,

$$D_{30} = 0.127 \text{ mm} \times 92^{\frac{36-30}{39}}$$

$$\text{or} \quad D_{30} = 0.127 \text{ mm} \times 92^{(6/39)}$$

$$\text{or} \quad D_{30} = 0.127 \text{ mm} \times 92^{(0.153846)}$$

$$\text{or} \quad D_{30} = 0.127 \text{ mm} \times 2.005$$

$$\therefore D_{30} = 0.254635 \text{ mm}$$

Similarly, the cross-sectional area of a wire can be calculated using the following equation,

$$D_n = 0.012668 \text{ mm}^2 \times 92^{\frac{36-n}{19.5}} \quad (1-45)$$

According to the number of cores, wires are categorized as **stranded wire** and **solid wire**. As shown in Fig.1-120 stranded wire is formed by twisting thin strands of wires and bundled together to form a cable-like solid wire. On the other hand, a solid wire is made up of a single strand. The main advantage of a stranded wire is its flexibility. So, stranded wire can be configured to meet the needs of a wide variety of applications. It is ideal for joining electronic circuit components in confined limited spaces. As shown in Fig.1-119(c), the AWG of stranded wire is also measured by the AWG disk in the same way.

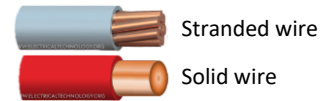


Fig.1-119: Stranded and solid core wire

### Example 1-17

A variable voltage source is connected to a 2 km long wire. If the wire number is 10 AWG, calculate the maximum voltage that can be applied to the wire. To verify the formula of Equ.(1-44), also calculate the diameter of the wire.

#### Solution:

From the AWG table [Table 1-2], we find that a 10 AWG wire has  $3.276392 \Omega/\text{km}$  and its ampacity is 15 A. Total resistance of the wire will be,

$$R_{10} = (3.276392 \Omega/\text{km}) \times (2 \text{ km}) \approx 6.55 \Omega$$

Maximum voltage

$$V_{max} = I_{10} R_{10} = (15 \text{ A}) \times (6.55 \Omega) = 98.25 \text{ V [Ans.]}$$

Using Equ.(1-44)

$$D_{10} = 0.127 \text{ mm} \times 92^{\frac{36-10}{39}}$$

$\therefore D_{10} \approx 2.588 \text{ mm}$  [Ans.] same as given in the AWG table

**Comments:** AWG and SWG numbering are different. So, give attention in using the wire gauge number.

## 1.27 Neon Lamp

A miniature gas discharge lamp in which the electrical discharge takes place through a mixture of gases containing a large proportion of neon gas is known as **neon lamp**. Electrons are emitted from cathode by subjecting the surface of the cathode to very high electric voltage so that the electrons are forcibly ejected from the orbits of the atoms against the nuclear centripetal forces.

### Construction of Neon Lamp

The construction of a typical neon lamp is shown in Fig.1-121.

The neon lamp has two closely spaced electrodes, but not touching each other, in the form of thick parallel wire or flat plates sealed within a soft glass envelope. The glass envelope is filled with the neon gas at a pressure of about 8 torr. Also, a series resistance of about 2 k $\Omega$  is mounted inside the glass envelope.

### Working of Neon Lamp

When a high electric power supply is connected to the neon lamp, the cathode terminal starts emitting electrons. These electrons get accelerated by the electric field and collide with the neon atoms. On collision, a part of the energy is transferred from the electrons to the neon atoms and the neon atoms get excited. When the excited neon atoms and ions come to their ground state, they give off light.

In the neon lamps, the cathode is heated for emission of electrons. For this purpose, these require a very high voltage depending on the distance between the electrodes. In order to start discharging through neon gas, it requires 1.5 to 2 times voltage of the normal operating voltage.

A very small lamp current will give flickering of the gas discharge and a high lamp current will give a bright stable discharge, but life will be shorter and blackening of the envelope will occur. Therefore, the lamp current

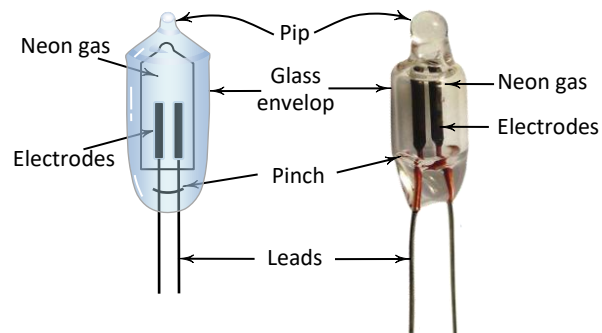


Fig.1-120: Construction of neon lamp

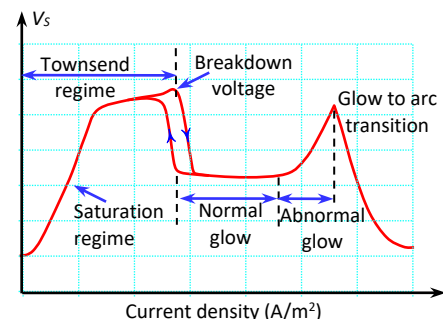


Fig.1-121: Characteristics of neon lamp

for a standard neon lamp is less than 1 mA or for a high brightness it is up to 2.5 mA.

When AC supply is used, each electrode will turn positive and negative alternatively. Hence, a continuous glow from the lamp is achieved. Therefore, these types of neon lamps are suitable for high operating voltage and low current applications. The V-I characteristics of a typical neon lamp is shown in Fig.1-122.

### Advantages and Disadvantages

The main advantages of neon lamps are: miniature configuration, less expensive, consumes less amount of electricity, long life (~5000 hours) and so on.

The main disadvantages of the neon lamps are: low efficiency, cannot be used for lighting purposes, small range of color spectrum.

### Applications of Neon Lamps

Neon lamps as used as- indicators, advertising purposes, decorative purposes etc.

## 1.28 Vacuum Tube to Semiconductor Devices

A vacuum tube is a type of glass tube from which all gaseous substances are expelled and the flow of electrons inside it is controlled by placing multiple electric grids in it [see Fig.1-123(a)]. In early electronic systems like- radio receivers, televisions, computers, etc. vacuum tubes were used as switches and amplifiers.

The journey of electronics began with the discovery of the **Edison Effect**. The Edison effect was the name given to a phenomenon that Edison observed in 1875 and refined later, in 1883, while he was trying to improve his new incandescent lamp. The effect was that, in a vacuum, electrons flow from a heated element like the filament of an incandescent lamp, to a cooler metal plate. Edison found no special value in the effect, but he patented it anyway. Today we call the effect the more descriptive term, **Thermionic Emission**.

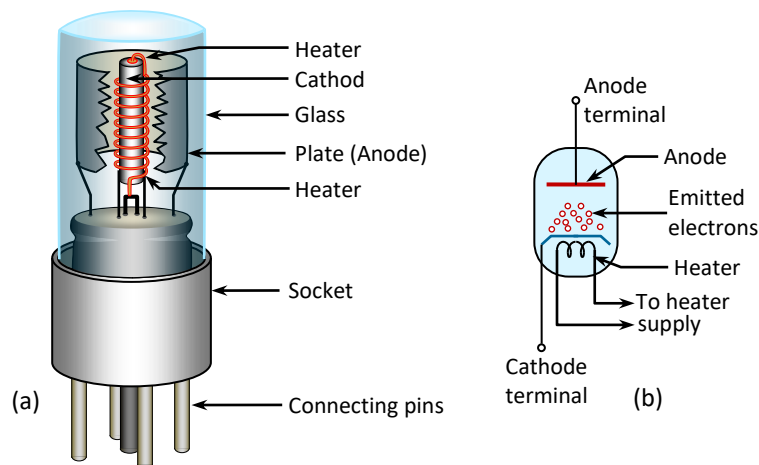


Fig.1-122: (a) Inner view of a vacuum tube, (b) Schematic symbol of a vacuum tube

Fleming, a British physicist, invented the first vacuum tube in 1904 using the Edison Effect. Fleming's vacuum tube acts as a rectifier (AC to DC converter). It had two electrodes [see Fig.1-123(b)], hence it was named **Vacuum Diode** or simply **Diode**. Then Forest invented another vacuum

tube called the triode, which had three electric grids. Many other scientists invented more vacuum tubes. The photographs of different types of vacuum tubes are shown in Fig.1-125.

In vacuum tubes, there was a heating arrangement at one end which heated the metal. The metal absorbed heat and emitted electrons. In the first half of the 20<sup>th</sup> century, radio broadcasting, long-distance telephone services, televisions, and the earliest electronic digital computers, all these technologies were implemented using vacuum tubes. Electronic computers, in particular, were the largest system made of vacuum tubes.

In the second half of the 20th century, in almost all practical cases, vacuum tubes were replaced by semiconductor devices called transistors. Even then, the use of vacuum tubes in image display devices such as televisions and computer monitors continued. These were called *cathode ray tubes* (CRT). CRTs are also used today as high-frequency transmitters in microwave reactors (a machine used for chemical reaction enhancement) and artificial satellites in space.

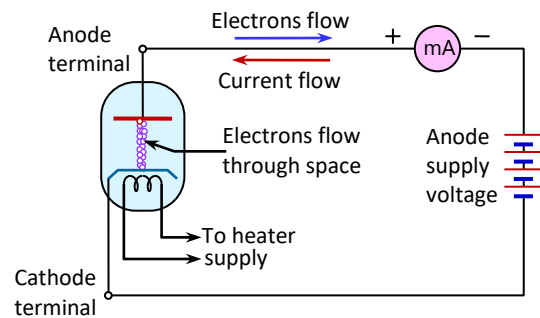


Fig.1-123: A circuit using vacuum

The birth history of semiconductor devices can be traced back to the invention of the rectifier (AC-DC converter) in 1874. Decades later, Bardeen and Brattain at Bell Laboratories in the US invented the point-contact transistor in 1947, and Shockley invented the junction transistor in 1948. This heralded the arrival of the transistor era. In 1946, the University of Pennsylvania in the US built a computer using vacuum tubes. The computer was so large that its vacuum tubes occupied the entire building, and it consumed a huge amount of electricity and produced a lot of heat. Later, the innovative transistor-calculator (computer) was developed, and since then computers have grown by leaps and bounds. In 1956, the Nobel Prize in Physics was awarded jointly to Shockley, Bardeen, and Brattain for their contribution to semiconductor research and the development of the transistor.



Fig.1-124: Photographs of different types of

The semiconductor industry grew rapidly following the invention of the transistor. In 1957, it already exceeded the scale of 100 million dollars. In 1959, the bipolar integrated circuit (ICs) was invented by Kilby of Texas Instruments and Noyce of Fairchild Semiconductor in the US. This invention had a major impact on the history of semiconductors, and it marked the dawn of the IC era. Being small in size and light in weight, the IC was widely used in a variety of electric appliances.

In 1967, Texas Instruments developed the electronic desktop calculator (the calculator) using IC. In Japan, electronic equipment manufacturers released calculators one after another, and fierce **calculator wars** continued until the end of the 1970s. IC integration advanced even further, and the large-scale integrated circuit (LSI) was developed. The technologies continued to advance. The VLSI (from 100 thousand to 10 million electronic components per chip) was developed in the 1980s, and the ULSI (more than 10 million electronic components per chip) was developed in the 1990s. In the 2000s, the system LSI (a multifunction LSI with multiple functions integrated into a single chip) was put into full-scale production. As IC progresses toward high performance and multiple functions, its field of application is expanding broadly. Semiconductors are now used in every corner of our society and support everyday life.



Fig.1-125: Magnetic effect of Loadstone

## 1.29 What is Electronics

**Electronics** is a branch of electrical engineering that deals with the flow of electrons through vacuum tubes or semiconductor devices. Currently, the functions of all vacuum tubes can be done using very small semiconductor devices, so relatively larger vacuum tubes are not used.

Among the semiconductor devices used in electronics are diodes, transistors, JFETs, MOSFETs, SCRs, and various types of ICs. The main goal of electrical and electronics engineering is to make all kinds of circuits and devices using these semiconductor components. We know that the contribution of computers, mobile phones, cars, etc. to the betterment of modern civilization is immense. On the other hand, computers, mobile phones are the contribution of electronics. Electronics has a special contribution to build cars, launches, steamers, factories, homes, etc.

## 1.30 History of Electronics:

The electronic world was not just the effort of some years or decades; rather it is the results of hard working of a number of great men since hundreds of years and more.

The formal beginning of electrical engineering goes back to 600 BC through the discovery of electrostatic attraction. **Thales of Miletus**, a Greek philosopher rubs together lodestone (naturally magnetized piece of the mineral magnetite) and amber (Amber is fossilized tree resin that has been appreciated for its color and natural beauty since Neolithic times) and he found that they attract.

In 1600 AD, an English scientist William Gilbert coined a new Latin word “electricus” from the Greek word for Amber “elektron” from which later we got the two the English words “Electric” and “Electricity”.



Fig.1-126: Amber (A hard translucent fossilized resin originating from extinct coniferous trees)

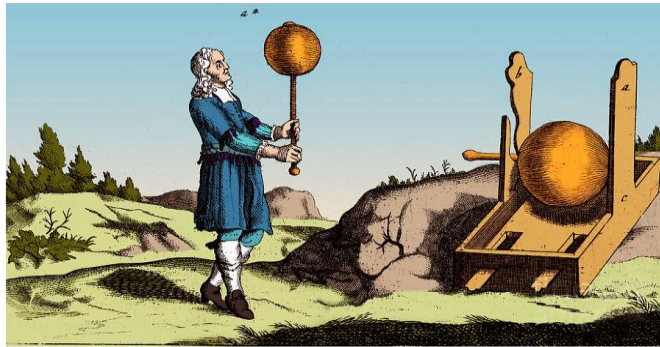


Fig.1-127: Otto von Guericke's electrostatic generator

In 1663 AD, German scientist **Otto Van Guericke** invented the first electric generator, which produced static electricity by applying friction against a revolving ball of sulfur (see Fig.1-128).

In 1750 AD, American scientist **Benjamin Franklin** proposed that flow of charges takes place during thunder and lightning.

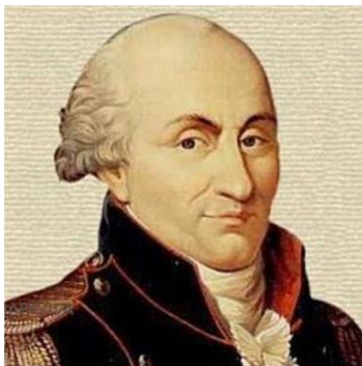


Fig.1-129(a): Charles-Augustin de Coulomb (1736-1806)

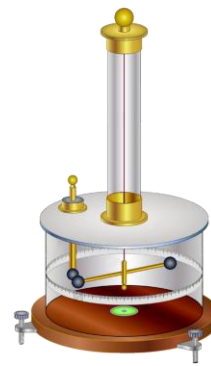


Fig.1-129(b): Torsion balance

In 1785 AD, a French physicist **Charles-Augustin de Coulomb** explained the interaction between electrically charged particles using a torsion balance (see Fig.1-130 ), which is now called, "electrostatic force" and the theory he proposed is named as "Coulombs Law".

**Luigi Aloisio Galvani**, an Italian physician, physicist, biologist, and philosopher, discovered animal electricity. In 1780, he discovered that the muscles of a dead frog's legs twitched when struck by an electrical spark.

In 1800 AD, an Italian physicist **Alessandro Volta** invented the first electric battery known as a **galvanic pile** (Fig.1-132). He discovered the existence of "potential difference" and charge flow occurs when there is a potential difference. The terminologies of "**Voltage**" for the potential difference and "**Volts**" for its unit are coined after his name.

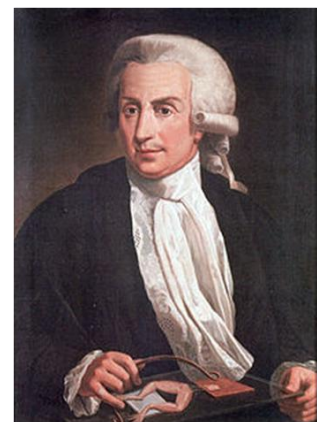


Fig.1-130: Luigi Galvani (1737-1798)

In 1820 AD, **Hans Christian Oersted**, a Danish physicist and chemist discovered that electric current in a wire can deflect a magnetized compass needle and hence laid the foundation of "Electromagnetism". In 1820 AD, **André-Marie Ampère** proposed the "**Ampere Force Law**" which describes the interaction between electricity and magnetism. The unit of current is named after him.

In 1827 AD, German physicist **George Simon Ohm** stated that every material has a certain kind of "**resistance**" that prevents the flow of charges through them. He proposed that the current flowing through a conductor between two points is directly proportional to the potential difference between the points- which is known as **Ohm's Law** after his name.

In 1831 AD, an English scientist **Michael Faraday** discovered "**Electromagnetic Induction**" which is the basic working principle of generators, motors and transformers. He proposed the "**Law of Electrolysis**" which is the basis of "**Batteries**" in 1834.

In 1876, **Alexander Graham Bell** obtained the patent for the first form of "**Telephone**". In 1879, an American inventor **Thomas Alva Edison** invented the most stable and reliable form of the **electric bulb** and patented it in the USA. Edison also invented the "Edison effect" presently the "Thermionic effect" (discussed earlier).

In 1887, German physicist **Henry Hertz** generated electromagnetic waves in his laboratory as postulated by **Maxwell**. He also proposed methods to transmit and receive electromagnetic wave which is used for radio, television, satellite, and telecommunication engineering.

In 1897, Italian physicist **Guglielmo Marconi** invented the first form of the radio and obtained a patent. He also

used an "Antenna" for the transmission of radio waves. **Sir Jagadish Chandra Bose** also worked on radio, microwave optics, and wireless signaling.

From 1880 to 1943, a Serbian American inventor, **Nikola Tesla** developed "**Alternating Current**" which is much cheaper to produce and transmit as compared to "Direct Current".

### Electronics with Vacuum Tubes

When the "Alternating Current" or AC became popular scientists began to think about converting AC to DC rather than generating DC which is expensive. Already we have discussed that, in 1904, English Physicist James Ambrose Fleming, invented the first vacuum tube (or vacuum diode) which is used as a rectifier, i.e. to convert AC to DC using the principle known as the "Edison Effect". And the History of Electronics begins here with the invention of this vacuum tube. In 1906, American inventor **Lee De Forest** and **Robert von Lieben** independently invented a three-terminal vacuum

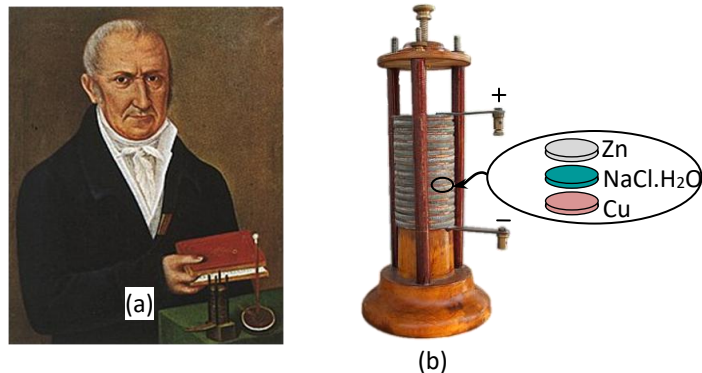


Fig.1-131 (a): Alessandro Volta (1745-1827)  
(b): Galvanic pile, the first battery

device called "**triode**" for first time which is used as an amplifier. Within 10 years, the device was used in radio transmitters and receivers as well as in systems for long-distance telephone calls.

Though the development of television was the result of work by many inventors, many historians credit **John Logic Baird**, a Scottish engineer, to produce a live, moving, grayscale television image. In 1928, **Philo Farnsworth** made the first public demonstration of a purely electronic television. During the 1930s, several countries began broadcasting, and after World War II it spread to millions of receivers, eventually worldwide. Since then, televisions and video displays have evolved from bulky electron tube technology to use more compact devices, such as plasma, LCD, and LED displays. Recently even lower power devices such as the organic light-emitting diode (OLED) displays, and it is most likely to replace the LCD and plasma technologies.



Fig.1-133(a): A replica of the first transistor, invented at Bell Labs on 23 December 1947



Fig.1-133(b): The same transistor, without glass cover

The German inventor, **Christian Hülsmeyer**, was the first to use radio waves to detect the presence of distant metallic objects. In 1904, he demonstrated the feasibility of detecting a ship in dense fog, but not yet its distance from the transmitter.

### Beginning of Modern Electronics

The discovery of the P-N junction is usually attributed to American physicist **Russell Shoemaker Ohl** of Bell Laboratories in 1939. However, **Vadim Evgenievich Lashkaryov**, a prominent soviet experimental physicist reported discovery of P-N junctions in cuprous oxide ( $\text{Cu}_2\text{O}$ ) and silver sulphide ( $\text{Ag}_2\text{S}$ ) photocells and selenium rectifiers in 1941.

In 1947, three researchers of Bell Labs at the USA- **John Bardeen**, **William Brattain**, and **William Shockley** discovered that when two gold point contacts were applied to a germanium crystal, an amplified signal is produced with greater power. In fact, it was the first bipolar transistor (Fig.1-134). "**Transistor**" term is coined by **John R. Pierce**. For their invention of transistor, they won the Nobel Prize in Physics in 1956.

Though were not able to practically build the device, the concept of a field-effect transistor (FET) was first

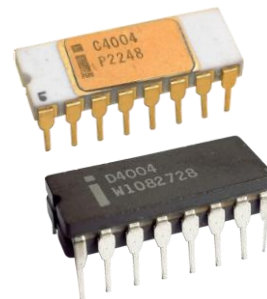


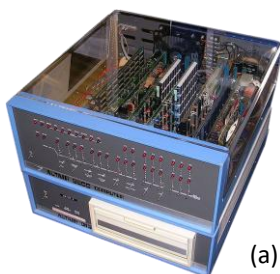
Fig.1-134: Photograph of the world first Microprocessor (Courtesy of the Intel Cop.)

patented by Austro-Hungarian physicist **Julius Edgar Lilienfeld** in 1925 and by **Oskar Heil** in 1934. The transistor effect of a field-effect transistor (FET) was observed and explained by **John Bardeen** and **Walter Houser Brattain** while working under William Shockley at Bell Labs in 1947.

In 1958, an American inventor **Jack Kilby** of Texas Instruments made the first **Integrated Circuit (IC)** that is a complete circuit in a single package. The MOSFET was invented by Mohamed M. Atalla and Dawon Kahng at Bell Labs in 1959, and first presented in 1960. It is the basic building block of modern electronics.

In 1968, **Gordon Moore** and **Robert Noyce** co-founded a new company- "Intel Corporation" to work in the field of micro electronics. In 1971, they invented the world's first microprocessor 4004 comprising of 2300 transistors in a single Si chip. The invention of microprocessors accelerated the revolutionary product "Computer" and "Computer Industry" around the globe.

Micro Instrumentation and Telemetry Systems (MITS) developed the Altair 8800 [Fig.1-136(a)] personal computer that used the new Intel 8080 microprocessor in 1975.



(a)



(b)

Fig.1-135: Photograph of the (a) First Altair 8800 PC (b) Apple PC

**Bill Gates** and **Paul Allen** write software for Altair using BASIC language. On 12 August 1981, IBM built the world first personal computer which could easily be used by new users and businesses.

Steve Jobs and his friend Steve Wozniak manufactured handmade personal computer **Apple I** in 1976 and they gave birth to a new computer industry "Apple Inc.".

Intel launched the world's first 16-bit microprocessor named 8086 in 1978. The world was revolving around 8-bit microprocessors till then. Intel 8088, released in 1979, was a slightly modified chip with an external 8-bit data bus (allowing the use of cheaper and fewer supporting ICs). Intel also introduced its first "**Microcontroller**" as Intel 8048 with a RAM and ROM added along with a processor on a single chip. Since then, a number of semiconductor device manufacturers developed different microprocessors, microcontrollers, and other integrated circuits that shift the electronics from discrete components to IC-based circuits.

### 1.31 Current Research on Electronics

For high-speed computation, it requires high-speed devices and hence high-speed transistors. Thus, the electronics industry is downscaling the sizes of the contemporary transistors constantly. This also saves power, cost, and chip size as well as enhances the functionality of the device. If the size of the MOSFET goes below 10 nm then MOSFET will be at its basic limits of operation. At this size, many physical complexities will arise. This leads to the development of **single electron transistors**

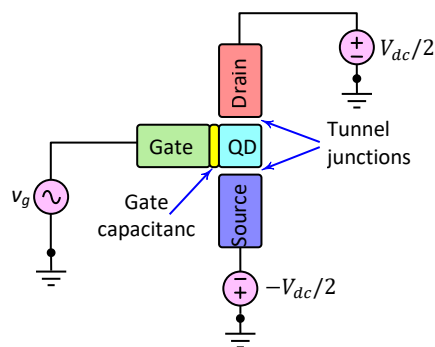


Fig.1-136: Schematic diagram of a single-electron transistor

(SET). The single electron transistor is a new type of switching device that uses controlled electron tunneling to amplify current. A tunnel junction consists of two pieces of metal separated by a very thin ( $\sim 1$  nm) insulator. The first semiconductor SET was fabricated accidentally in 1989 by **Scott-Thomas** et al. in narrow Si field effect transistors. Shortly thereafter Meirav et al. made controlled devices.

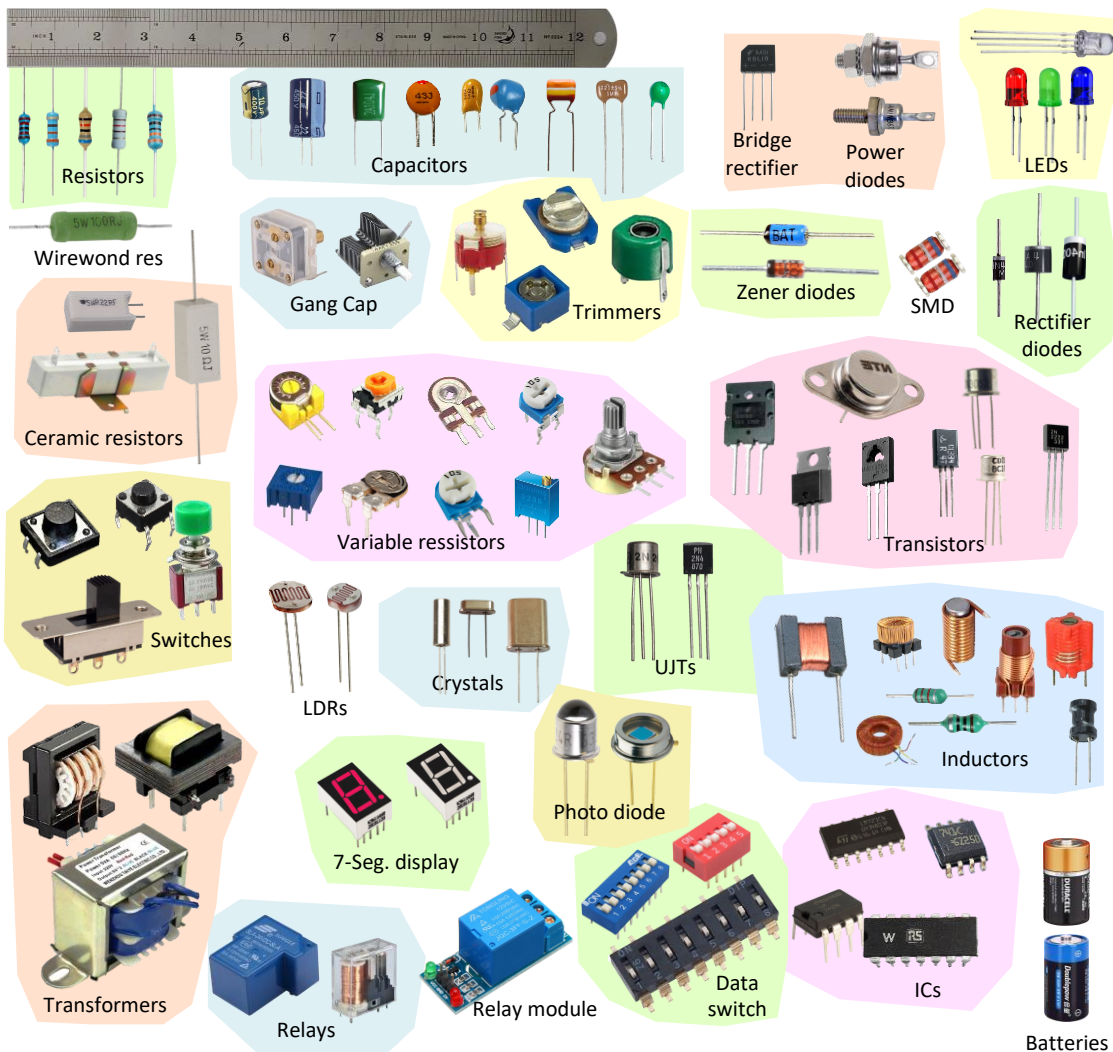


Fig.1-137 : Photographs of the components used in analog electronics lab

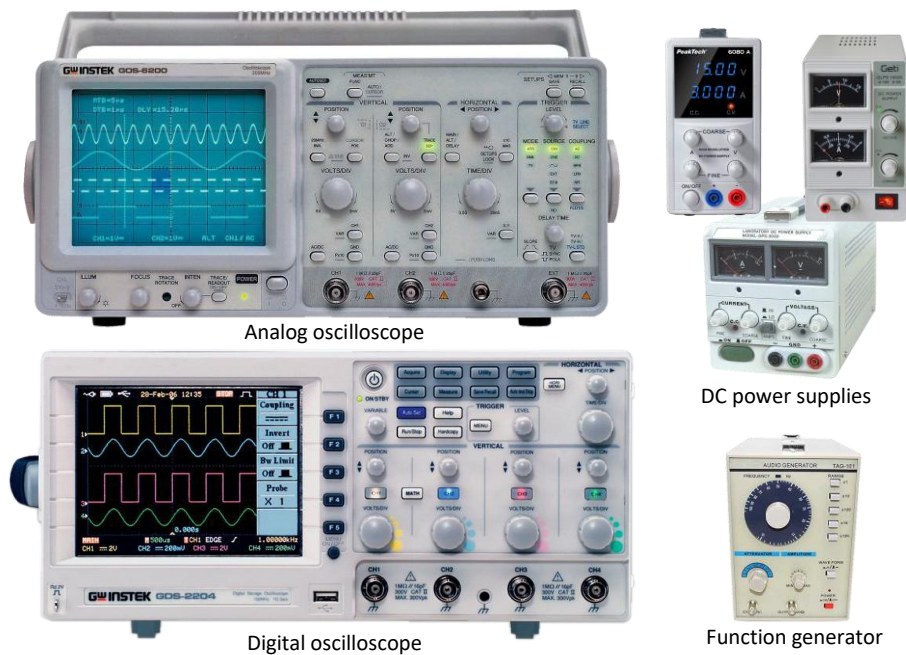


Fig.1-138: Photographs of some common equipment used in the electronics lab



Fig.1-139: Photographs of different tools and meters used in Analog Electronics Lab

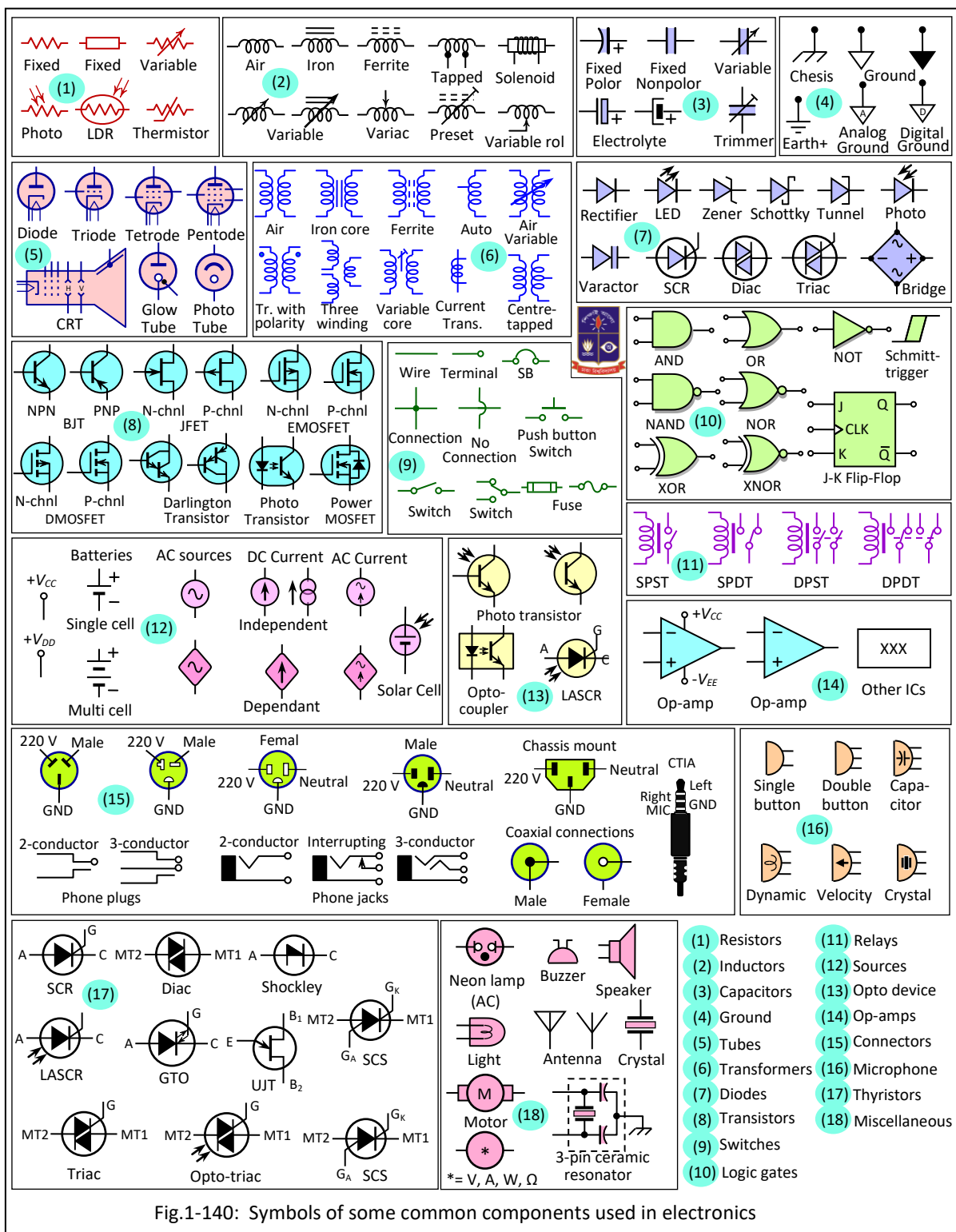


Fig.1-140: Symbols of some common components used in electronics

### 1.32 Sample Questions

1. Explain different kinds of formulas
2. What is approximation in electronics? Describe different types of approximations.
3. What is potential difference? Explain water tank analogy for potential difference.
4. What is AC and DC current?
5. Describe different types of voltage sources.
6. Describe different types of current sources.
7. What is stiff voltage source and stiff current source?
8. Describe different mechanisms of charge carriers movement.
9. What is displacement current? How displacement current is produced in a capacitor?
10. Give the classifications of resistors.
11. Describe carbon composition resistors.
12. Describe ceramic composition resistors.
13. Describe wire wound resistors.
14. Explain how film resistors are made.
15. What are the differences between metal film and metal oxide film resistors?
16. What are the advantages of SMD resistors?
17. Describe the construction and working procedure of a potentiometer.
18. Describe the working principle of a light dependant resistor.
19. Explain how color code is used to represent the value of resistance.
20. Explain how to determine the equivalent resistance of series connected resistors.
21. What will be equivalent power ratings of series/parallel connected resistors.
22. What is self inductance and mutual inductance?
23. Describe the construction of toroidal inductor and mention the advantages of this type of inductors.
24. What are the advantages and disadvantages of iron core and ferrite core inductors?
25. Describe the construction and working process of slug type variable inductors.
26. Describe the color coding method to represent the values of inductors.
27. Explain how to determine the equivalent inductance of series connected inductors.
28. Explain how to determine the equivalent inductance of parallel connected inductors.
29. Explain how to determine the equivalent inductance of series connected inductors with mutual coupling.
30. Explain how to determine the equivalent inductance of parallel connected inductors with mutual coupling.
31. Describe the theory of coupled inductors. Derive the expression for the input impedance when a load resistance is connected at the output coil.
32. Describe the construction and operation process of a parallel plate capacitor.
33. Give the general classifications of capacitors.
34. Describe the construction of Al electrolytic capacitors.
35. Why electrolytic and tantalum capacitors have polarity?
36. Describe the construction of tantalum electrolytic capacitors.
37. Describe the construction of super capacitors. Write some major applications of super capacitors.
38. What are the advantages of super capacitor as storage device over batteries?
39. What are the main disadvantages of super capacitors?

40. Describe the construction of ceramic capacitors. What are the main advantages of ceramic capacitors?
41. Describe different types of film capacitors.
42. Describe the construction of multi-layer SMD capacitor. What are the advantages of SMD capacitors?
43. Explain the working principle of variable capacitors.
44. What is trimmer capacitor? Describe its construction and operation process.
45. Draw the equivalent model of a capacitor.
46. Describe numeric coding system for representing the values of non-polar capacitors.
47. Describe color coding system for representing the values of non-polar capacitors.
48. Describe how to calculate the equivalent capacitance of series connected capacitors.
49. Describe how to determine the equivalent capacitance of parallel connected capacitors. Why the equivalent capacitance increases when connected in parallel?
50. What are the differences between analog and digital multimeter?
51. Explain how to measure the voltage and current in a circuit using a multimeter.
52. Explain how to measure the resistance of a resistor using multimeter.
53. What is sinusoidal signal? Define amplitude, frequency, and phase for this signal.
54. What is ramp voltage? Describe different types of ramp voltage.
55. What is saw-tooth wave? Describe different types of saw-tooth waves with figures.
56. What is triangular wave? Describe different types of triangular waves.
57. What is square wave? Describe different types of square waves.
58. Describe different types of pulses- steps and spikes? Describe with diagrams.
59. Describe the construction and working procedure of breadboard.
60. What is noise? What are internal and external noises? Describe different types of external noises.
61. What is thermal noise? Draw the equivalent circuit of thermal noise and describe.
62. What is internal noise? Describe different types of internal noises.
63. What is shot noise? Explain with a diagram.
64. What do you mean by transit-time noise? Explain the effect of transit-time on a signal.
65. What is white noise? Why is it called so? Is there any applications of white noise-explain.
66. Describe the terms: Signal-to-Noise Ratio (SNR), Noise Figure, and Noise Factor.
67. What is electromagnetic relay? Describe the construction and working principle of an electromagnetic relay.
68. Describe about the classifications of electromagnetic relays.
69. What is wire gauge? What are the types of standard used for measuring the ampacity of wire?
70. What is stranded and non-stranded wire? Compare their advantages and disadvantages.
71. Describe the construction and operation of a neon lamp.
72. Write some important applications of neon lamp.
73. Draw the characteristics of a neon lamp and describe different regions.
74. What is vacuum tube? Describe the construction and operation of a simple diode vacuum tube.
75. What is electronics? Write a brief history of electronics.

## 1.33 Exercises

1. The open circuit (output) voltage of a voltage source is 16 V. If a  $120\ \Omega$  load is connected at the output the voltage falls to 15.4 V. Determine the internal resistance of the voltage source.
2. The open circuit (output) voltage of a voltage source is 24 V. If the internal resistance of the source is  $10\ \Omega$ , what will be the output voltage when a  $220\ \Omega$  load is connected at the output.
3. The rated current capacity of a current source is 3 A, but when a  $30\ \Omega$  load resistance is connected at the output, 2.95 A current flows through it. Determine the internal resistance of the current source.
4. 1.2 A current flows through a  $330\ \Omega$  load resistance when connected at the output. The internal resistance of the current source is  $10\ \text{k}\Omega$ . Calculate the short-circuit current of the current source.
5. 10 mA current is flowing a Cu conductor. For this current, determine the drift velocity of electrons flowing through the conductor. Assume, one mole of Cu has a mass of  $63.5 \times 10^{-3}\ \text{kg}$ , Avogadro's number  $N_A = 6.02 \times 10^{23}$ , and the density of copper  $\rho = 8920\ \text{kg m}^{-3}$ .
6. A parallel plate capacitor of plate area  $50\ \text{cm}^2$ , and plate separation of 0.1 mm, is connected to a voltage source. In  $1\ \mu\text{s}$ , the voltage across the capacitor rises to 200 V. Calculate the displacement current. Assume the dielectric constant of the capacitor is  $\epsilon_0 = 8.85 \times 10^{-12}$ . [Ans. 88.5 mA]
7. The resistivity of Nichrome wire is  $\rho = \Omega \cdot \text{m}$ . Calculate the resistance of a 2 km long Nichrome wire with the diameter  $2\ \text{mm}^2$ .
8. As shown in Fig.1-142, 16 V is applied at terminal 3 of a potentiometer with terminal 1 grounded. The output voltage, at terminal 2, becomes 6 V when the wiper (terminal 2) is placed exactly at the middle of the resistive track. If the resistance of the pot is  $20\ \text{k}\Omega$ , determine the individual resistances of the two halves of the potentiometer. Is the resistive track linear?
9. For the following resistors, determine the value of resistance.

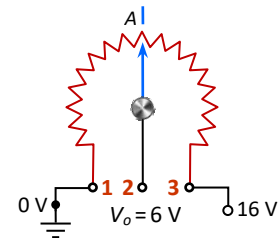


Fig.1-141: Figure for Ex.8

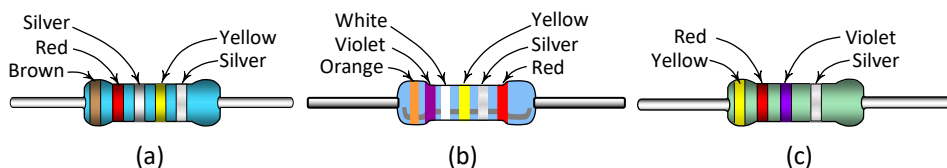


Fig.1-142: Resistors for Ex. 9

10. Suppose three resistors  $R_1$ ,  $R_2$ , and  $R_3$  are connected in series. The values of the resistors are  $4.7\ \text{k}\Omega \pm 10\%$ ,  $3.3\ \text{k}\Omega \pm 5\%$ , and  $2.2\ \text{k}\Omega \pm 2\%$ , respectively. Calculate the value of equivalent resistor with its tolerance. Also calculate the equivalent power of the combination.

11. Suppose three resistors  $R_1$ ,  $R_2$ , and  $R_3$  are connected in parallel. The values of the resistors are  $10\text{ k}\Omega \pm 5\%$ ,  $7.4\text{ k}\Omega \pm 2\%$ , and  $12\text{ k}\Omega \pm 1\%$ , respectively. Calculate the value of equivalent resistor with its tolerance. Also calculate the equivalent power of the combination.
12. Suppose  $R_1 = 100\ \Omega$ ,  $1\text{ W}$  and  $R_2 = 20\text{ k}\Omega$ ,  $2\text{ W}$ . If the resistors are connected in parallel, what will be power capacity of the combination?
13. Repeat Ex.12 for series combination.
14. Determine the inductance value of the inductor shown in Fig.1-144.

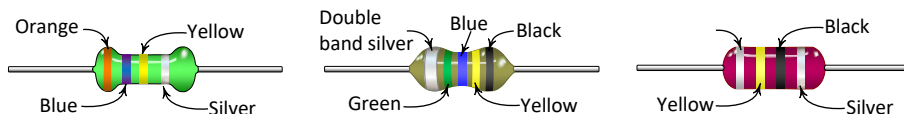


Fig.1-143: Inductors for Ex. 14

15. Find the equivalent inductance of the circuit of Fig.1-145. Assume,  $L_1 = 1.5\text{ H}$ ,  $L_2 = 2.2\text{ H}$ ,  $L_3 = 5.5\text{ H}$ ,  $M_{12} = 1.2\text{ H}$ ,  $M_{23} = 1\text{ H}$ , and  $M_{13} = 0.75\text{ H}$ .

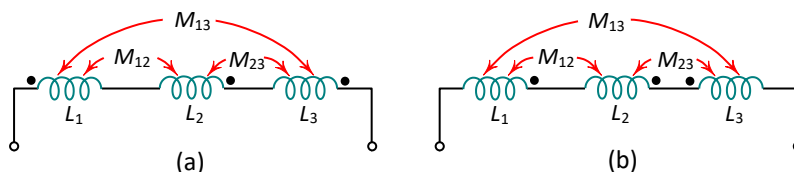


Fig.1-144: Series connected inductors for Ex.15 and 16

16. The values of two mutually coupled inductors are  $L_1 = 3\text{ H}$ ,  $L_2 = 6\text{ H}$ . If they are connected like i) Fig.1-145(a) and ii) Fig.1-145(b) determine the equivalent inductance. Assume,  $M_{12} = 2.5\text{ H}$ .

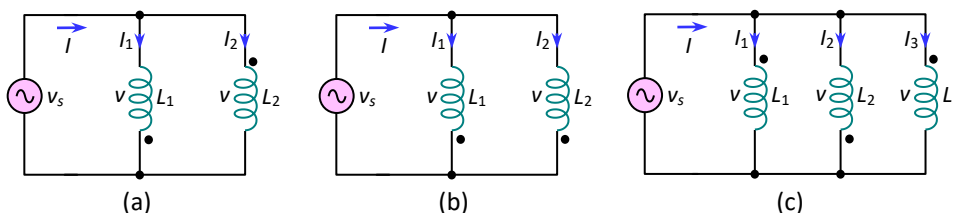


Fig.1-145: Parallel connected inductors

17. In Fig.1-146(c) the values of mutually coupled inductors are  $L_1 = 4\text{ H}$ ,  $L_2 = 2\text{ H}$ , and  $L_3 = 2.5\text{ H}$ . Assuming,  $M_{12} = 1.5\text{ H}$ ,  $M_{23} = 2.0\text{ H}$ , and  $M_{23} = 0.5\text{ H}$ , determine the equivalent inductance.

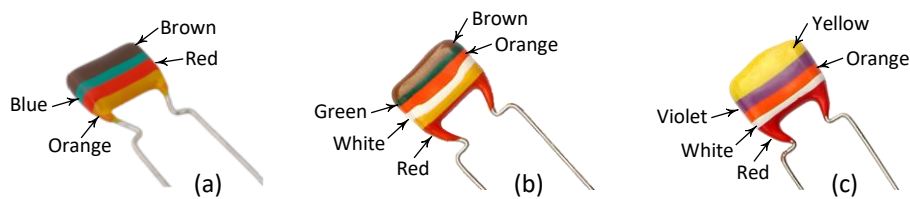


Fig.1-146: Series connected inductors for Ex. 18

18. Determine the value of capacitance of the capacitor shown in Fig.1-147. Calculate the value of reactance ( $X_C$ ) of each of them if the capacitors are used for 60 Hz AC voltage.

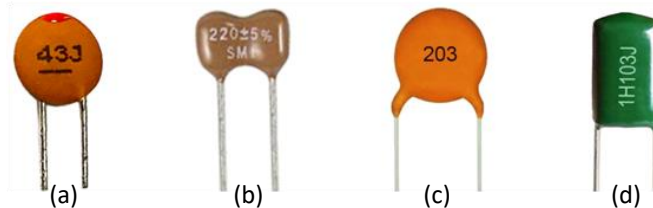


Fig.1-147: Series connected inductors for Ex. 19

19. Determine the value of capacitance of the capacitor shown in Fig.1-147. Calculate the reactance ( $X_C$ ) of each capacitors if they are used for 1 kHz AC voltage.
20. Suppose three capacitors have the ratings as  $C_1$ : 47  $\mu\text{F}$ , 25 V,  $C_2$ : 33  $\mu\text{F}$ , 16 V,  $C_3$ : 22  $\mu\text{F}$ , 50 V. Determine the equivalent capacitance and tolerance voltage when the capacitors are connected in series.
21. Suppose three capacitors have the ratings as  $C_1$ : 10  $\mu\text{F}$ , 16 V,  $C_2$ : 33  $\mu\text{F}$ , 20 V,  $C_3$ : 47  $\mu\text{F}$ , 35 V. Determine the equivalent capacitance and tolerance voltage when the capacitors are connected in parallel.
22. An AC voltage is represented as  $v_s = 12 \sin(100\pi t + 30^\circ)$ . Determine the amplitude, peak-to-peak value, frequency, and phase angle. Draw the graph of this AC voltage.
23. An AC current is represented as  $i_s = 110 \sin(120\pi t + 60^\circ)$ . Determine the amplitude, peak-to-peak value, frequency, and phase angle. Draw the graph for this AC current.
24. An amplifier has a bandwidth of 200 kHz with 2.2 k $\Omega$  as the input resistance. Determine the RMS value of the thermal noise at  $T = 27^\circ\text{C}$ .
25. The input signal power of an amplifier is 150  $\mu\text{W}$  and that of noise is 1.5  $\mu\text{W}$ . If the signal power at the output is 2.5 W, and the noise power is 120 mW, calculate the noise factor and noise figure of the amplifier.
26. A variable voltage source is connected to a 2 km long wire. If the wire number is 10 AWG, calculate the maximum voltage that can be supplied by the wire. Also calculate the diameter of the wire and check it from Table 1-2.